

PERFORMANCE OF EMPIRICAL RISK MINIMIZATION FOR PRINCIPAL COMPONENT REGRESSION

Christian Brownlees[†] Guðmundur Stefán Guðmundsson^{‡,*} Yaping Wang[†]

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Abstract

This paper studies the predictive performance of empirical risk minimization for principal component regression. Our analysis accommodates the leading eigenvalues of the predictor covariance matrix growing either linearly or sublinearly with the number of predictors. Additionally, we allow for both light-tailed and heavy-tailed data. Our main result establishes that, under appropriate conditions, empirical risk minimization for principal component regression is consistent for prediction and achieves near-optimal performance.

Keywords: empirical risk minimization, principal component regression, time series, oracle inequality

JEL: C13, C14, C22, C55

[†] Department of Economics and Business, Universitat Pompeu Fabra and Barcelona GSE;
e-mail: christian.brownlees@upf.edu, yaping.wang@upf.edu.

[‡] ACE and Department of Economics and Business Economics, Aarhus University;
e-mail: gsgudmundsson@econ.au.dk.

* Corresponding author.

1 Introduction

Principal component regression (PCR) is a regression methodology with a long and well-established tradition that can be traced back to at least Hotelling (1957) and Kendall (1957). In a nutshell, PCR is used to forecast a prediction target of interest on the basis of the principal components extracted from a potentially large set of predictors. PCR is closely related to diffusion index forecasting models, which are one of the leading forecasting methodologies in macroeconomics (Stock and Watson, 2012).

In this paper, we study the properties of PCR from a learning theory perspective. Our central contribution is establishing nonasymptotic prediction performance guarantees for PCR. Our main theorems may be interpreted as nonasymptotic analogues of the classic asymptotic results on the prediction properties of PCR obtained in Stock and Watson (2002a). An important feature of our analysis is that we consider both the case in which the largest eigenvalues of the predictor covariance matrix grow linearly with the number of predictors and the case in which they grow sublinearly. Moreover, we allow for both light-tailed and heavy-tailed data. Our framework builds upon assumptions and proof strategies developed in the factor model literature (Bai and Ng, 2002; Bai, 2003; Fan, Liao, and Mincheva, 2011, 2013; Onatski, 2012).

PCR may be described as a two-step procedure. Let $\mathcal{D}_T = \{(Y_t, \mathbf{X}_t')'\}_{t=1}^T$ be a stationary sequence of zero-mean random vectors taking values in $\mathcal{Y} \times \mathcal{X} \subset \mathbb{R} \times \mathbb{R}^p$. The goal is to forecast the prediction target Y_t using the p -dimensional vector of predictors $\mathbf{X}_t = (X_{1t}, \dots, X_{pt})'$. The first step of PCR consists in computing the $T \times K$ principal components matrix $\hat{\mathbf{P}} = (\hat{\mathbf{P}}_1, \dots, \hat{\mathbf{P}}_T)'$ associated with the $T \times p$ predictor matrix $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_T)'$, for some appropriate choice of K . This may be defined as the solution of the constrained least squares problem

$$(\hat{\mathbf{B}}, \hat{\mathbf{P}}) = \underset{\substack{\mathbf{B} \in \mathbb{R}^{p \times K} \\ \mathbf{P} \in \mathbb{R}^{T \times K}}}{\operatorname{argmin}} \|\mathbf{X} - \mathbf{P}\mathbf{B}'\|_F^2 \quad \text{s.t.} \quad \frac{1}{T}\mathbf{P}'\mathbf{P} = \mathbf{I}_K, \frac{1}{p}\mathbf{B}'\mathbf{B} \text{ is diagonal},$$

where $\|\cdot\|_F$ denotes the Frobenius norm. It is well known that $\hat{\mathbf{P}}$ is given by $\sqrt{T}$ times the first K eigenvectors of the matrix $\mathbf{X}\mathbf{X}'$. It is useful to remark here that this allows us to

express the vector of predictors $\mathbf{X}_t$ as a linear combination of the K principal components $\widehat{\mathbf{P}}_t$ plus a residual vector $\widehat{\mathbf{u}}_t$, that is

$$\mathbf{X}_t = \widehat{\mathbf{B}}\widehat{\mathbf{P}}_t + \widehat{\mathbf{u}}_t, \quad (1)$$

where $\widehat{\mathbf{u}}_t = \mathbf{X}_t - \widehat{\mathbf{B}}\widehat{\mathbf{P}}_t$. The second step of PCR consists in computing the $K \times 1$ least squares coefficient vector $\widehat{\boldsymbol{\theta}}$ associated with the regression of the $T \times 1$ target variable vector $\mathbf{Y} = (Y_1, \dots, Y_T)'$ on the principal components matrix $\widehat{\mathbf{P}}$. This is the solution to the least squares problem

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \mathbb{R}^K}{\operatorname{argmin}} \|\mathbf{Y} - \widehat{\mathbf{P}}\boldsymbol{\theta}\|_2^2,$$

where $\|\cdot\|_2$ denotes the Euclidean norm. It is straightforward to check that $\widehat{\boldsymbol{\theta}} = \widehat{\mathbf{P}}'\mathbf{Y}/T$. We remark that in many empirical forecasting applications it is common to augment PCR with lags of the dependent variable. We conjecture that the analysis and results of this paper can be extended to such settings, but leave the formal analysis for future research.

PCR can be formulated as a regularized empirical risk minimization procedure.¹ Consider the class of linear prediction rules $\boldsymbol{\theta}'\mathbf{X}_t$, indexed by $\boldsymbol{\theta} \in \mathbb{R}^p$. Then PCR can be cast as the regularized empirical risk minimization problem given by

$$\widehat{\boldsymbol{\theta}} \in \underset{\boldsymbol{\theta} \in \mathbb{R}^p}{\operatorname{argmin}} R_T(\boldsymbol{\theta}) \quad \text{s.t.} \quad \widehat{\mathbf{V}}_R'\boldsymbol{\theta} = \mathbf{0}, \quad (2)$$

where

$$R_T(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^T (Y_t - \boldsymbol{\theta}'\mathbf{X}_t)^2 = \frac{1}{T} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\theta}\|_2^2,$$

is the empirical risk and

$$\widehat{\mathbf{V}}_R = (\widehat{\mathbf{v}}_{K+1}, \dots, \widehat{\mathbf{v}}_p),$$

where $\widehat{\mathbf{v}}_i$ is the eigenvector associated with the i -th largest eigenvalue of the sample covariance matrix of the predictors $\widehat{\boldsymbol{\Sigma}} = \mathbf{X}'\mathbf{X}/T$. We emphasize that the minimizer in (2) is not necessarily unique, and if more than one prediction rule achieves the minimum we

¹We remark that in this paper we use the expression “empirical risk minimization for principal component regression” only for simplicity. A more appropriate name for the procedure we study would be “regularized empirical risk minimization based on principal component analysis”.

may pick one arbitrarily. In what follows we refer to $\hat{\boldsymbol{\theta}}$ as a (regularized) empirical risk minimizer (ERM). It is straightforward to verify that $\hat{\boldsymbol{\theta}}' \mathbf{X}_t = \hat{\boldsymbol{\vartheta}}' \hat{\mathbf{P}}_t$, which implies that the PCR predictions can be equivalently expressed as linear predictions based on constrained least squares.

One of the classical objectives in learning theory is to establish predictive performance guarantees of statistical learning procedures relative to a suitable benchmark. Define the out-of-sample principal components $\hat{\mathbf{P}}_{T+1} = \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\mathbf{V}}_K' \mathbf{X}_{T+1}$ with $\hat{\mathbf{V}}_K = (\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_K)$ and $\hat{\boldsymbol{\Lambda}}_K = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_K)$ where $\hat{\lambda}_1, \dots, \hat{\lambda}_K$ are the K largest eigenvalues of $\hat{\boldsymbol{\Sigma}}$. We can then define the predictive performance of PCR as its conditional risk, that is

$$R(\hat{\boldsymbol{\vartheta}}) = \mathbb{E} \left[(Y_{T+1} - \hat{\boldsymbol{\vartheta}}' \hat{\mathbf{P}}_{T+1})^2 \mid \mathcal{D}_T \right] . \quad (3)$$

The performance measure in (3) can be interpreted as the risk of the ERM obtained from the “training” sample $\mathcal{D}_T$ evaluated on the “validation” observation $(Y_{T+1}, \mathbf{X}_{T+1})'$. Define the optimal risk for the class of linear prediction rules as

$$R^* = \min_{\boldsymbol{\theta} \in \mathbb{R}^p} \mathbb{E} \left[(Y_{T+1} - \boldsymbol{\theta}' \mathbf{X}_{T+1})^2 \right] .$$

Our aim is to find a pair $(B_T(p, K), \delta_T)$ such that $B_T(p, K) \rightarrow 0$ and $\delta_T \rightarrow 0$ as $T \rightarrow \infty$ for which

$$R(\hat{\boldsymbol{\vartheta}}) \leq R^* + B_T(p, K) , \quad (4)$$

holds with probability at least $1 - \delta_T$ for any T sufficiently large. The inequality in (4) is commonly referred to as an *oracle inequality*. Oracle inequalities such as (4) provide nonasymptotic guarantees on the performance of empirical risk minimization and imply that the ERM performs asymptotically as well as the best linear predictor.

This paper is related to various strands of the literature. First, the literature on diffusion index forecasting models, which includes Stock and Watson (2002a), Stock and Watson (2002b), Bai and Ng (2006), Boivin and Ng (2006) and Fan, Masini, and Medeiros (2023). In particular, our work is close in spirit to the important contribution of Fan *et al.* (2023) that studies the properties of a large class of high-dimensional models, which in-

cludes factor models. In contrast to Fan *et al.* (2023), we allow the growth rate of the largest eigenvalues of the covariance matrix of the predictors to range from sublinear to linear. Specifically, our largest eigenvalues are proportional to p^α for $\alpha \in (0, 1]$, whereas Fan *et al.* (2023) focus on the case of $\alpha = 1$. Moreover, we derive explicit conditions under which principal component regression attains the oracle rate K/T , up to logarithmic factors. We further establish a matching lower bound (up to logarithmic factors), showing that this rate is essentially tight. Second, our work is also related to the literature on approximate factor models, principal component analysis and spiked covariance models, which includes Wang and Fan (2017), Donoho, Gavish, and Johnstone (2018), Forni, Hallin, Lippi, and Reichlin (2000, 2005), Bai and Ng (2019), Bai and Li (2012, 2016), Lam, Yao, and Bathia (2011), Fan, Liao, and Wang (2016), Gonçalves and Peron (2020), Barigozzi and Cho (2020), Su and Wang (2017) and Uematsu and Yamagata (2022). Third, the literature on the small-ball method, which includes Lecué and Mendelson (2016), Lecué and Mendelson (2017), Mendelson (2018) and Lecué and Mendelson (2018). Fourth, the literature on empirical risk minimization for linear regression, which includes Birge and Massart (1998) and Tsybakov (2003) among others. In particular, our contribution is close to the subset of the literature that deals with dependent data, as in Jiang and Tanner (2010); Brownlees and Guðmundsson (2025); Brownlees and Llorens-Terrazas (2024). Relative to Brownlees and Guðmundsson (2025), who also study the performance of a linear regression procedure, the main novelty lies in the fact that in this paper the regressors (i.e. the estimated principal components) are themselves subject to estimation error, which introduces a substantial additional layer of technical complexity into the analysis.

The rest of the paper is structured as follows. Section 2 introduces additional basic notation, assumptions and preliminary results. Section 3 establishes a bound on the performance of empirical risk minimization for thin-tailed data, which is the main result of the paper, as well as a lower bound. Section 4 contains the proof of the main result. Section 5 contains an extension of our main result that allows for heavy-tailed data. Concluding remarks follow in Section 6. All remaining proofs are given in the appendix

or the online appendix.

2 Notation, Preliminaries and Assumptions

In this section we lay out the assumptions required for our analysis. We first introduce some basic notation. For a generic vector $\mathbf{x} \in \mathbb{R}^d$ we define $\|\mathbf{x}\|_r$ as $(\sum_{i=1}^d |x_i|^r)^{1/r}$ for $1 \leq r < \infty$ and $\max_{i=1,\dots,d} |x_i|$ for $r = \infty$. For a generic random variable $X \in \mathbb{R}$ we define $\|X\|_{L_r}$ as $(\mathbb{E}[|X|^r])^{1/r}$ for $1 \leq r < \infty$ and $\inf\{a : \mathbb{P}(|X| > a) = 0\}$ for $r = \infty$. For a positive semi-definite matrix $\mathbf{M}$ we use $\mathbf{M}^{1/2}$ to denote the positive semi-definite square root matrix of $\mathbf{M}$ and $\mathbf{M}^{-1/2}$ to denote the generalized inverse of $\mathbf{M}^{1/2}$. For a generic matrix $\mathbf{M}$ we use $\|\mathbf{M}\|_2$ to denote the spectral norm of $\mathbf{M}$.

We begin with a preliminary lemma that establishes the population analogue of the principal components decomposition of the prediction vector $\mathbf{X}_t$ introduced in (1).

Lemma 1. *Let Y_t be a random variable such that $\|Y_t\|_{L_2} < \infty$ and $\mathbf{X}_t$ be a zero-mean p -dimensional random vector such that $\|X_{it}\|_{L_2} < \infty$ for $i = 1, \dots, p$. For any $K \in \{1, \dots, p\}$, define $\mathbf{\Lambda}_K = \text{diag}(\lambda_1, \dots, \lambda_K)$, $\mathbf{\Lambda}_R = \text{diag}(\lambda_{K+1}, \dots, \lambda_p)$, $\mathbf{V}_K = (\mathbf{v}_1, \dots, \mathbf{v}_K)$ and $\mathbf{V}_R = (\mathbf{v}_{K+1}, \dots, \mathbf{v}_p)$ where $\lambda_1, \dots, \lambda_p$ and $\mathbf{v}_1, \dots, \mathbf{v}_p$ denote the non-increasing sequence of eigenvalues of $\mathbf{\Sigma} = \mathbb{E}(\mathbf{X}_t \mathbf{X}_t')$ and the corresponding sequence of eigenvectors.*

Then, (i) it holds that

$$\mathbf{X}_t = \mathbf{B}\mathbf{P}_t + \mathbf{u}_t,$$

where $\mathbf{B} = \mathbf{V}_K \mathbf{\Lambda}_K^{1/2}$, $\mathbf{P}_t = \mathbf{\Lambda}_K^{-1/2} \mathbf{V}_K' \mathbf{X}_t$ and $\mathbf{u}_t = \mathbf{V}_R (\mathbf{V}_R' \mathbf{V}_R)^{-1} \mathbf{V}_R' \mathbf{X}_t$, with $\mathbf{B}'\mathbf{B}$ diagonal, $\mathbb{E}(\mathbf{P}_t \mathbf{P}_t') = \mathbf{I}_K$, $\mathbb{E}(\mathbf{u}_t \mathbf{u}_t') = \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R'$ and $\mathbb{E}(\mathbf{P}_t \mathbf{u}_t') = \mathbf{0}_{K \times p}$.

(ii) Let $\boldsymbol{\theta}^ \in \mathbb{R}^p$ be a vector of coefficients associated with the best linear predictor of Y_t based on $\mathbf{X}_t$. Then, it holds that*

$$\mathbf{X}_t' \boldsymbol{\theta}^* = \mathbf{P}_t' \boldsymbol{\vartheta}^* + \mathbf{u}_t' \boldsymbol{\gamma}^*,$$

where $\boldsymbol{\vartheta}^ = \mathbf{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}^*$ and $\boldsymbol{\gamma}^* = \mathbf{V}_R (\mathbf{V}_R' \mathbf{V}_R)^{-1} \mathbf{V}_R' \boldsymbol{\theta}^*$.*

(iii) $\boldsymbol{\vartheta}^$ is unique provided that $\lambda_K > 0$, and $\mathbf{u}_t' \boldsymbol{\gamma}^*$ is unique almost surely.*

$$(iv) \|\boldsymbol{\vartheta}^*\|_2 \leq \|Y_t\|_{L_2} \text{ and } \|\boldsymbol{\Lambda}_R^{1/2} \mathbf{V}_R' \boldsymbol{\gamma}^*\|_2 \leq \|Y_t\|_{L_2}.$$

Part (i) of the lemma states that $\mathbf{X}_t$ can be decomposed into two parts: a common component consisting of the product of the matrix of coefficients $\mathbf{B}$ with the random vector $\mathbf{P}_t$, and the residual random vector $\mathbf{u}_t$, where $\mathbf{P}_t$ and $\mathbf{u}_t$ are orthogonal. We call $\mathbf{P}_t$ the population principal components and $\mathbf{u}_t$ the idiosyncratic component. Part (ii) implies that the best linear predictor for Y_t can alternatively be represented as a function of the population principal components $\mathbf{P}_t$ and the idiosyncratic component $\mathbf{u}_t$ with the coefficient vectors $\boldsymbol{\vartheta}^*$ and $\boldsymbol{\gamma}^*$. Note that since $\mathbf{P}_t$ and $\mathbf{u}_t$ are orthogonal, $\boldsymbol{\vartheta}^*$ may be interpreted as the best linear predictor based on the population principal components $\mathbf{P}_t$. We remark that, clearly, the matrix $\mathbf{V}_R(\mathbf{V}_R' \mathbf{V}_R)^{-1} \mathbf{V}_R'$ may be simplified to $\mathbf{V}_R \mathbf{V}_R'$ in the definitions of $\mathbf{u}_t$ and $\boldsymbol{\gamma}^*$, but we prefer the former as it emphasizes that this is a projection matrix. Moreover, note that the decomposition established in part (ii) holds for any vector in $\mathbb{R}^p$, but for our purposes it suffices to focus on a vector of coefficients associated with the best linear predictor of Y_t based on $\mathbf{X}_t$. Part (iii) implies that, despite the fact that the vector of coefficients of the best linear predictor is not unique in general, $\boldsymbol{\vartheta}^*$ is unique when $\lambda_K > 0$, and $\mathbf{u}_t' \boldsymbol{\gamma}^*$ is unique almost surely.

We now lay out the assumptions of our analysis. We say that the d -dimensional random vector $\mathbf{U}$ is sub-Gaussian with parameter $C_U > 0$ if, for any $\varepsilon > 0$ and some constant $C_0 > 0$, it holds that

$$\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{U}| > \varepsilon) \leq C_0 \exp(-C_U \varepsilon^2).$$

For a univariate random variable U this is equivalent to $\mathbb{P}(|U| > \varepsilon) \leq C_0 \exp(-C_U \varepsilon^2)$.

Assumption 1 (Distribution). *There exists a constant C_Z such that Y_t and $\mathbf{Z}_t = \boldsymbol{\Sigma}^{-1/2} \mathbf{X}_t$ where $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{X}_t \mathbf{X}_t')$ are sub-Gaussian with parameter C_Z .*

Assumption 1 requires the prediction target Y_t and the standardized predictors $\mathbf{Z}_t$ to have sub-Gaussian tails, so that the tails of the data decay exponentially. We remark that such a condition is fairly standard in the analysis of large-dimensional factor models (Fan *et al.*, 2011). In Section 5 we relax this assumption and study the performance of

PCR under heavy-tailed data, at the cost of more involved proofs.

Let $\mathcal{F}_{-\infty}^t$ and $\mathcal{F}_{t+l}^\infty$ be the σ -algebras generated by $\{(Y_s, \mathbf{X}'_s)' : -\infty \leq s \leq t\}$ and $\{(Y_s, \mathbf{X}'_s)' : t+l \leq s \leq \infty\}$ respectively for some $t \in \mathbb{Z}$ and define the β -mixing coefficients

$$\beta(l) = \frac{1}{2} \sup_{\{A_i\}_{i=1}^I, \{B_j\}_{j=1}^J} \sum_{i=1}^I \sum_{j=1}^J |\mathbb{P}(A_i \cap B_j) - \mathbb{P}(A_i)\mathbb{P}(B_j)|,$$

where the supremum is taken over all finite measurable partitions $\{A_i\}_{i=1}^I$ and $\{B_j\}_{j=1}^J$ such that $A_i \in \mathcal{F}_{-\infty}^t$ and $B_j \in \mathcal{F}_{t+l}^\infty$.

Assumption 2 (Dependence). (i) *There exist constants $C_\beta > 0$ and $r_\beta \geq 1$ such that the β -mixing coefficients satisfy $\beta(l) \leq \exp(-C_\beta l^{r_\beta})$.* (ii) *Let $\varepsilon_t = Y_t - \mathbf{X}'_t \boldsymbol{\theta}^*$ where $\boldsymbol{\theta}^*$ is a vector coefficients associated with the best linear predictor of Y_t based on $\mathbf{X}_t$. The process $\{(Y_t, \mathbf{X}'_t)'\}$ satisfies $\mathbb{E}[\varepsilon_{T+1} \mathbf{X}_{T+1} \mid \mathcal{D}_T] = \mathbf{0}$, $\mathbb{E}[\varepsilon_{T+1}^2 \mid \mathcal{D}_T] = \mathbb{E}[\varepsilon_t^2]$ and $\mathbb{E}[\mathbf{X}_{T+1} \mathbf{X}'_{T+1} \mid \mathcal{D}_T] = \mathbb{E}[\mathbf{X}_t \mathbf{X}'_t]$.*

Part (i) of Assumption 2 states that the process $\{(Y_t, \mathbf{X}'_t)'\}$ is absolutely regular and has geometrically decaying β -mixing coefficients. We recall that β -mixing implies α -mixing, which is a fairly standard assumption in the analysis of large-dimensional time series models (Jiang and Tanner, 2010; Fan *et al.*, 2011; Kock and Callot, 2015). We require β -mixing to use the matrix Bernstein inequality of Banna, Merlevède, and Youssef (2016) which allows us to establish predictive performance guarantees for PCR for heavy-tailed data.² Part (ii) of Assumption 2 imposes three conditions: a conditional martingale difference sequence condition on the score of the best linear predictor, conditional homoskedasticity on the prediction error of the best linear predictor, and a conditional homoskedasticity condition on the predictors. These conditions play a key role in decomposing the predictive performance (see Lemma 2 below), allowing the analysis of the out-of-sample risk of the ERM to proceed analogously to the i.i.d. case despite the dependent data setting. Notably, part (ii) plays no role in controlling the estimation error of PCR.

²We remark that Theorem 1 can be derived under geometric α -mixing. However, for simplicity, we prefer to rely on a single notion of dependence throughout the paper.

Assumption 3 (Eigenvalues). *There is an integer $K \in \{1, \dots, p\}$, a constant $\alpha \in (0, 1]$ and a sequence of non-increasing nonnegative constants $c_1, \dots, c_p$ with $c_K > 0$ such that $\lambda_i = c_i p^\alpha$ for $i = 1, \dots, K$ and $\lambda_i = c_i$ for $i = K + 1, \dots, p$.*

Assumption 3 allows the first K eigenvalues of the covariance matrix Σ to diverge with the number of predictors p , where the rate of divergence is determined by α . We distinguish between different regimes that depend on the value of this constant. We label $\alpha = 1$ the strong signal regime, $\alpha \in (1/2, 1)$ the semi-strong signal regime, and $\alpha \in (0, 1/2]$ the semi-weak signal regime. We point out that here K is assumed known and that there is a large literature devoted to the estimation of this quantity (Bai and Ng, 2002; Amengual and Watson, 2007; Onatski, 2010; Lam and Yao, 2012; Ahn and Horenstein, 2013; Yu, He, and Zhang, 2019). It is important to emphasize that the assumption allows for the non-diverging eigenvalues of Σ to be zero, so our framework allows Σ to be singular.

Assumption 4 (Dimensions). *(i) There are constants $C_p \geq 1$ and $r_p \in [0, \bar{r}_p)$ such that $p = \lfloor C_p T^{r_p} \rfloor$ where $\bar{r}_p = r_\beta / (1 + (1 - \alpha)r_\beta)$; (ii) There are constants $C_p \geq C_K \geq 1$ and $r_K \in [0, \bar{r}_K)$ such that $K = \lfloor C_K T^{r_K} \rfloor$ where $\bar{r}_K = r_\beta / (1 + r_\beta)$; and (iii) r_p and r_K additionally satisfy $r_K + r_p(1 - \alpha) < 1$ and $r_K \leq r_p$.*

Assumption 4 permits both the predictor dimension and the number of retained principal components to increase with the sample size T . Their admissible growth rates depend on the rate of decay of the temporal dependence, as measured by r_β , and on the signal strength, measured by α . The weaker the dependence and the stronger the signal, the larger the numbers of allowed predictors and principal components. The assumption permits the number of predictors to exceed the sample size, but requires the number of principal components to be smaller than T .

3 Performance of Empirical Risk Minimization

We now state our main result on the performance of PCR for thin-tailed data. The theorem establishes a nonasymptotic oracle inequality that bounds the excess risk of PCR relative to the best linear predictor based on $\mathbf{X}_t$.

Theorem 1. *Suppose Assumptions 1–4 are satisfied.*

Then for every $\eta > 0$ there exist constants $C_1, C_2 > 0$ such that, for all T sufficiently large

$$\begin{aligned} R(\hat{\boldsymbol{\theta}}) - R^* &\leq 2(\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^* \\ &\quad + C_1 \left(1 + \frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T} \\ &\quad + C_2 \frac{K \log(T)}{T}, \end{aligned}$$

holds with probability at least $1 - T^{-\eta}$.

The theorem establishes a regret bound on the excess risk of PCR relative to the best linear predictor that can be obtained on the basis of the predictors $\mathbf{X}_t$. The gap is made up of three terms. The first accounts for the approximation error of PCR and the last two account for the estimation error. The approximation error measures the gap between the performance of the best linear predictor based on the population principal components $\mathbf{P}_t$ and the best linear predictor based on the predictors $\mathbf{X}_t$. The estimation error measures the gap between the performance of PCR relative to the best linear predictor based on the population principal components $\mathbf{P}_t$.

A natural question that arises from Theorem 1 is whether the rate for the estimation error is sharp. The following theorem establishes a lower bound for the performance of PCR under a Gaussian data-generating process.

Theorem 2. *Suppose $\{(\mathbf{X}_t', \varepsilon_t)'\}$ is an i.i.d. sequence of Gaussian vectors such that $\mathbf{X}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, with ε_t independent of $\mathbf{X}_t$. Let there be an integer $K \in \{1, \dots, p\}$, a constant $\alpha \in (0, 1]$ and a sequence of non-increasing strictly positive constants $c_1, \dots, c_K$ such that $\lambda_i = c_i p^\alpha$ for $i = 1, \dots, K$ and $\lambda_i = 1$ for $i = K + 1, \dots, p$. Suppose moreover that it holds that*

$$Y_t = \mathbf{X}_t' \boldsymbol{\theta}^* + \varepsilon_t, \quad t = 1, \dots, T,$$

where $\boldsymbol{\theta}^* \in \text{span}(\mathbf{V}_R)$ and thus satisfies $\boldsymbol{\vartheta}^* = \boldsymbol{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}^* = \mathbf{0}$ and $\boldsymbol{\gamma}^* = \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\theta}^* = \boldsymbol{\theta}^*$. Finally, suppose that Assumption 4 is satisfied.

Then we have that for every $\eta > 0$, and all sufficiently large T

$$R(\hat{\boldsymbol{\vartheta}}) - R^* \geq \frac{1}{6} \left(\|\boldsymbol{\gamma}^*\|_2^2 + \sigma^2 \frac{K}{T} \right)$$

holds with probability at least $1 - T^{-\eta} - e^{-K/9}$.

The theorem establishes that, under a specific Gaussian data-generating process, the estimation error of PCR is bounded from below by a term of order K/T with positive probability. It is straightforward to verify that the assumptions of the theorem are compatible with Assumptions 1–4.

A number of remarks on the performance of PCR are in order. First, we discuss under which conditions the estimation error rate established in Theorem 1 is, in some appropriate sense, optimal. For ease of exposition we assume that the approximation error is zero for the remainder of this paragraph. Theorem 2 establishes that the lower bound on the performance of PCR is of the order K/T . This coincides with the optimal rate that the least squares estimator could achieve if the population principal components were observed (Tsybakov, 2003). This motivates us to label the K/T rate as the optimal rate. Inspection of Theorem 1 reveals that the estimation error rate depends crucially on the signal strength α . When $\alpha \in [1/2, 1]$, the near-optimal rate $(K \log(T))/T$ is achieved provided that the number of predictors is not too large in the sense that

$$r_p \leq \frac{r_\beta}{1 + 2(1 - \alpha)r_\beta}.$$

This condition is, for example, satisfied when p is fixed ($r_p = 0$), and more generally, the admissible range of r_p is increasing in both α and r_β . Notice that the condition simplifies to $r_p < r_\beta$ when $\alpha = 1$, which is already guaranteed by Assumption 4. When $\alpha \in (0, 1/2)$, the signal is too weak for the near-optimal rate to be achieved unless p is fixed. This is because the factor $p^{1-2\alpha}$ in the estimation error diverges with p , and the bound is strictly slower than $(K \log(T))/T$ as long as p is growing. In this regime PCR remains consistent

for prediction under Assumption 4, but at a rate that is penalized by the weakness of the signal.

Second, it is insightful to provide an alternative representation of the approximation error of PCR. This may be equivalently expressed as

$$(\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^* = (\boldsymbol{\theta}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\theta}^* .$$

This highlights that the approximation error of PCR is small when the projection of the best linear predictor $\boldsymbol{\theta}^*$ on the subspace spanned by the population eigenvectors $\mathbf{v}_{K+1}, \dots, \mathbf{v}_p$ is small. Equivalently, if the contribution of the idiosyncratic component vector $\mathbf{u}_t$ is negligible then PCR has a negligible approximation error. Clearly, when $\boldsymbol{\gamma}^* = \mathbf{0}$ the best linear predictor based on $\mathbf{X}_t$ coincides with the best linear predictor based on $\mathbf{P}_t$ and the approximation error is zero. We remark that part (iv) of Lemma 1 shows that the approximation error is bounded by a constant that does not depend on K, p or T .

Finally, we note that the bound in Theorem 1 provides some theoretical support for the empirical finding of Boivin and Ng (2006) that adding more predictors to a diffusion index model does not always improve forecast performance. When $\alpha \in (0, 1)$, Assumption 4 restricts the admissible growth rate of p relative to T more severely than in the strong signal setting $\alpha = 1$, illustrating how expanding the predictor set indiscriminately can be detrimental to forecast performance.

4 Proof of Main Result

The proof of Theorem 1 begins with a decomposition of an upper bound of the excess risk of the empirical risk minimizer. Define the approximate rotation matrix $\mathbf{H} = \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\mathbf{V}}_K' \mathbf{V}_K \boldsymbol{\Lambda}_K^{1/2}$ (Bai and Ng, 2002) and the infeasible PCR estimator $\tilde{\boldsymbol{\vartheta}} = \operatorname{argmin}_{\boldsymbol{\vartheta} \in \mathbb{R}^K} \frac{1}{T} \sum_{t=1}^T (Y_t - \mathbf{P}_t' \boldsymbol{\vartheta})^2$. The following lemma provides a useful bound for the excess risk.

Lemma 2. *Suppose that $\{(Y_t, \mathbf{X}_t')'\}_t$ is a stationary sequence taking values in $\mathcal{Y} \times \mathcal{X} \subset \mathbb{R} \times \mathbb{R}^p$ such that $\mathbb{E}(Y_t^2) < \infty$ and $\mathbb{E}(X_{it}^2) < \infty$ for all $i = 1, \dots, p$. Let $\varepsilon_t = Y_t - \mathbf{X}_t' \boldsymbol{\theta}^*$*

where $\boldsymbol{\theta}^*$ is a vector coefficients associated with the best linear predictor of Y_t based on $\mathbf{X}_t$. Suppose $\{(Y_t, \mathbf{X}_t')'\}$ satisfies $\mathbb{E}[\varepsilon_{T+1} \mathbf{X}_{T+1} \mid \mathcal{D}_T] = \mathbf{0}$, $\mathbb{E}[\varepsilon_{T+1}^2 \mid \mathcal{D}_T] = \mathbb{E}[\varepsilon_t^2]$ and $\mathbb{E}[\mathbf{X}_{T+1} \mathbf{X}_{T+1}' \mid \mathcal{D}_T] = \mathbb{E}[\mathbf{X}_t \mathbf{X}_t']$.

Then, for all $T \geq K$ it holds that

$$\begin{aligned} R(\hat{\boldsymbol{\vartheta}}) - R^* &\leq 2\|\hat{\boldsymbol{\vartheta}}\|_2^2 \mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) + 2\|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_2^2 \\ &\quad + 2\|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2^2 + 2(\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^*. \end{aligned} \quad (5)$$

Lemma 2 implies that in order to control the excess risk of PCR it suffices to provide appropriate bounds on the terms on the right hand side of (5). The following three propositions establish such bounds.

The first two propositions control the terms $\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T)$ and $\|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_2^2$. We remark that these two terms capture the risk of PCR arising from the estimation of the principal components.

Proposition 1. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) \leq C \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$.

Proposition 2. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_2^2 \leq C \left(1 + \frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$.

The next proposition controls the term $\|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2^2$. We remark that this term may be interpreted as the risk of the least squares estimator of PCR based on the population principal components.

Proposition 3. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2^2 \leq C \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$.

The claim of Theorem 1 then follows from Propositions 1 to 3, along with Proposition A.7 which establishes that for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large, $\|\hat{\boldsymbol{\vartheta}}\|_2^2 < C$ holds with probability at least $1 - T^{-\eta}$, after a straightforward application of the implication rule and the union bound.

5 Performance of Empirical Risk Minimization for Heavy-tailed Data

The analysis in the preceding sections assumes that the data are light-tailed, in the sense that Y_t and the standardized predictors $\mathbf{Z}_t$ have sub-Gaussian tails. In this section we relax this condition and study the performance of PCR under heavy-tailed data. The main result of this section establishes a bound on the predictive performance of PCR analogous to Theorem 1 under a polynomial tail condition.

To this end we introduce modified versions of the distributional and dimensionality assumptions.

Assumption 1* (Distribution). *Let $\mathbf{Z}_t = \boldsymbol{\Sigma}^{-1/2} \mathbf{X}_t$. There exists a constant $m > 4$ and a positive constant C_Z such that $\mathbb{E}(|Y_t|^m) \leq C_Z$ and $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}(|\mathbf{v}' \mathbf{Z}_t|^m) \leq C_Z$.*

Assumption 1* is a fairly weak polynomial tail assumption. Note that we assume that all possible one-dimensional marginals of the standardized predictors $\mathbf{Z}_t$ have polynomial tails.

Assumption 4* (Dimensions). *Let $0 < \kappa < \min\{1 - 2/m, (m - 4)/2\}$ be a constant.*

(i) There are constants $C_p \geq 1$ and $r_p \in [0, \bar{r}_p]$ such that $p = \lfloor C_p T^{r_p} \rfloor$ where $(1 - \alpha)\bar{r}_p = 1 - 2/m - \kappa$ if $\alpha \in (0, 1)$ and $\bar{r}_p = \infty$ if $\alpha = 1$; (ii) There are constants $C_p \geq C_K \geq 1$

and $r_K \in [0, \bar{r}_K)$ such that $K = \lfloor C_K T^{r_K} \rfloor$ where $\bar{r}_K = 1 - 2/m - \kappa$; and (iii) r_p and r_K additionally satisfy $r_K + r_p(1 - \alpha) < 1$ and $r_K \leq r_p$.

Assumption 4* states the allowed growth regimes for p and K . In particular, the larger m is, the higher the admissible growth rates. The dependence of the growth rate of p on α is analogous to that in the thin-tailed setting. Note that the growth rates here are restricted by m , whereas in Assumption 4 they are restricted by r_β .

We next state our result on the performance of PCR for heavy-tailed data.

Theorem 3. *Suppose Assumptions 1*, 2, 3 and 4* are satisfied.*

Then for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exist constants $C_1, C_2 > 0$ such that, for all T sufficiently large

$$\begin{aligned} R(\hat{\boldsymbol{\vartheta}}) - R^* &\leq 2(\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^* \\ &\quad + C_1 \left(1 + \sqrt{\frac{p \log(T)}{T}} + \frac{p \log(T)^3}{T^{1-2/m-\delta}} \right) \times \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T} \\ &\quad + C_2 \frac{K \log(T)}{T}, \end{aligned}$$

with probability at least $1 - T^{-\eta}$.

A number of remarks on Theorem 3 are in order. The lower bound established in Theorem 2 continues to hold in the present setting, as it is derived under a Gaussian data-generating process which satisfies Assumption 1* and is consistent with Assumption 4*. The optimal rate therefore remains K/T , as in the light-tailed case. The key difference between the bounds obtained in Theorems 3 and 1 is that the former incurs a penalty based on the number of moments m , whereas the latter is penalized by r_β . If we set $\delta = c/m$ for some $\kappa > c > 2\eta$ and let m and r_β be large, the theorems give essentially the same conclusions.³ Similar to the thin-tailed setting, the near-optimal rate $(K \log(T))/T$ is obtained without further restrictions if $\alpha = 1$. If $\alpha \in [1/2, 1)$, it is obtained when $r_p < (1 - 2/m - \delta)/(2(1 - \alpha))$, which is slightly stricter than what Assumption 4* stipulates. If $\alpha \in (0, 1/2)$ the signal is too weak for optimality unless p is fixed.

³Note that the theorems respectively require m and r_β to be finite.

6 Conclusions

This paper develops a nonasymptotic theory of prediction for principal component regression. Our main result is an oracle inequality bounding the excess risk of principal component regression relative to the best linear predictor. Under appropriate conditions, the oracle inequality shows that principal component regression achieves near-optimal performance up to a logarithmic factor. Several features of the analysis merit attention. The framework accommodates both linearly and sublinearly growing leading eigenvalues of the predictor covariance matrix, indexed by the signal strength parameter $\alpha \in (0, 1]$. Signal strength governs not only the convergence rate of the estimation error but also the admissible growth rate of the predictor dimension: stronger signals permit larger predictor sets while preserving near-optimal performance, whereas weaker signals impose tighter dimension restrictions. The analysis also covers a broad class of tail behaviours, encompassing both thin-tailed and heavy-tailed distributions. Our results provide theoretical grounding for the empirical finding of Boivin and Ng (2006) that indiscriminately expanding the predictor set can be detrimental to forecast accuracy.

A Proofs

We recall that $\Sigma = \mathbf{V}_K \mathbf{\Lambda}_K \mathbf{V}_K' + \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R'$ where $\mathbf{V}_K = (\mathbf{v}_1, \dots, \mathbf{v}_K)$, $\mathbf{V}_R = (\mathbf{v}_{K+1}, \dots, \mathbf{v}_p)$, $\mathbf{\Lambda}_K = \text{diag}(\lambda_1, \dots, \lambda_K)$ and $\mathbf{\Lambda}_R = \text{diag}(\lambda_{K+1}, \dots, \lambda_p)$. Analogously, we have that $\widehat{\Sigma} = \widehat{\mathbf{V}}_K \widehat{\mathbf{\Lambda}}_K \widehat{\mathbf{V}}_K' + \widehat{\mathbf{V}}_R \widehat{\mathbf{\Lambda}}_R \widehat{\mathbf{V}}_R'$ where $\widehat{\mathbf{V}}_K = (\widehat{\mathbf{v}}_1, \dots, \widehat{\mathbf{v}}_K)$, $\widehat{\mathbf{V}}_R = (\widehat{\mathbf{v}}_{K+1}, \dots, \widehat{\mathbf{v}}_p)$, $\widehat{\mathbf{\Lambda}}_K = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_K)$ and $\widehat{\mathbf{\Lambda}}_R = \text{diag}(\hat{\lambda}_{K+1}, \dots, \hat{\lambda}_p)$. For a random variable X , we introduce the shorthand $\|X\|_{L_2, \mathcal{D}_T} = \sqrt{\mathbb{E}(X^2 | \mathcal{D}_T)}$ for convenience.

Proof of Lemma 1. (i) Note that by definition

$$\mathbf{B}\mathbf{P}_t + \mathbf{u}_t = \mathbf{V}_K \mathbf{V}_K' \mathbf{X}_t + \mathbf{V}_R \mathbf{V}_R' \mathbf{X}_t = \mathbf{V}_K \mathbf{V}_K' \mathbf{X}_t + (\mathbf{I}_p - \mathbf{V}_K \mathbf{V}_K') \mathbf{X}_t = \mathbf{X}_t.$$

We also note that $\mathbf{B}'\mathbf{B} = \mathbf{\Lambda}_K^{1/2} \mathbf{V}_K' \mathbf{V}_K \mathbf{\Lambda}_K^{1/2} = \mathbf{\Lambda}_K$. Furthermore, as $\Sigma = \mathbf{V}_K \mathbf{\Lambda}_K \mathbf{V}_K' +$

$\mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R'$, and $\mathbf{V}_K' \mathbf{V}_R = \mathbf{0}_{K \times (p-K)}$, we immediately obtain

$$\begin{aligned}\mathbb{E}(\mathbf{P}_t \mathbf{P}_t') &= \mathbf{\Lambda}_K^{-1/2} \mathbf{V}_K' \mathbf{\Sigma} \mathbf{V}_K \mathbf{\Lambda}_K^{-1/2} = \mathbf{\Lambda}_K^{-1/2} \mathbf{V}_K' (\mathbf{V}_K \mathbf{\Lambda}_K \mathbf{V}_K' + \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R') \mathbf{V}_K \mathbf{\Lambda}_K^{-1/2} = \mathbf{I}_K \\ \mathbb{E}(\mathbf{u}_t \mathbf{u}_t') &= \mathbf{V}_R \mathbf{V}_R' (\mathbf{V}_K \mathbf{\Lambda}_K \mathbf{V}_K' + \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R') \mathbf{V}_R \mathbf{V}_R' = \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R' \\ \mathbb{E}(\mathbf{P}_t \mathbf{u}_t') &= \mathbf{\Lambda}_K^{-1/2} \mathbf{V}_K' (\mathbf{V}_K \mathbf{\Lambda}_K \mathbf{V}_K' + \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R') \mathbf{V}_R \mathbf{V}_R' = \mathbf{0}_{K \times p} .\end{aligned}$$

(ii) It follows from part (i) that

$$\mathbf{X}_t' \boldsymbol{\theta}^* = \mathbf{P}_t' \mathbf{B}' \boldsymbol{\theta}^* + \mathbf{u}_t' \boldsymbol{\theta}^* .$$

The claim then follows from noting that $\mathbf{P}_t' \mathbf{B}' \boldsymbol{\theta}^* = \mathbf{P}_t' \mathbf{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}^* = \mathbf{P}_t' \boldsymbol{\vartheta}^*$ and

$$\mathbf{u}_t' \boldsymbol{\theta}^* = \mathbf{X}_t' \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\theta}^* = \mathbf{X}_t' \mathbf{V}_R \mathbf{V}_R' \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\theta}^* = \mathbf{u}_t' \boldsymbol{\gamma}^* .$$

(iii) First note that if $\boldsymbol{\theta}_1^*$ and $\boldsymbol{\theta}_2^*$ are two distinct vectors of coefficients associated with the best linear predictor, we have

$$\mathbb{E}[(\mathbf{X}_t'(\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*))^2] = (\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*)' \mathbf{\Sigma} (\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*) = 0 ,$$

so that $\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^* \in \ker(\mathbf{\Sigma})$. Since we assume $\lambda_K > 0$, we have that $\text{span}(\mathbf{V}_K)$ is orthogonal to $\ker(\mathbf{\Sigma})$. Hence, for any two minimisers $\mathbf{V}_K'(\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*) = \mathbf{0}$, and thus

$$\boldsymbol{\vartheta}_1^* = \mathbf{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}_1^* = \mathbf{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}_2^* = \boldsymbol{\vartheta}_2^* .$$

Moreover, $\mathbf{V}_R \mathbf{V}_R'$ is an orthogonal projector onto the complement space of $\text{span}(\mathbf{V}_K)$, and thus $\mathbf{V}_R \mathbf{V}_R'(\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*) = \boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*$. As a consequence

$$\mathbb{E}[(\mathbf{u}_t' \boldsymbol{\gamma}_1^* - \mathbf{u}_t' \boldsymbol{\gamma}_2^*)^2] = \mathbb{E}[(\mathbf{X}_t' \mathbf{V}_R \mathbf{V}_R'(\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*))^2] = \mathbb{E}[(\mathbf{X}_t'(\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_2^*))^2] = 0 ,$$

so the random variable $\mathbf{u}_t' \boldsymbol{\gamma}^*$ is almost surely unique.

(iv) Note that the normal equations after pre-multiplying with $(\boldsymbol{\theta}^*)'$ yield $\mathbb{E}[(\boldsymbol{\theta}^*)' \mathbf{X}_t \mathbf{X}_t' \boldsymbol{\theta}^*] =$

$\mathbb{E}[(\boldsymbol{\theta}^*)' \mathbf{X}_t Y_t]$. Then by Cauchy-Schwarz

$$\mathbb{E}[(\mathbf{X}_t' \boldsymbol{\theta}^*)^2] = \mathbb{E}[(\mathbf{X}_t' \boldsymbol{\theta}^*) Y_t] \leq \mathbb{E}[(\mathbf{X}_t' \boldsymbol{\theta}^*)^2]^{1/2} \mathbb{E}[Y_t^2]^{1/2},$$

which in turn implies $\mathbb{E}[(\mathbf{X}_t' \boldsymbol{\theta}^*)^2]^{1/2} \leq \mathbb{E}[Y_t^2]^{1/2} \leq C$. But we also have

$$\mathbb{E}[(\mathbf{X}_t' \boldsymbol{\theta}^*)^2] = (\boldsymbol{\theta}^*)' \boldsymbol{\Sigma} \boldsymbol{\theta}^* = \|\boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^*\|_2^2,$$

which then yields $\|\boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^*\|_2 \leq C$. Notice next that

$$\begin{aligned} \|\boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^*\|_2^2 &= (\boldsymbol{\theta}^*)' (\mathbf{V}_K \boldsymbol{\Lambda}_K \mathbf{V}_K' + \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R') \boldsymbol{\theta}^* \\ &= \|\boldsymbol{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}^*\|_2^2 + \|\boldsymbol{\Lambda}_R^{1/2} \mathbf{V}_R' \boldsymbol{\theta}^*\|_2^2. \end{aligned}$$

Using the definitions $\boldsymbol{\vartheta}^* = \boldsymbol{\Lambda}_K^{1/2} \mathbf{V}_K' \boldsymbol{\theta}^*$ and $\boldsymbol{\gamma}^* = \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\theta}^*$, and the fact that

$$\|\boldsymbol{\Lambda}_R^{1/2} \mathbf{V}_R' \boldsymbol{\theta}^*\|_2^2 = (\boldsymbol{\theta}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\theta}^* = (\boldsymbol{\theta}^*)' \mathbf{V}_R \mathbf{V}_R' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\theta}^* = (\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^*,$$

we obtain

$$\|\boldsymbol{\Sigma}^{1/2} \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\vartheta}^*\|_2^2 + (\boldsymbol{\gamma}^*)' \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}_R' \boldsymbol{\gamma}^* \leq C^2,$$

which proves the claim. $\square$

Proof of Lemma 2. Consider the excess risk decomposition given by

$$\begin{aligned} R(\hat{\boldsymbol{\vartheta}}) - R^* &= \|Y_{T+1} - \hat{\mathbf{P}}_{T+1}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \mathbb{E}[(Y_{T+1} - \mathbf{X}_{T+1}' \boldsymbol{\theta}^*)^2] \\ &= \|Y_{T+1} - \hat{\mathbf{P}}_{T+1}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{X}_{T+1}' \boldsymbol{\theta}^*\|_{L_2, \mathcal{D}_T}^2 \\ &= \|Y_{T+1} - \hat{\mathbf{P}}_{T+1}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{P}_{T+1}' \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &\quad + \|Y_{T+1} - \mathbf{P}_{T+1}' \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{P}_{T+1}' \tilde{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &\quad + \|Y_{T+1} - \mathbf{P}_{T+1}' \tilde{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{P}_{T+1}' \boldsymbol{\vartheta}^*\|_{L_2, \mathcal{D}_T}^2 \\ &\quad + \|Y_{T+1} - \mathbf{P}_{T+1}' \boldsymbol{\vartheta}^*\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{X}_{T+1}' \boldsymbol{\theta}^*\|_{L_2, \mathcal{D}_T}^2 \\ &\equiv S_1 + S_2 + S_3 + S_4, \end{aligned} \tag{A.1}$$

where the first equality follows as $\mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^*)^2 \mid \mathcal{D}_T] = \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^*)^2]$. We remark that all quantities in this expression exist with high probability for sufficiently large T . We begin by applying the projection theorem to the last two terms to obtain

$$S_3 = \|\mathbf{P}'_{T+1}\tilde{\boldsymbol{\vartheta}} - \mathbf{P}'_{T+1}\boldsymbol{\vartheta}^*\|_{L_2, \mathcal{D}_T}^2 = \|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2^2, \quad (\text{A.2})$$

$$S_4 = \|\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*\|_{L_2, \mathcal{D}_T}^2. \quad (\text{A.3})$$

Next we note that for the first term we have that

$$\begin{aligned} S_1 &= \|Y_{T+1} - \hat{\mathbf{P}}'_{T+1}\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &= \|Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^* + \mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \hat{\mathbf{P}}'_{T+1}\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^* + \mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &= \|\mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \hat{\mathbf{P}}'_{T+1}\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|\mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2, \end{aligned}$$

by adding and subtracting, expanding the squares and applying the projection theorem.

Adding and subtracting again we obtain

$$\begin{aligned} S_1 &= \|\mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}} - (\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|\mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[(\mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}})(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] \\ &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[\mathbf{X}'_{T+1}\boldsymbol{\theta}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] \\ &\quad + 2\mathbb{E}[\mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T]. \end{aligned} \quad (\text{A.4})$$

To deal with the cross terms, we will use the fact that

$$\mathbb{E}[\hat{\mathbf{P}}'_{T+1}\mathbf{P}'_{T+1} \mid \mathcal{D}_T] = \hat{\boldsymbol{\Lambda}}_K^{-1/2}\hat{\mathbf{V}}'_K\Sigma\mathbf{V}_K\boldsymbol{\Lambda}_K^{-1/2} = \hat{\boldsymbol{\Lambda}}_K^{-1/2}\hat{\mathbf{V}}'_K\mathbf{V}_K\boldsymbol{\Lambda}_K^{1/2} = \mathbf{H}, \quad (\text{A.5})$$

which follows by the definitions of $\mathbf{P}_{T+1}$, $\hat{\mathbf{P}}_{T+1}$, the structure of $\Sigma = \mathbb{E}[\mathbf{X}_{T+1}\mathbf{X}'_{T+1} \mid \mathcal{D}_T]$ and the definition of $\mathbf{H}$. Now use this fact to see that the second cross term is zero

$$\begin{aligned} 2\mathbb{E}[\mathbf{P}'_{T+1}\mathbf{H}'\hat{\boldsymbol{\vartheta}}(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] &= 2\hat{\boldsymbol{\vartheta}}'\mathbf{H}\mathbb{E}[\mathbf{P}_{T+1}\hat{\mathbf{P}}'_{T+1} \mid \mathcal{D}_T]\hat{\boldsymbol{\vartheta}} - 2\hat{\boldsymbol{\vartheta}}'\mathbf{H}\mathbb{E}[\mathbf{P}_{T+1}\mathbf{P}'_{T+1} \mid \mathcal{D}_T]\mathbf{H}'\hat{\boldsymbol{\vartheta}} \\ &= 2\hat{\boldsymbol{\vartheta}}'\mathbf{H}\mathbf{H}'\hat{\boldsymbol{\vartheta}} - 2\hat{\boldsymbol{\vartheta}}'\mathbf{H}\mathbf{H}'\hat{\boldsymbol{\vartheta}} = 0. \end{aligned}$$

Going back to (A.4) and using the fact that Lemma 1 implies $\mathbf{X}'_{T+1}\boldsymbol{\theta}^* = \mathbf{P}'_{T+1}\boldsymbol{\vartheta}^* + \mathbf{u}'_{T+1}\boldsymbol{\gamma}^*$, we thus obtain

$$\begin{aligned} S_1 &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[\mathbf{X}'_{T+1}\boldsymbol{\theta}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] \\ &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[\mathbf{P}'_{T+1}\boldsymbol{\vartheta}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] \\ &\quad - 2\mathbb{E}[\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T]. \end{aligned}$$

We use (A.5) again to show that the second cross term is zero

$$-2\mathbb{E}[\mathbf{P}'_{T+1}\boldsymbol{\vartheta}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] = -2(\boldsymbol{\vartheta}^*)'\mathbb{E}[\mathbf{P}_{T+1}\hat{\mathbf{P}}'_{T+1} \mid \mathcal{D}_T]\hat{\boldsymbol{\vartheta}} + 2(\boldsymbol{\vartheta}^*)'\mathbb{E}[\mathbf{P}_{T+1}\mathbf{P}'_{T+1} \mid \mathcal{D}_T]\mathbf{H}'\hat{\boldsymbol{\vartheta}} = 0,$$

and therefore have obtained

$$S_1 = \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T].$$

Now use (A.3) and consider the sum of the first and fourth terms

$$\begin{aligned} S_1 + S_4 &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - 2\mathbb{E}[\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} \mid \mathcal{D}_T] + \|\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*\|_{L_2, \mathcal{D}_T}^2 \\ &= \|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}} - \mathbf{u}'_{T+1}\boldsymbol{\gamma}^*\|_{L_2, \mathcal{D}_T}^2 \\ &\leq 2\|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 + 2\|\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*\|_{L_2, \mathcal{D}_T}^2, \end{aligned}$$

where we used the fact that $(a - b)^2 \leq 2a^2 + 2b^2$. Now note that $\|\mathbf{u}'_{T+1}\boldsymbol{\gamma}^*\|_{L_2, \mathcal{D}_T}^2 = (\boldsymbol{\gamma}^*)'\mathbf{V}_R\boldsymbol{\Lambda}_R\mathbf{V}_R'\boldsymbol{\gamma}^*$. Moreover, by Cauchy-Schwarz

$$\|(\hat{\mathbf{P}}'_{T+1} - \mathbf{P}'_{T+1}\mathbf{H}')\hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 = \mathbb{E}[((\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1})'\hat{\boldsymbol{\vartheta}})^2 \mid \mathcal{D}_T] \leq \|\hat{\boldsymbol{\vartheta}}\|_2^2 \mathbb{E}[\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T].$$

We therefore have

$$S_1 + S_4 \leq 2\|\hat{\boldsymbol{\vartheta}}\|_2^2 \mathbb{E}[\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T] + 2(\boldsymbol{\gamma}^*)'\mathbf{V}_R\boldsymbol{\Lambda}_R\mathbf{V}_R'\boldsymbol{\gamma}^*. \quad (\text{A.6})$$

For the second term, the following identity will be useful. For any $\mathbf{a} \in \mathbb{R}^K$, we have

$Y_{T+1} - \mathbf{P}'_{T+1} \mathbf{a} = Y_{T+1} - \mathbf{P}'_{T+1} \boldsymbol{\vartheta}^* - \mathbf{P}'_{T+1} (\mathbf{a} - \boldsymbol{\vartheta}^*)$. By the projection theorem, the two right hand side terms are orthogonal so the Pythagorean identity gives

$$\|Y_{T+1} - \mathbf{P}'_{T+1} \mathbf{a}\|_{L_2, \mathcal{D}_T}^2 = \|Y_{T+1} - \mathbf{P}'_{T+1} \boldsymbol{\vartheta}^*\|_{L_2, \mathcal{D}_T}^2 + \|\mathbf{P}'_{T+1} (\mathbf{a} - \boldsymbol{\vartheta}^*)\|_{L_2, \mathcal{D}_T}^2. \quad (\text{A.7})$$

Applying (A.7) first with $\mathbf{a} = \mathbf{H}' \hat{\boldsymbol{\vartheta}}$ and then $\mathbf{a} = \tilde{\boldsymbol{\vartheta}}$ gives

$$\begin{aligned} S_2 &= \|Y_{T+1} - \mathbf{P}'_{T+1} \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 - \|Y_{T+1} - \mathbf{P}'_{T+1} \tilde{\boldsymbol{\vartheta}}\|_{L_2, \mathcal{D}_T}^2 \\ &= \|\mathbf{P}'_{T+1} (\mathbf{H}' \hat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*)\|_{L_2, \mathcal{D}_T}^2 - \|\mathbf{P}'_{T+1} (\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*)\|_{L_2, \mathcal{D}_T}^2 \\ &= \|\mathbf{H}' \hat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2^2 - \|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2^2, \end{aligned}$$

where the final step follows as $\mathbb{E}[\mathbf{P}_{T+1} \mathbf{P}'_{T+1} \mid \mathcal{D}_T] = \mathbf{I}_K$. Adding and subtracting yields

$$\begin{aligned} S_2 &= \|\mathbf{H}' \hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}} + \tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2^2 - \|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2^2 \\ &= \|\mathbf{H}' \hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}}\|_2^2 + 2(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*)' (\mathbf{H}' \hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}}) \\ &\leq \|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_2^2 + 2\|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2 \|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_2 \\ &\leq 2\|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}' \hat{\boldsymbol{\vartheta}}\|_2^2 + \|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2^2, \end{aligned} \quad (\text{A.8})$$

where the inequalities are Cauchy-Schwarz followed by $2ab \leq a^2 + b^2$.

The claim then follows by inserting (A.2), (A.6) and (A.8) into (A.1). $\square$

A.1 Proof of Theorem 1

A.1.1 Proof of Proposition 1

Proof of Proposition 1. Begin by noting that as

$$\mathbf{H} \mathbf{P}_{T+1} = \hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K \mathbf{V}_K \Lambda_K^{1/2} \Lambda_K^{-1/2} \mathbf{V}'_K \mathbf{X}_{T+1} = \hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K \mathbf{V}_K \mathbf{V}'_K \mathbf{X}_{T+1},$$

we have

$$\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1} = \hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K (\mathbf{I}_p - \mathbf{V}_K \mathbf{V}'_K) \mathbf{X}_{T+1} = \hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K \mathbf{V}_R \mathbf{V}'_R \mathbf{X}_{T+1} = \hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K \mathbf{u}_{T+1},$$

where the last step follows as $\mathbf{u}_{T+1} = \mathbf{V}_R \mathbf{V}_R' \mathbf{X}_{T+1}$. Thus

$$\|\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1}\|_2^2 = \mathbf{u}_{T+1}' \hat{\mathbf{V}}_K \hat{\mathbf{\Lambda}}_K^{-1} \hat{\mathbf{V}}_K' \mathbf{u}_{T+1}.$$

We may rewrite this as

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) = \text{tr}(\hat{\mathbf{\Lambda}}_K^{-1} \hat{\mathbf{V}}_K' \mathbb{E}(\mathbf{u}_{T+1} \mathbf{u}_{T+1}') \hat{\mathbf{V}}_K) = \text{tr}(\hat{\mathbf{\Lambda}}_K^{-1} \hat{\mathbf{V}}_K' \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R' \hat{\mathbf{V}}_K),$$

as $\mathbb{E}(\mathbf{u}_{T+1} \mathbf{u}_{T+1}') = \mathbf{V}_R \mathbf{\Lambda}_R \mathbf{V}_R'$. Using the definition $\mathbf{S} = \mathbf{V}_R' \hat{\mathbf{V}}_K$, we may upper bound with

$$\text{tr}(\hat{\mathbf{\Lambda}}_K^{-1} \mathbf{S}' \mathbf{\Lambda}_R \mathbf{S}) \leq \frac{1}{\hat{\lambda}_K} \text{tr}(\mathbf{\Lambda}_R \mathbf{S} \mathbf{S}') \leq \frac{\lambda_{K+1}}{\hat{\lambda}_K} \text{tr}(\mathbf{S}' \mathbf{S}),$$

where the first inequality follows as $\text{tr}(\mathbf{A} \mathbf{B}) \leq \|\mathbf{A}\|_2 \text{tr}(\mathbf{B})$ for two symmetric non-negative definite conformable matrices $\mathbf{A}, \mathbf{B}$, and the second because the diagonal of $\mathbf{S} \mathbf{S}'$ is a sum of squares and thus non-negative. As a consequence, we obtain

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) \leq \frac{\lambda_{K+1}}{\hat{\lambda}_K} \text{tr}(\mathbf{S}' \mathbf{S}) = \frac{\lambda_{K+1}}{\hat{\lambda}_K} \|\mathbf{S}\|_F^2 = \frac{\lambda_{K+1}}{\hat{\lambda}_K} \|\mathbf{V}_R' \hat{\mathbf{V}}_K\|_F^2.$$

Let $\eta' = \eta + \delta$ with $\delta > 0$. Proposition A.1 implies that for all sufficiently large T , we have $\hat{\lambda}_K \geq c_K p^\alpha / 2$ with probability at least $1 - T^{-\eta'}$. This together with Assumption 3 implies that there exists a constant $C_1 > 0$ such that

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) \leq C_1 \frac{1}{p^\alpha} \|\mathbf{V}_R' \hat{\mathbf{V}}_K\|_F^2,$$

with probability at least $1 - T^{-\eta'}$. Proposition A.6 then implies that there exists a constant $C > 0$ such that, for sufficiently large T

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H} \mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) \leq C \frac{p^{1-\alpha}}{p^\alpha} \frac{K \log(T)}{T} = C \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - 2T^{-\eta'} \geq 1 - T^{-\eta}$, which proves the claim. $\square$

Proposition A.1. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$, for all*

T sufficiently large

$$\begin{aligned}
(i) \quad & \hat{\lambda}_K \geq \frac{c_K}{2} p^\alpha, \\
(ii) \quad & \|\mathbf{V}'_R(\hat{\Sigma} - \Sigma)\mathbf{V}_R\|_2 \leq \frac{\hat{\lambda}_K - \lambda_{K+1}}{2}, \\
(iii) \quad & \|\hat{\mathbf{V}}_K \hat{\mathbf{V}}'_K - \mathbf{V}_K \mathbf{V}'_K\|_2 \leq \sqrt{5/6}, \\
(iv) \quad & \lambda_{\min}(\hat{\Lambda}_K^{-1/2} \hat{\mathbf{V}}'_K \Sigma \hat{\mathbf{V}}_K \hat{\Lambda}_K^{-1/2}) \geq 1/2,
\end{aligned}$$

hold with probability at least $1 - T^{-\eta}$.

Proof. (i) Using the representation of Lemma 1 we get

$$\begin{aligned}
\hat{\Sigma} - \Sigma &= \frac{1}{T} \sum_{t=1}^T (\mathbf{B}\mathbf{P}_t + \mathbf{u}_t)(\mathbf{B}\mathbf{P}_t + \mathbf{u}_t)' - \mathbb{E}(\mathbf{X}_t \mathbf{X}_t') \\
&= \frac{1}{T} \mathbf{B} \sum_{t=1}^T (\mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K) \mathbf{B}' + \frac{1}{T} \sum_{t=1}^T (\mathbf{u}_t \mathbf{u}_t' - \mathbb{E}(\mathbf{u}_t \mathbf{u}_t')) + \frac{1}{T} \sum_{t=1}^T (\mathbf{B}\mathbf{P}_t \mathbf{u}_t') + \frac{1}{T} \sum_{t=1}^T (\mathbf{u}_t \mathbf{P}_t' \mathbf{B}') \\
&= \mathbf{D}_1 + \mathbf{D}_2 + \mathbf{D}_3 + \mathbf{D}_4.
\end{aligned} \tag{A.9}$$

This, in turn, implies that $\mathbb{P}(\|\hat{\Sigma} - \Sigma\|_2 > \varepsilon) \leq \sum_{i=1}^4 \mathbb{P}(\|\mathbf{D}_i\|_2 > \varepsilon_i)$, or

$$\mathbb{P}\left(\|\hat{\Sigma} - \Sigma\|_2 \leq \varepsilon\right) \geq 1 - \sum_{i=1}^4 \mathbb{P}(\|\mathbf{D}_i\|_2 > \varepsilon_i), \tag{A.10}$$

for some $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4 > 0$ such that $\varepsilon = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4$. Let $\eta' = \eta + \delta > 0$ with $\delta > 0$.

Proposition A.3 and Assumption 3 imply that there is a constant $C_1 > 0$ such that

$$\|\mathbf{D}_1\|_2 \leq \left\| \frac{1}{T} \sum_{t=1}^T (\mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K) \right\|_2 \|\mathbf{B}\mathbf{B}'\|_2 \leq p^\alpha C_1 \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} = \varepsilon_1, \tag{A.11}$$

holds with probability at least $1 - T^{-\eta'}$. By Proposition A.2, there exists a constant $C_2 > 0$ such that

$$\|\mathbf{D}_2\|_2 \leq C_2 \left(\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) = \varepsilon_2, \tag{A.12}$$

holds with probability at least $1 - T^{-\eta'}$. Proposition A.4, Assumption 3 then imply that there is a constant $C_3 > 0$ such that

$$\|\mathbf{D}_3\|_2 \leq \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right\|_2 \|\mathbf{B}\|_2 \leq p^{\alpha/2} C_3 \sqrt{\frac{pK \log T}{T}} = \varepsilon_3, \quad (\text{A.13})$$

holds with probability at least $1 - T^{-\eta'}$. Last, note that $\|\mathbf{D}_3\|_2 = \|\mathbf{D}_4\|_2$. Combining the inequalities in (A.10), (A.11), (A.12), (A.13) we get

$$\mathbb{P} \left(\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2 \leq \varepsilon \right) \geq 1 - 4T^{-\eta'} \geq 1 - T^{-\eta},$$

where ε is defined as

$$\begin{aligned} \varepsilon = & p^\alpha C_1 \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} + C_2 \left(\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \\ & + 2p^{\alpha/2} C_3 \sqrt{\frac{pK \log T}{T}}. \end{aligned}$$

Claim (i) follows from noting that

$$\begin{aligned} \varepsilon = & p^\alpha C_1 \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} + p^\alpha C_2 \left(\frac{1}{p^\alpha} \frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \frac{1}{p^\alpha} \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \\ & + 2p^\alpha C_3 \sqrt{\frac{p^{1-\alpha} K \log T}{T}}, \end{aligned}$$

so that by Assumption 4, for all T sufficiently large, $\varepsilon \leq c_K p^\alpha / 2$. Then Assumption 3 implies

$$\hat{\lambda}_K \geq \lambda_K - |\lambda_K - \hat{\lambda}_K| \geq c_K p^\alpha - \|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2 \geq c_K p^\alpha - \varepsilon = \frac{c_K}{2} p^\alpha,$$

where the second inequality follows from Weyl's inequality.

(ii) Note that $\hat{\lambda}_K - \lambda_{K+1} \geq C' p^\alpha$ for some constant $C' > 0$ by Assumption 3. The claim then follows from Assumption 4 as $\|\mathbf{V}'_R(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_R\|_2 \leq \|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2 \leq \varepsilon$.

(iii) follows by the Davis-Kahan theorem of Yu, Wang, and Samworth (2015, Theorem

2), which implies

$$\|\widehat{\mathbf{V}}_K \widehat{\mathbf{V}}_K' - \mathbf{V}_K \mathbf{V}_K'\|_2 \leq \frac{2\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2}{c_K p^\alpha - c_{K+1}} \leq \frac{2\varepsilon}{c_K p^\alpha - c_{K+1}} \leq \sqrt{5/6},$$

where the final step follows for large enough T .

(iv) Consider the matrix

$$\widehat{\boldsymbol{\Lambda}}_K^{-1/2} \widehat{\mathbf{V}}_K' \boldsymbol{\Sigma} \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K^{-1/2} - \mathbf{I}_K = \widehat{\boldsymbol{\Lambda}}_K^{-1/2} \widehat{\mathbf{V}}_K' (\boldsymbol{\Sigma} - \widehat{\boldsymbol{\Sigma}}) \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K^{-1/2},$$

which follows as $\widehat{\mathbf{V}}_K' \widehat{\boldsymbol{\Sigma}} \widehat{\mathbf{V}}_K = \widehat{\boldsymbol{\Lambda}}_K$. Then

$$\|\widehat{\boldsymbol{\Lambda}}_K^{-1/2} \widehat{\mathbf{V}}_K' \boldsymbol{\Sigma} \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K^{-1/2} - \mathbf{I}_K\|_2 \leq \|\widehat{\boldsymbol{\Lambda}}_K^{-1/2}\|_2^2 \|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2 = \frac{\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2}{\widehat{\lambda}_K} \leq \frac{2\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2}{c_K p^\alpha}.$$

It then follows that

$$\lambda_{\min}(\widehat{\boldsymbol{\Lambda}}_K^{-1/2} \widehat{\mathbf{V}}_K' \boldsymbol{\Sigma} \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K^{-1/2}) \geq 1 - \|\widehat{\boldsymbol{\Lambda}}_K^{-1/2} \widehat{\mathbf{V}}_K' \boldsymbol{\Sigma} \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K^{-1/2} - \mathbf{I}_K\|_2 \geq 1 - \frac{2\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_2}{c_K p^\alpha} \geq 1/2,$$

by Weyl's inequality, for large enough T . $\square$

Proposition A.2. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t' - \mathbb{E}(\mathbf{u}_t \mathbf{u}_t') \right\|_2 \leq C \left(\frac{(p + \log(T))^{\frac{r_\beta + 1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta + 1}{r_\beta}}}{T}} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. When $c_{K+1} = 0$ the claim is trivial as $\mathbf{u}_t = \mathbf{0}$ almost surely, so we may assume otherwise. We check the conditions of Lemma A.2 for $\mathbf{u}_t$. Let $\boldsymbol{\Sigma}_u = \mathbb{E}(\mathbf{u}_t \mathbf{u}_t') = \mathbf{V}_R \mathbf{V}_R' \boldsymbol{\Sigma} \mathbf{V}_R \mathbf{V}_R'$. Lemma A.1 establishes that $\boldsymbol{\Sigma}_u^{-1/2} \mathbf{u}_t$ is sub-Gaussian. Additionally, standard properties of strong mixing processes imply that $\mathbf{u}_t$ inherits the mixing properties of the sequence $\{(Y_t, \mathbf{X}_t)'\}_{t=1}^T$ from Assumption 2. Finally, Assumption 4 implies that $p = \lfloor C_p T^{r_p} \rfloor$ where

$r_p < r_\beta$. We may then apply Lemma A.2 to obtain, for all T sufficiently large

$$\mathbb{P} \left(\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t' - \boldsymbol{\Sigma}_u \right\|_2 > \|\boldsymbol{\Sigma}_u\|_2 C_1 \left(\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \right) \leq T^{-\eta},$$

for some constant $C_1 > 0$. The claim then follows by noting that Assumption 3 implies that $\|\boldsymbol{\Sigma}_u\|_2 = c_{K+1}$ and setting $C = C_1 c_{K+1}$. $\square$

Proposition A.3. *Suppose Assumptions 1, 2 and 4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\begin{aligned} (i) \quad & \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2 \leq C_2 \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}}, \\ (ii) \quad & \lambda_{\min} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) \geq \frac{1}{2}, \end{aligned}$$

hold with probability at least $1 - T^{-\eta}$.

Proof. (i) Following similar arguments as in the proof of Proposition A.2, Lemma A.1 establishes that $\mathbf{P}_t$ is sub-Gaussian. Standard properties of strong mixing processes imply that $\mathbf{P}_t$ inherits the mixing properties from Assumption 2. Finally, Assumption 4 implies that $K = \lfloor C_K T^{r_K} \rfloor$ where $r_K < r_\beta$. We may then apply Lemma A.2 and Assumption 4 to obtain for some constant $C_1 > 0$, for all T sufficiently large

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2 & \leq C_1 \left(\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \\ & \leq C_2 \sqrt{\frac{(K + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}}, \end{aligned} \tag{A.14}$$

with probability at least $1 - T^{-\eta}$.

(ii) By Weyl's inequality

$$\lambda_{\min} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) \geq 1 - \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2.$$

The claim then follows from (A.14) and Assumption 4, since for every $\eta > 0$, for all T sufficiently large, $\|(1/T) \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K\|_2 \leq 1/2$ with probability at least $1 - T^{-\eta}$. $\square$

Proposition A.4. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}_t' \right\|_F \leq C \sqrt{\frac{pK \log(T)}{T}},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. We assume that c_{K+1} is positive (when c_{K+1} is zero the claim is trivial). Let $V_{ij,t}$ denote $P_{it}u_{jt}$ and $\mathbf{V}_{j,t}$ denote the j -th column of $\mathbf{P}_t \mathbf{u}_t'$ where $1 \leq i \leq K$ and $1 \leq j \leq p$. We begin by showing that, for each j , the sequence of K -dimensional random vectors $\{\mathbf{V}_{j,t}\}_{t=1}^T$ satisfies the conditions required by Lemma A.3. Lemma A.1 establishes that $\mathbf{P}_t$ and $\mathbf{u}_t$ are sub-Gaussian random vectors. Standard results on sub-Gaussian random variables then imply that for some $C'_m > 0$

$$\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{V}_{j,t}' \mathbf{v}| > \varepsilon) \leq \sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{P}_t' \mathbf{v}| |u_{jt}| > \varepsilon) \leq \exp(-C'_m \varepsilon),$$

and Lemma 1 implies that $\mathbb{E}(\mathbf{V}_{j,t}) = \mathbb{E}(\mathbf{P}_t u_{jt}) = \mathbf{0}$, so $\mathbf{V}_{j,t}$ is a zero-mean sub-exponential vector. Moreover, standard properties of strong mixing processes imply that $\{\mathbf{V}_{j,t}\}_{t=1}^T$ inherits the mixing properties of the sequence $\{(Y_t, \mathbf{X}_t)'\}_{t=1}^T$ from Assumption 2. Lastly, Assumption 4 implies that $K = \lfloor C_K T^{r_K} \rfloor$ where $r_K < 1$. We may then apply Lemma A.3 to the sequence $\{\mathbf{V}_{j,t}\}_{t=1}^T$. Choosing $\eta' = \eta + r_p + \delta$ with $\delta > 0$, there exists a $C_{\eta'} > 0$ such that, for all T sufficiently large

$$\mathbb{P} \left(\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_{j,t} \right\|_2 > C_{\eta'} \sqrt{\frac{K \log(T)}{T}} \right) \leq T^{-\eta'}.$$

The claim then follows as

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}_t' \right\|_F^2 = \sum_{j=1}^p \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_{j,t} \right\|_2^2 \leq p \max_{1 \leq j \leq p} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_{j,t} \right\|_2^2,$$

which implies by Assumption 4

$$\begin{aligned}
\mathbb{P} \left(\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right\|_F^2 > C_{\eta'}^2 \frac{pK \log(T)}{T} \right) &\leq \mathbb{P} \left(p \max_{1 \leq j \leq p} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_{j,t} \right\|_2^2 > C_{\eta'}^2 \frac{pK \log(T)}{T} \right) \\
&\leq p \max_{1 \leq j \leq p} \mathbb{P} \left(\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_{j,t} \right\|_2 > C_{\eta'} \sqrt{\frac{K \log(T)}{T}} \right) \\
&\leq \frac{p}{T^{\eta'}} = C_p \frac{T^{r_p}}{T^{\eta+r_p+\delta}} = C_p T^{-\eta-\delta} \leq T^{-\eta}.
\end{aligned}$$

□

Proposition A.5. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\mathbf{V}'_R(\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_K\|_F^2 \leq Cp^{1+\alpha} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. Consider the decomposition $\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma} = \mathbf{D}_1 + \mathbf{D}_2 + \mathbf{D}_3 + \mathbf{D}_4$ from (A.9). Projecting it into the relevant block, we obtain

$$\mathbf{V}'_R(\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_K = \sum_{j=1}^4 \mathbf{V}'_R \mathbf{D}_j \mathbf{V}_K = \mathbf{V}'_R \mathbf{D}_4 \mathbf{V}_K,$$

because $\mathbf{V}'_R \mathbf{D}_1 = \mathbf{0}$ and $\mathbf{V}'_R \mathbf{D}_3 = \mathbf{0}$ as $\mathbf{B} = \mathbf{V}_K \boldsymbol{\Lambda}_K^{1/2}$, and $\mathbf{D}_2 \mathbf{V}_K = \mathbf{0}$ as $\mathbf{u}_t = \mathbf{V}_R \mathbf{V}'_R \mathbf{X}_t$ and $\mathbb{E}[\mathbf{u}_t \mathbf{u}'_t] = \mathbf{V}_R \boldsymbol{\Lambda}_R \mathbf{V}'_R$, all of which follows from Lemma 1. Using $\mathbf{B}' \mathbf{V}_K = \boldsymbol{\Lambda}_K^{1/2}$ again yields

$$\mathbf{V}'_R(\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_K = \mathbf{V}'_R \left(\frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{P}'_t \mathbf{B}' \right) \mathbf{V}_K = \mathbf{V}'_R \left(\frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{P}'_t \right) \boldsymbol{\Lambda}_K^{1/2}.$$

Taking squared Frobenius norms and upper bounding gives

$$\|\mathbf{V}'_R(\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_K\|_F^2 = \left\| \mathbf{V}'_R \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right) \boldsymbol{\Lambda}_K^{1/2} \right\|_F^2 \leq c_1 p^\alpha \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right\|_F^2,$$

as $\lambda_1 = c_1 p^\alpha$ by Assumption 3. Then by Proposition A.4, for every $\eta > 0$ there exists a

constant $C > 0$ such that, for all T sufficiently large

$$\|\mathbf{V}'_R(\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\mathbf{V}_K\|_F^2 \leq Cp^{1+\alpha} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$ which proves the claim. $\square$

Proposition A.6. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\mathbf{V}'_R \widehat{\mathbf{V}}_K\|_F^2 \leq Cp^{1-\alpha} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. Let $\mathbf{S} \equiv \mathbf{V}'_R \widehat{\mathbf{V}}_K$ denote the matrix of interest. Our argument is inspired by the beginning of the proof of Theorem 2 in Yu *et al.* (2015), although we apply it to different matrices. We also note that our object of interest, $\mathbf{S}$, appears in their equation A4. Start from

$$\widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K = \widehat{\boldsymbol{\Sigma}} \widehat{\mathbf{V}}_K = (\boldsymbol{\Sigma} + (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})) \widehat{\mathbf{V}}_K.$$

Multiplying by $\mathbf{V}'_R$, we obtain

$$\mathbf{V}'_R \widehat{\mathbf{V}}_K \widehat{\boldsymbol{\Lambda}}_K = \mathbf{V}'_R \boldsymbol{\Sigma} \widehat{\mathbf{V}}_K + \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) \widehat{\mathbf{V}}_K = \boldsymbol{\Lambda}_R \mathbf{V}'_R \widehat{\mathbf{V}}_K + \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) \widehat{\mathbf{V}}_K,$$

or

$$\mathbf{S} \widehat{\boldsymbol{\Lambda}}_K - \boldsymbol{\Lambda}_R \mathbf{S} = \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) \widehat{\mathbf{V}}_K.$$

As $\mathbf{V}_K \mathbf{V}'_K + \mathbf{V}_R \mathbf{V}'_R = \mathbf{I}_p$, we may continue with

$$\begin{aligned} \mathbf{S} \widehat{\boldsymbol{\Lambda}}_K - \boldsymbol{\Lambda}_R \mathbf{S} &= \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) (\mathbf{V}_K \mathbf{V}'_K + \mathbf{V}_R \mathbf{V}'_R) \widehat{\mathbf{V}}_K \\ &= \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) \mathbf{V}_K \mathbf{V}'_K \widehat{\mathbf{V}}_K + \mathbf{V}'_R (\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}) \mathbf{V}_R \mathbf{S}. \end{aligned}$$

Taking the Frobenius norm on both sides, using the triangle inequality and the fact that

$\|\mathbf{V}'_K \widehat{\mathbf{V}}_K\|_2 \leq 1$ yields

$$\|\mathbf{S}\widehat{\mathbf{\Lambda}}_K - \mathbf{\Lambda}_R \mathbf{S}\|_F \leq \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_K\|_F + \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_R\|_2 \|\mathbf{S}\|_F, \quad (\text{A.15})$$

where we also used the fact that $\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_F$ for two conformable matrices $\mathbf{A}$ and $\mathbf{B}$. This Sylvester-type equation is an analogue to equation A2 in Yu *et al.* (2015). Now we lower bound the left hand side. Notice that the (i, j) -th entry of the $(p-K) \times K$ matrix $\mathbf{S}\widehat{\mathbf{\Lambda}}_K - \mathbf{\Lambda}_R \mathbf{S}$ is $S_{ij}(\hat{\lambda}_j - \lambda_{K+i})$, where S_{ij} is the (i, j) -entry of $\mathbf{S}$. Then we see that the square of the left hand side quantity from (A.15) is

$$\|\mathbf{S}\widehat{\mathbf{\Lambda}}_K - \mathbf{\Lambda}_R \mathbf{S}\|_F^2 = \sum_{i=1}^{p-K} \sum_{j=1}^K S_{ij}^2 (\hat{\lambda}_j - \lambda_{K+i})^2 \geq (\hat{\lambda}_K - \lambda_{K+1})^2 \|\mathbf{S}\|_F^2,$$

as $\hat{\lambda}_j - \lambda_{K+i} \geq \hat{\lambda}_K - \lambda_{K+1}$. Combining this with (A.15), we obtain

$$(\hat{\lambda}_K - \lambda_{K+1}) \|\mathbf{S}\|_F \leq \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_K\|_F + \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_R\|_2 \|\mathbf{S}\|_F,$$

which implies

$$\|\mathbf{S}\|_F \leq \frac{\|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_K\|_F}{(\hat{\lambda}_K - \lambda_{K+1}) - \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_R\|_2}. \quad (\text{A.16})$$

Let $\eta' = \eta + \delta$ with $\delta > 0$. Proposition A.1 implies that for all sufficiently large T , we have $\hat{\lambda}_K \geq c_K p^\alpha / 2$ and $\|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_R\|_2 \leq (\hat{\lambda}_K - \lambda_{K+1})/2$ with probability at least $1 - T^{-\eta'}$. In particular, this implies

$$(\hat{\lambda}_K - \lambda_{K+1}) - \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_R\|_2 \geq \frac{\hat{\lambda}_K - \lambda_{K+1}}{2}.$$

Therefore, there exists a constant $C_1 > 0$ such that, for T sufficiently large, we obtain from (A.16)

$$\|\mathbf{S}\|_F^2 \leq \frac{4C_1}{p^{2\alpha}} \|\mathbf{V}'_R(\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma})\mathbf{V}_K\|_F^2,$$

with probability at least $1 - T^{-\eta'}$. Finally, by Proposition A.5, there exists a constant

$C > 0$ such that, for all T sufficiently large

$$\|\mathbf{S}\|_F^2 = \|\mathbf{V}'_R \widehat{\mathbf{V}}_K\|_F^2 \leq C \frac{p^{1+\alpha}}{p^{2\alpha}} \frac{K \log(T)}{T} = C \frac{1}{p^{\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - 2T^{-\eta'} \geq 1 - T^{-\eta}$. $\square$

Proposition A.7. *Suppose Assumptions 1 and 2 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\hat{\boldsymbol{\theta}}\|_2^2 \leq \frac{1}{T} \|\mathbf{Y}\|_2^2 \leq C,$$

holds with probability at least $1 - T^{-\eta}$.

Proof. First note that we have

$$\|\hat{\boldsymbol{\theta}}\|_2^2 = \left\| \frac{1}{T} \widehat{\mathbf{P}}' \mathbf{Y} \right\|_2^2 \leq \left\| \sqrt{\frac{1}{T}} \widehat{\mathbf{P}}' \right\|_2^2 \left\| \sqrt{\frac{1}{T}} \mathbf{Y} \right\|_2^2 = \frac{1}{T} \|\mathbf{Y}\|_2^2.$$

The claim then follows from Assumption 1 and Lemma A.3, which implies that for every $\eta > 0$, there exists a constant $C > 0$ such that, for all T sufficiently large

$$\frac{1}{T} \|\mathbf{Y}\|_2^2 = \mathbb{E}[Y_t^2] + \frac{1}{T} \sum_{t=1}^T (Y_t^2 - \mathbb{E}[Y_t^2]) \leq C,$$

with probability at least $1 - T^{-\eta}$. $\square$

A.1.2 Proof of Proposition 2

Proof of Proposition 2. Let $\eta' = \eta + \delta$ with $\delta > 0$. By Proposition A.3, for all T sufficiently large

$$\lambda_{\min} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) \geq \frac{1}{2},$$

holds with probability at least $1 - T^{-\eta'}$. By Proposition A.7, there exists a constant C_1 such that, for all T sufficiently large, $\|\mathbf{Y}\|_2^2/T \leq C_1$ holds with probability at least

$1 - T^{-\eta'}$. Then by Proposition A.8, there exists a constant $C_2 > 0$ such that

$$\|\tilde{\boldsymbol{\theta}} - \mathbf{H}'\hat{\boldsymbol{\theta}}\|_2^2 \leq 4\lambda_{\min}\left(\frac{\mathbf{P}'\mathbf{P}}{T}\right)^{-1} \frac{\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2^2}{T} \frac{\|\mathbf{Y}\|_2^2}{T} \leq C_2 \frac{\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2^2}{T},$$

with probability at least $1 - 2T^{-\eta'}$. Recall that $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T)'$, and note that $\mathbf{P} = \mathbf{X}\mathbf{V}_K\boldsymbol{\Lambda}_K^{-1/2}$ and $\hat{\mathbf{P}} = \mathbf{X}\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2}$. The proof is based on an argument very similar in nature to the one used in the proof of Proposition 1. Note that

$$\begin{aligned} \hat{\mathbf{P}} - \mathbf{P}\mathbf{H}' &= \mathbf{X}\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2} - \mathbf{X}\mathbf{V}_K\boldsymbol{\Lambda}_K^{1/2}\boldsymbol{\Lambda}_K^{-1/2}\mathbf{V}_K'\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2} \\ &= \mathbf{X}(\mathbf{I}_p - \mathbf{V}_K\mathbf{V}_K')\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2} \\ &= \mathbf{X}\mathbf{V}_R\mathbf{V}_R'\mathbf{V}_R\mathbf{V}_R'\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2} \\ &= \mathbf{U}\mathbf{V}_R\mathbf{V}_R'\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2}, \end{aligned}$$

where $\mathbf{U} = \mathbf{X}\mathbf{V}_R\mathbf{V}_R'$. We then have

$$\frac{1}{T}\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_F^2 = \frac{1}{T}\|\mathbf{U}\mathbf{V}_R\mathbf{V}_R'\hat{\mathbf{V}}_K\hat{\boldsymbol{\Lambda}}_K^{-1/2}\|_F^2 \leq \left\|\frac{1}{\sqrt{T}}\mathbf{U}\right\|_2^2 \|\mathbf{V}_R'\hat{\mathbf{V}}_K\|_F^2 \|\hat{\boldsymbol{\Lambda}}_K^{-1/2}\|_2^2. \quad (\text{A.17})$$

We then bound each term. First, we have $\|\mathbf{U}/\sqrt{T}\|_2^2 = \|\mathbf{U}'\mathbf{U}/T\|_2 = \|\sum_{t=1}^T \mathbf{u}_t\mathbf{u}_t'/T\|_2$ and

$$\left\|\frac{1}{T}\sum_{t=1}^T \mathbf{u}_t\mathbf{u}_t'\right\|_2 \leq \left\|\frac{1}{T}\sum_{t=1}^T \mathbf{u}_t\mathbf{u}_t' - \mathbb{E}(\mathbf{u}_t\mathbf{u}_t')\right\|_2 + \|\mathbb{E}(\mathbf{u}_t\mathbf{u}_t')\|_2 = \left\|\frac{1}{T}\sum_{t=1}^T \mathbf{u}_t\mathbf{u}_t' - \mathbb{E}(\mathbf{u}_t\mathbf{u}_t')\right\|_2 + C_{K+1}.$$

by Assumption 3. Proposition A.2 implies that there exists a constant $C_3 > 0$ such that, for all T sufficiently large

$$\left\|\frac{1}{\sqrt{T}}\mathbf{U}\right\|_2^2 \leq C_3 \left(1 + \frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}}\right),$$

holds with probability at least $1 - T^{-\eta'}$. Proposition A.1 and Assumption 3 imply that there exists a constant $C_4 > 0$ such that, for all T sufficiently large

$$\|\widehat{\mathbf{\Lambda}}_K^{-1/2}\|_2^2 = \frac{1}{\widehat{\lambda}_K} \leq C_4 p^{-\alpha},$$

with probability at least $1 - T^{-\eta'}$. Finally, Proposition A.6 implies that there exists a constant $C_5 > 0$ such that, for all T sufficiently large

$$\|\mathbf{V}'_R \widehat{\mathbf{V}}_K\|_F^2 \leq C_5 p^{1-\alpha} \frac{K \log(T)}{T},$$

with probability at least $1 - T^{-\eta'}$. Combining these three bounds with (A.17) then yields

$$\frac{1}{T} \|\widehat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_F^2 \leq C_6 \left(1 + \frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T} + \sqrt{\frac{(p + \log(T))^{\frac{r_\beta+1}{r_\beta}}}{T}} \right) \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

for some constant $C_6 > 0$, with probability at least $1 - 3T^{-\eta'}$. This then implies that there exists a constant $C > 0$ such that the claim holds with probability at least $1 - 5T^{-\eta'} \geq 1 - T^{-\eta}$. $\square$

Proposition A.8. *Assume that $\mathbf{P}$ has full column rank. Then*

$$\|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_2^2 \leq 4\lambda_{\min} \left(\frac{\mathbf{P}'\mathbf{P}}{T} \right)^{-1} \frac{\|\widehat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2^2}{T} \frac{\|\mathbf{Y}\|_2^2}{T}.$$

Proof. Recall that $\tilde{\boldsymbol{\vartheta}} = \operatorname{argmin}_{\boldsymbol{\vartheta} \in \mathbb{R}^K} \|\mathbf{Y} - \mathbf{P}\boldsymbol{\vartheta}\|_2^2$ and $\hat{\boldsymbol{\vartheta}} = \widehat{\mathbf{P}}'\mathbf{Y}/T$, because $\widehat{\mathbf{P}}'\widehat{\mathbf{P}}/T = \mathbf{I}_K$. We thus have $\tilde{\boldsymbol{\vartheta}} = (\mathbf{P}'\mathbf{P}/T)^{-1}\mathbf{P}'\mathbf{Y}/T$. Define the projectors $\Pi_{\widehat{\mathbf{P}}} = \widehat{\mathbf{P}}(\widehat{\mathbf{P}}'\widehat{\mathbf{P}})^{-1}\widehat{\mathbf{P}}' = \frac{1}{T}\widehat{\mathbf{P}}\widehat{\mathbf{P}}'$ and $\Pi_{\mathbf{P}} = \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}'$. Then consider the fitted values for $\hat{\boldsymbol{\vartheta}}$ and $\tilde{\boldsymbol{\vartheta}}$

$$\begin{aligned} \widehat{\mathbf{P}}\hat{\boldsymbol{\vartheta}} &= \frac{1}{T}\widehat{\mathbf{P}}\widehat{\mathbf{P}}'\mathbf{Y} = \Pi_{\widehat{\mathbf{P}}}\mathbf{Y}, \\ \mathbf{P}\tilde{\boldsymbol{\vartheta}} &= \mathbf{P} \left(\frac{1}{T}\mathbf{P}'\mathbf{P} \right)^{-1} \frac{1}{T}\mathbf{P}'\mathbf{Y} = \Pi_{\mathbf{P}}\mathbf{Y}. \end{aligned}$$

Then we have by addition and subtraction

$$\begin{aligned}
\mathbf{P}(\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}) &= \Pi_{\mathbf{P}}\mathbf{Y} - \mathbf{P}\mathbf{H}'\hat{\boldsymbol{\vartheta}} \\
&= \Pi_{\mathbf{P}}\mathbf{Y} - \hat{\mathbf{P}}\hat{\boldsymbol{\vartheta}} + \hat{\mathbf{P}}\hat{\boldsymbol{\vartheta}} - \mathbf{P}\mathbf{H}'\hat{\boldsymbol{\vartheta}} \\
&= (\Pi_{\mathbf{P}} - \Pi_{\hat{\mathbf{P}}})\mathbf{Y} + (\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}')\hat{\boldsymbol{\vartheta}}.
\end{aligned}$$

As a consequence we can upper bound its norm as

$$\|\mathbf{P}(\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}})\|_2 \leq \|\Pi_{\mathbf{P}} - \Pi_{\hat{\mathbf{P}}}\|_2 \|\mathbf{Y}\|_2 + \|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2 \frac{\|\mathbf{Y}\|_2}{\sqrt{T}}, \quad (\text{A.18})$$

where we used Proposition A.7. We now apply Theorem 2.5.1 from Golub and Van Loan (2013). In their notation, let $S_1 = \text{span}(\mathbf{P})$ and $S_2 = \text{span}(\hat{\mathbf{P}})$. We may then choose $\mathbf{Z}_1 = \hat{\mathbf{P}}/\sqrt{T}$ so that $\mathbf{Z}_1'\mathbf{Z}_1 = \mathbf{I}_K$ and $\mathbf{W}_1 = \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1/2}$ so that $\mathbf{W}_1'\mathbf{W}_1 = \mathbf{I}_K$. We then have

$$\mathbf{W}_2\mathbf{W}_2' = \mathbf{I}_T - \mathbf{W}_1\mathbf{W}_1' = \mathbf{I}_T - \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1/2}(\mathbf{P}'\mathbf{P})^{-1/2}\mathbf{P}' = \mathbf{I}_T - \Pi_{\mathbf{P}}.$$

Their Theorem 2.5.1 along with equation 2.5.1 then yields

$$\|\Pi_{\mathbf{P}} - \Pi_{\hat{\mathbf{P}}}\|_2 = \|\mathbf{W}_2'\mathbf{Z}_1\|_2 = \|\mathbf{W}_2\mathbf{W}_2'\mathbf{Z}_1\|_2 = \left\| (\mathbf{I}_T - \Pi_{\mathbf{P}}) \frac{1}{\sqrt{T}} \hat{\mathbf{P}} \right\|_2.$$

Since $\Pi_{\mathbf{P}}$ is the orthogonal projector onto the column space of $\mathbf{P}$, $\mathbf{I}_T - \Pi_{\mathbf{P}}$ is the orthogonal projector onto its orthogonal complement. Moreover, $\text{span}(\mathbf{P}\mathbf{H}') \subseteq \text{span}(\mathbf{P})$, so it follows that $(\mathbf{I}_T - \Pi_{\mathbf{P}})\mathbf{P}\mathbf{H}' = \mathbf{0}$. This means that

$$\left\| (\mathbf{I}_T - \Pi_{\mathbf{P}}) \frac{1}{\sqrt{T}} \hat{\mathbf{P}} \right\|_2 = \left\| (\mathbf{I}_T - \Pi_{\mathbf{P}}) \frac{\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'}{\sqrt{T}} \right\|_2 \leq \frac{\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2}{\sqrt{T}}.$$

We can then continue from (A.18)

$$\|\mathbf{P}(\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}})\|_2 \leq \|\Pi_{\mathbf{P}} - \Pi_{\hat{\mathbf{P}}}\|_2 \|\mathbf{Y}\|_2 + \|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2 \frac{\|\mathbf{Y}\|_2}{\sqrt{T}} \leq 2\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2 \frac{\|\mathbf{Y}\|_2}{\sqrt{T}}.$$

Finally, we translate from the fitted values to the coefficients. For every $\mathbf{a} \in \mathbb{R}^K$

$$\frac{1}{T} \|\mathbf{P}\mathbf{a}\|_2^2 = \mathbf{a}' \left(\frac{\mathbf{P}'\mathbf{P}}{T} \right) \mathbf{a} \geq \lambda_{\min} \left(\frac{\mathbf{P}'\mathbf{P}}{T} \right) \|\mathbf{a}\|_2^2.$$

Applying this with $\mathbf{a} = \tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}$ gives

$$\lambda_{\min} \left(\frac{\mathbf{P}'\mathbf{P}}{T} \right)^{1/2} \|\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}}\|_2 \leq \frac{1}{\sqrt{T}} \|\mathbf{P}(\tilde{\boldsymbol{\vartheta}} - \mathbf{H}'\hat{\boldsymbol{\vartheta}})\|_2 \leq 2 \frac{\|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_2}{\sqrt{T}} \frac{\|\mathbf{Y}\|_2}{\sqrt{T}}.$$

□

A.1.3 Proof of Proposition 3

Proof of Proposition 3. The normal equations for $\tilde{\boldsymbol{\vartheta}}$ imply

$$\left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) (\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*) = \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t (Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*). \quad (\text{A.19})$$

Let $\eta' = \eta + \delta$ with $\delta > 0$. Proposition A.3 guarantees that $\lambda_{\min} \left(\sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' / T \right) \geq 1/2$ with probability at least $1 - T^{-\eta'}$. As a consequence, (A.19) and Proposition A.9 imply that

$$\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^*\|_2 \leq 2 \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t (Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \right\|_2 \leq 2C \sqrt{\frac{K \log(T)}{T}},$$

with probability at least $1 - 2T^{-\eta'} \geq 1 - T^{-\eta}$. The claim then follows after squaring both sides. □

Proposition A.9. *Suppose Assumptions 1–4 are satisfied. Then for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t (Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \right\|_2 \leq C \sqrt{\frac{K \log(T)}{T}},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. Define $\mathbf{W}_t = (Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t$. We first verify that $\mathbf{W}_t$ satisfies the conditions of Lemma A.3. First, $\mathbb{E}[\mathbf{W}_t] = \mathbb{E}[(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t] = \mathbf{0}$, since $\boldsymbol{\vartheta}^*$ is the population least squares

coefficient associated with the regression of Y_t on $\mathbf{P}_t$. Assumption 1 and Lemma A.1 imply that $\mathbf{W}_t$ is sub-exponential. Moreover, $\{\mathbf{W}_t\}$ inherits the mixing properties of $\{(Y_t, \mathbf{X}_t')'\}$ from Assumption 2. Assumption 4 ensures that the remaining conditions of Lemma A.3 are satisfied. The lemma then implies that for every $\eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large,

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \right\|_2 \leq C \sqrt{\frac{K \log T}{T}},$$

with probability at least $1 - T^{-\eta}$ which establishes the claim. $\square$

A.2 Proof of Theorem 2

Proof of Theorem 2. The proof proceeds in four steps. First we decompose the excess risk into an estimation term and a bias term. Second, we lower bound the bias term. Third, we lower bound the estimation term conditional on the training regressors with high probability. Finally, we remove the conditioning and complete the argument.

Step 1. We begin by introducing the oracle PCR estimator on the sample eigenvectors $\hat{\mathbf{V}}_K$

$$\tilde{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \text{span}(\hat{\mathbf{V}}_K)}{\text{argmin}} R(\boldsymbol{\theta}).$$

Define $R(\boldsymbol{\theta}) = \mathbb{E} [(Y_{T+1} - \mathbf{X}_{T+1}' \boldsymbol{\theta})^2 \mid \mathcal{D}_T]$ for an arbitrary $\boldsymbol{\theta} \in \mathbb{R}^p$ and note that

$$R(\boldsymbol{\theta}^*) = \min_{\boldsymbol{\theta} \in \mathbb{R}^p} \mathbb{E} [(Y_{T+1} - \mathbf{X}_{T+1}' \boldsymbol{\theta})^2 \mid \mathcal{D}_T] = R^*.$$

We may then decompose the excess risk as

$$R(\hat{\boldsymbol{\theta}}) - R^* = (R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}})) + (R(\tilde{\boldsymbol{\theta}}) - R^*),$$

where the first term is the estimation error within the class of PCR estimators using $\hat{\mathbf{V}}_K$ (i.e. relative to the PCR oracle on the sample eigenvectors), and the second term is the bias relative to the oracle.

Step 2. We next lower bound the bias term. For any $\boldsymbol{\theta} \in \text{span}(\widehat{\mathbf{V}}_K)$, we have

$$\begin{aligned}
R(\boldsymbol{\theta}) - R^* &= \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta})^2 \mid \mathcal{D}_T] - \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^*)^2 \mid \mathcal{D}_T] \\
&= \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^* + \mathbf{X}'_{T+1}\boldsymbol{\theta}^* - \mathbf{X}'_{T+1}\boldsymbol{\theta})^2 \mid \mathcal{D}_T] - \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^*)^2 \mid \mathcal{D}_T] \\
&= \mathbb{E}[(\mathbf{X}'_{T+1}(\boldsymbol{\theta} - \boldsymbol{\theta}^*))^2 \mid \mathcal{D}_T] \\
&= (\boldsymbol{\theta} - \boldsymbol{\theta}^*)' \boldsymbol{\Sigma} (\boldsymbol{\theta} - \boldsymbol{\theta}^*),
\end{aligned}$$

which follows as $Y_{T+1} - \mathbf{X}'_{T+1}\boldsymbol{\theta}^* = \varepsilon_{T+1}$, $\mathbb{E}[\varepsilon_{T+1}\mathbf{X}_{T+1} \mid \mathcal{D}_T] = \mathbf{0}$ and $\mathbb{E}[\mathbf{X}_{T+1}\mathbf{X}'_{T+1} \mid \mathcal{D}_T] = \boldsymbol{\Sigma}$. As a consequence

$$R(\tilde{\boldsymbol{\theta}}) - R^* = \inf_{\boldsymbol{\theta} \in \text{span}(\widehat{\mathbf{V}}_K)} (\boldsymbol{\theta} - \boldsymbol{\theta}^*)' \boldsymbol{\Sigma} (\boldsymbol{\theta} - \boldsymbol{\theta}^*) \geq \inf_{\boldsymbol{\theta} \in \text{span}(\widehat{\mathbf{V}}_K)} \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2,$$

because

$$(\boldsymbol{\theta} - \boldsymbol{\theta}^*)' \boldsymbol{\Sigma} (\boldsymbol{\theta} - \boldsymbol{\theta}^*) = (\boldsymbol{\theta} - \boldsymbol{\theta}^*)' \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}' (\boldsymbol{\theta} - \boldsymbol{\theta}^*) \geq \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2.$$

Now define the projectors $\boldsymbol{\Pi} = \mathbf{V}_K \mathbf{V}'_K$ and $\widehat{\boldsymbol{\Pi}} = \widehat{\mathbf{V}}_K \widehat{\mathbf{V}}'_K$. Then by the projection theorem

$$\inf_{\boldsymbol{\theta} \in \text{span}(\widehat{\mathbf{V}}_K)} \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 = \|(\mathbf{I}_p - \widehat{\boldsymbol{\Pi}})\boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}^*\|_2^2 - \|\widehat{\boldsymbol{\Pi}}\boldsymbol{\theta}^*\|_2^2,$$

as this is the shortest distance from $\boldsymbol{\theta}^*$ to the subspace $\text{span}(\widehat{\mathbf{V}}_K)$, and the second step is by the Pythagorean theorem since $\widehat{\boldsymbol{\Pi}}$ is an orthogonal projector. In our setting, $\boldsymbol{\theta}^* = \mathbf{V}_R \mathbf{V}'_R \boldsymbol{\theta}^* = \boldsymbol{\gamma}^*$, so $\boldsymbol{\Pi}\boldsymbol{\theta}^* = \mathbf{0}$, so that $\widehat{\boldsymbol{\Pi}}\boldsymbol{\theta}^* = (\widehat{\boldsymbol{\Pi}} - \boldsymbol{\Pi})\boldsymbol{\theta}^*$ and thus

$$\|\widehat{\boldsymbol{\Pi}}\boldsymbol{\theta}^*\|_2 \leq \|\widehat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}\|_2 \|\boldsymbol{\theta}^*\|_2.$$

Putting things together, we obtain

$$R(\tilde{\boldsymbol{\theta}}) - R^* \geq \inf_{\boldsymbol{\theta} \in \text{span}(\widehat{\mathbf{V}}_K)} \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 \geq (1 - \|\widehat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}\|_2^2) \|\boldsymbol{\theta}^*\|_2^2 \geq (1 - \|\widehat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}\|_2^2) \|\boldsymbol{\gamma}^*\|_2^2.$$

Step 3. We next turn to lower bounding the estimation term. We have

$$\begin{aligned}
R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) &= \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\hat{\boldsymbol{\theta}})^2 \mid \mathcal{D}_T] - \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\tilde{\boldsymbol{\theta}})^2 \mid \mathcal{D}_T] \\
&= \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\tilde{\boldsymbol{\theta}} + \mathbf{X}'_{T+1}\tilde{\boldsymbol{\theta}} - \mathbf{X}'_{T+1}\hat{\boldsymbol{\theta}})^2 \mid \mathcal{D}_T] - \mathbb{E}[(Y_{T+1} - \mathbf{X}'_{T+1}\tilde{\boldsymbol{\theta}})^2 \mid \mathcal{D}_T] \\
&= \mathbb{E}[(\mathbf{X}'_{T+1}\tilde{\boldsymbol{\theta}} - \mathbf{X}'_{T+1}\hat{\boldsymbol{\theta}})^2 \mid \mathcal{D}_T] \\
&= (\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}})' \boldsymbol{\Sigma} (\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}),
\end{aligned}$$

by the projection theorem. Now consider $\tilde{\boldsymbol{\theta}}$, the oracle PCR estimator using $\hat{\mathbf{V}}_K$. Because every $\boldsymbol{\theta} \in \text{span}(\hat{\mathbf{V}}_K)$ has the form $\boldsymbol{\theta} = \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2} \boldsymbol{\vartheta}$, there exists a corresponding oracle coefficient $\tilde{\boldsymbol{\vartheta}} \in \mathbb{R}^K$ such that $\tilde{\boldsymbol{\theta}} = \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2} \tilde{\boldsymbol{\vartheta}}$. We can thus continue to obtain

$$\begin{aligned}
R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) &= (\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}})' \boldsymbol{\Sigma} (\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}) \\
&= (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}})' \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\mathbf{V}}_K' \boldsymbol{\Sigma} \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2} (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}}) \\
&\equiv (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}})' \mathbf{G} (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}}),
\end{aligned} \tag{A.20}$$

as $\hat{\boldsymbol{\theta}} = \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\boldsymbol{\vartheta}}$, because $\hat{\boldsymbol{\theta}}_{PCR} \in \text{span}(\hat{\mathbf{V}}_K)$, and we define $\mathbf{G} = \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\mathbf{V}}_K' \boldsymbol{\Sigma} \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2}$.

To keep the proof self-contained, recall that $\hat{\mathbf{P}}_t = \hat{\boldsymbol{\Lambda}}_K^{-1/2} \hat{\mathbf{V}}_K' \mathbf{X}_t$ and $\hat{\mathbf{P}} = (\hat{\mathbf{P}}_1, \dots, \hat{\mathbf{P}}_T)'$. We also have from the paper that $\hat{\mathbf{P}}' \hat{\mathbf{P}} / T = \mathbf{I}_K$ and $\hat{\boldsymbol{\vartheta}} = \hat{\mathbf{P}}' \mathbf{Y} / T$. We therefore obtain

$$\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}} = \frac{1}{T} \hat{\mathbf{P}}' \mathbf{Y} - \frac{1}{T} \hat{\mathbf{P}}' \hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}} = \frac{1}{T} \hat{\mathbf{P}}' (\mathbf{Y} - \hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}}). \tag{A.21}$$

Now rewrite the vector

$$\mathbf{Y} - \hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}} = \mathbf{X} \boldsymbol{\theta}^* + \boldsymbol{\varepsilon} - \hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}} = \mathbf{X} (\boldsymbol{\theta}^* - \tilde{\boldsymbol{\theta}}) + \boldsymbol{\varepsilon},$$

because $\hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}} = \mathbf{X} \hat{\mathbf{V}}_K \hat{\boldsymbol{\Lambda}}_K^{-1/2} \tilde{\boldsymbol{\vartheta}} = \mathbf{X} \tilde{\boldsymbol{\theta}}$. Then we obtain from (A.21)

$$\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}} = \frac{1}{T} \hat{\mathbf{P}}' (\mathbf{Y} - \hat{\mathbf{P}} \tilde{\boldsymbol{\vartheta}}) = \frac{1}{T} \hat{\mathbf{P}}' \mathbf{X} (\boldsymbol{\theta}^* - \tilde{\boldsymbol{\theta}}) + \frac{1}{T} \hat{\mathbf{P}}' \boldsymbol{\varepsilon} \equiv \mathbf{a} + \mathbf{Z},$$

where we defined $\mathbf{a} = \hat{\mathbf{P}}' \mathbf{X} (\boldsymbol{\theta}^* - \tilde{\boldsymbol{\theta}}) / T$ and $\mathbf{Z} = \hat{\mathbf{P}}' \boldsymbol{\varepsilon} / T$. Combining this with (A.20), we

obtain

$$R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) = (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}})' \mathbf{G} (\hat{\boldsymbol{\vartheta}} - \tilde{\boldsymbol{\vartheta}}) = (\mathbf{a} + \mathbf{Z})' \mathbf{G} (\mathbf{a} + \mathbf{Z}).$$

Let $\mathcal{D}_{T,X} = \{\mathbf{X}_s\}_{s=1}^T$ be the regressors from the training sample. Conditioning on $\mathcal{D}_{T,X}$, the vector $\mathbf{a}$ is deterministic, while $\mathbf{Z}$ satisfies

$$\mathbf{Z} \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma^2}{T} \mathbf{I}_K\right), \quad (\text{A.22})$$

because $\text{Var}(\mathbf{Z} \mid \mathcal{D}_{T,X}) = \text{Var}(\hat{\mathbf{P}}' \boldsymbol{\varepsilon} / T \mid \mathcal{D}_{T,X}) = \sigma^2 \hat{\mathbf{P}}' \hat{\mathbf{P}} / T^2$. In other words, lower bounding the excess risk boils down to lower bounding $(\mathbf{a} + \mathbf{Z})' \mathbf{G} (\mathbf{a} + \mathbf{Z})$, which is a quadratic form in a shifted Gaussian vector if we condition on $\mathcal{D}_{T,X}$. Because $\mathbf{G}$ is non-negative definite, we use Anderson's shifted ball inequality (Anderson, 1955) to obtain for any $r > 0$

$$\mathbb{P}\left(R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) \geq r \mid \mathcal{D}_{T,X}\right) = \mathbb{P}\left((\mathbf{a} + \mathbf{Z})' \mathbf{G} (\mathbf{a} + \mathbf{Z}) \geq r \mid \mathcal{D}_{T,X}\right) \geq \mathbb{P}\left(\mathbf{Z}' \mathbf{G} \mathbf{Z} \geq r \mid \mathcal{D}_{T,X}\right).$$

Define $\lambda_{\min}(\mathbf{G}) \geq 0$ as the smallest eigenvalue of $\mathbf{G}$, so that $\mathbf{Z}' \mathbf{G} \mathbf{Z} \geq \lambda_{\min}(\mathbf{G}) \|\mathbf{Z}\|_2^2$.

By (A.22) we have

$$\|\mathbf{Z}\|_2^2 = \frac{\sigma^2}{T} \left\| \mathbf{Z} \sqrt{\frac{T}{\sigma^2}} \right\|_2^2 \sim \frac{\sigma^2}{T} \chi_K^2,$$

which implies

$$\mathbb{P}\left(R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) \geq r \mid \mathcal{D}_T\right) \geq \mathbb{P}\left(\lambda_{\min}(\mathbf{G}) \frac{\sigma^2}{T} \chi_K^2 \geq r \mid \mathcal{D}_{T,X}\right).$$

We may then apply the Laurent-Massart lower-tail bound for chi-square variables (Laurent and Massart, 2000) for $0 \leq x < K/4$

$$\mathbb{P}\left(\chi_K^2 \leq K - 2\sqrt{Kx}\right) \leq e^{-x}.$$

Take $x = K/9$ and $r = \lambda_{\min}(\mathbf{G}) \sigma^2 K / (3T)$ to obtain

$$\mathbb{P}\left(R(\hat{\boldsymbol{\theta}}) - R(\tilde{\boldsymbol{\theta}}) \geq \lambda_{\min}(\mathbf{G}) \frac{\sigma^2 K}{3T} \mid \mathcal{D}_{T,X}\right) \geq \mathbb{P}\left(\chi_K^2 \geq \frac{K}{3}\right) \geq 1 - e^{-K/9}.$$

Finally, add back the bias term to obtain

$$\mathbb{P} \left(R(\hat{\boldsymbol{\theta}}) - R^* \geq (1 - \|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}\|_2^2) \|\boldsymbol{\gamma}^*\|_2^2 + \lambda_{\min}(\mathbf{G}) \frac{\sigma^2 K}{3} \frac{K}{T} \middle| \mathcal{D}_{T,X} \right) \geq 1 - e^{-K/9}.$$

Step 4. Proposition A.1 implies that for every η , and for sufficiently large T , we have $\|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}\|_2 \leq \sqrt{5/6}$ and $\lambda_{\min}(\mathbf{G}) \geq 1/2$ with probability at least $1 - T^{-\eta}$. Let $\mathcal{E}_T$ denote this event. On $\mathcal{E}_T$, we thus obtain

$$\mathbb{P} \left(R(\hat{\boldsymbol{\theta}}) - R^* \geq \frac{1}{6} \left(\|\boldsymbol{\gamma}^*\|_2^2 + \sigma^2 \frac{K}{T} \right) \middle| \mathcal{D}_{T,X} \right) \geq 1 - e^{-K/9}.$$

Next define the event

$$\mathcal{A}_T = \left\{ R(\hat{\boldsymbol{\theta}}) - R^* \geq \frac{1}{6} \left(\|\boldsymbol{\gamma}^*\|_2^2 + \sigma^2 \frac{K}{T} \right) \right\}.$$

We have shown $\mathbb{P}(\mathcal{A}_T \mid \mathcal{D}_{T,X}) \geq 1 - e^{-K/9}$ on the event $\mathcal{E}_T$, or equivalently that

$$\mathbf{1}_{\mathcal{E}_T} \mathbb{P}(\mathcal{A}_T \mid \mathcal{D}_{T,X}) \geq \mathbf{1}_{\mathcal{E}_T} (1 - e^{-K/9}), \quad (\text{A.23})$$

almost surely. We may write

$$\mathbf{1}_{\mathcal{E}_T} \mathbb{P}(\mathcal{A}_T \mid \mathcal{D}_{T,X}) = \mathbf{1}_{\mathcal{E}_T} \mathbb{E}[\mathbf{1}_{\mathcal{A}_T} \mid \mathcal{D}_{T,X}] = \mathbb{E}[\mathbf{1}_{\mathcal{E}_T} \mathbf{1}_{\mathcal{A}_T} \mid \mathcal{D}_{T,X}],$$

where the second step follows because $\mathbf{1}_{\mathcal{E}_T}$ is measurable with respect to the sigma algebra generated by $\mathcal{D}_{T,X}$. Plugging this into the left hand side of (A.23) and taking expectations gives

$$\mathbb{E}[\mathbb{E}[\mathbf{1}_{\mathcal{E}_T} \mathbf{1}_{\mathcal{A}_T} \mid \mathcal{D}_{T,X}]] = \mathbb{E}[\mathbf{1}_{\mathcal{E}_T} \mathbf{1}_{\mathcal{A}_T}] = \mathbb{P}(\mathcal{A}_T \cap \mathcal{E}_T),$$

by the law of iterated expectations. It follows that

$$\mathbb{P}(\mathcal{A}_T \cap \mathcal{E}_T) \geq \mathbb{E}[\mathbf{1}_{\mathcal{E}_T}] (1 - e^{-K/9}) = \mathbb{P}(\mathcal{E}_T) (1 - e^{-K/9}).$$

Finally, since $\mathbb{P}(\mathcal{A}_T) \geq \mathbb{P}(\mathcal{A}_T \cap \mathcal{E}_T)$ and $\mathbb{P}(\mathcal{E}_T) \geq 1 - T^{-\eta}$ by Proposition A.1 , we have

$$\mathbb{P}(\mathcal{A}_T) \geq 1 - T^{-\eta} - e^{-K/9},$$

which completes the proof. $\square$

A.3 Proof of Theorem 3

Proof of Theorem 3. We give a sketch of the proof. Since $m\delta/2 > \eta > 0$, fix $\eta' = (m\delta/2 + \eta)/2$, so that $\eta < \eta' < m\delta/2$. Inspecting the proof of Proposition A.1 shows that its claims (i) and (ii) go through as before using Propositions A.10, A.11 and A.12 under Assumption 4*, that is, for all T sufficiently large

$$\hat{\lambda}_K \geq \frac{c_K}{2} p^\alpha,$$

holds with probability at least $1 - T^{-\eta'}$.

As Proposition A.12 obtains the same rate as Proposition A.4, adapting Propositions A.5, A.6 and 1 yields the same rate as in the thin-tailed case. There thus exists a constant $C' > 0$ such that, for all T sufficiently large

$$\mathbb{E}(\|\hat{\mathbf{P}}_{T+1} - \mathbf{H}\mathbf{P}_{T+1}\|_2^2 \mid \mathcal{D}_T) \leq C' \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta'}$.

Similarly, inspecting the proof of Proposition 2 and using Proposition A.10 shows that there exists a constant $C'' > 0$ such that, for all T sufficiently large

$$\frac{1}{T} \|\hat{\mathbf{P}} - \mathbf{P}\mathbf{H}'\|_F^2 \leq C'' \left(1 + \sqrt{\frac{p \log(T)}{T}} + \frac{p \log(T)^3}{T^{1-2/m-\delta}} \right) \times \frac{1}{p^{2\alpha-1}} \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta'}$.

Proposition A.13 adapts Proposition A.9 to the heavy-tailed setting. Using it along with Proposition A.11 in the proof of Proposition 3 then shows that there exists a constant

$C''' > 0$ such that, for all T sufficiently large

$$\|\boldsymbol{\vartheta}^* - \tilde{\boldsymbol{\vartheta}}\|_2^2 \leq C''' \frac{K \log(T)}{T},$$

holds with probability at least $1 - T^{-\eta'}$. Finally, Proposition A.14 bounds $\|\hat{\boldsymbol{\vartheta}}\|_2^2$ with a constant with probability at least $1 - T^{-\eta'}$. Taking all of these facts together then implies the claim. $\square$

Proposition A.10. *Suppose Assumptions 1*, 2, 3 and 4* are satisfied. Then for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t' - \boldsymbol{\Sigma}_u \right\|_2 \leq C \left(\sqrt{\frac{p \log(T)}{T}} + \frac{p \log(T)^3}{T^{1-2/m-\delta}} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. When $c_{K+1} = 0$ the claim is trivial as $\mathbf{u}_t = \mathbf{0}$ almost surely, so we may assume otherwise. Note that by part (iii) of Lemma A.4 we have $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{v}' \boldsymbol{\Sigma}_u^{-1/2} \mathbf{u}_t|^m] \leq C_Z$. Moreover, $\mathbf{u}_t$ inherits the mixing properties of $\mathbf{X}_t$ from Assumption 2. We may thus apply Lemma A.7 to $\mathbf{u}_t$. Since Assumption 3 implies $\|\boldsymbol{\Sigma}_u\|_2 = c_{K+1}$, we have that for every $\nu > (1 + \eta)/m$ there exists a constant $C > 0$ such that, for all sufficiently large T

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t' - \boldsymbol{\Sigma}_u \right\|_2 \leq C \left(\sqrt{\frac{p(\log(2p) + \log(T))}{T}} + T^{2\nu} \frac{p \log(T)^2 (\log(2p) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$. Set $\nu = 1/m + \delta/2$. This is an admissible choice since $m\delta/2 > \eta > 0$ implies $\nu = 1/m + \delta/2 > (1 + \eta)/m$. We then have that for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C_1 > 0$ such that, for all T sufficiently large

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t' - \boldsymbol{\Sigma}_u \right\|_2 &\leq C_1 \left(\sqrt{\frac{p(\log(2p) + \log(T))}{T}} + T^{2/m+\delta} \frac{p \log(T)^2 (\log(2p) + \log(T))}{T} \right) \\ &\leq C \left(\sqrt{\frac{p \log(T)}{T}} + \frac{p \log(T)^3}{T^{1-2/m-\delta}} \right), \end{aligned}$$

which follows for some constant $C > 0$, as $\log(2p) + \log(T) \leq C_3 \log(T)$ and $\log(T)^2(\log(2p) + \log(T)) \leq C_4 \log(T)^3$, for some constants $C_3, C_4 > 0$. $\square$

Proposition A.11. *Suppose Assumptions 1*, 2, 3 and 4* are satisfied. Then for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$(i) \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2 \leq C \left(\sqrt{\frac{K \log(T)}{T}} + \frac{K \log(T)^3}{T^{1-2/m-\delta}} \right)$$

$$(ii) \lambda_{\min} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) \geq \frac{1}{2},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. First note that part (ii) of Lemma A.4 implies that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{v}' \mathbf{P}_t|^m] \leq C_Z$ and $\mathbf{P}_t$ inherits the mixing properties of $\mathbf{X}_t$ from Assumption 2. We may then apply Lemma A.7 which implies that for every $\nu > (1 + \eta)/m$ there exists a constant $C > 0$ such that, for all sufficiently large T

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2 \leq C \left(\sqrt{\frac{K(\log(2K) + \log(T))}{T}} + T^{2\nu} \frac{K \log(T)^2(\log(2K) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$. We may then follow the same approach as in the proof of Proposition A.10 to obtain that for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$, there exists a constant $C > 0$ such that, for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2 \leq C \left(\sqrt{\frac{K \log(T)}{T}} + \frac{K \log(T)^3}{T^{1-2/m-\delta}} \right),$$

with probability at least $1 - T^{-\eta}$. This establishes the first claim. By Weyl's inequality

$$\lambda_{\min} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' \right) \geq 1 - \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K \right\|_2.$$

The second claim then follows from the first by Assumption 4*, since for every $m\delta/2 > \eta > 0$ and for all T sufficiently large, $\|(1/T) \sum_{t=1}^T \mathbf{P}_t \mathbf{P}_t' - \mathbf{I}_K\|_2 \leq 1/2$ with probability at

least $1 - T^{-\eta}$. □

Proposition A.12. *Suppose Assumptions 1*, 2, 3 and 4* are satisfied. Then for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right\|_F \leq C \sqrt{\frac{Kp \log(T)}{T}},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. We assume that c_{K+1} is positive, since if c_{K+1} is zero the claim is trivial. Let $V_t = \text{vec}(\mathbf{P}_t \mathbf{u}'_t)$. We begin by showing that the sequence of (Kp) -dimensional random vectors $\{\mathbf{V}_t\}_{t=1}^T$ satisfies the conditions required by Lemma A.9. First, $\mathbb{E}[\mathbf{V}_t] = \mathbf{0}$. Second, we have $\|\mathbf{V}_t\|_2^2 = \|\mathbf{P}_t \mathbf{u}'_t\|_F^2 = \|\mathbf{P}_t\|_2^2 \|\mathbf{u}_t\|_2^2$. Then by Cauchy-Schwarz and parts (ii) and (iv) of Lemma A.4, there exists a constant $C_1 > 0$ such that

$$\mathbb{E}[\|\mathbf{V}_t\|_2^{m/2}] = \mathbb{E}[\|\mathbf{P}_t\|_2^{m/2} \|\mathbf{u}_t\|_2^{m/2}] \leq \mathbb{E}[\|\mathbf{P}_t\|_2^m]^{1/2} \mathbb{E}[\|\mathbf{u}_t\|_2^m]^{1/2} \leq C_1 (Kp)^{m/4}.$$

Moreover, $\{\mathbf{V}_t\}_{t=1}^T$ inherits the mixing properties of the sequence $\{(Y_t, \mathbf{X}_t)'\}_{t=1}^T$ from Assumption 2.

We may then apply Lemma A.9 with its moment parameter equal to $m/2$ to the sequence $\{\mathbf{V}_t\}_{t=1}^T$. The lemma implies that for every $\eta > 0$ and every $\nu > 2(1 + \eta)/m$ there exist constants $C_1, C_2 > 0$ such that, for all sufficiently large T

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_t \right\|_2 &\leq C_1 \sqrt{Kp} \left(\sqrt{\frac{\log(2Kp) + \log(T)}{T}} + T^\nu \frac{\log(T)^2 (\log(2Kp) + \log(T))}{T} \right) \\ &\leq C_2 \left(\sqrt{\frac{Kp \log(T)}{T}} + \frac{\sqrt{Kp} \log(T)^3}{T^{1-\nu}} \right), \end{aligned}$$

with probability at least $1 - T^{-\eta}$. Notice that the former term dominates if $\nu < 1/2$. Take $\nu = 2/m + \delta$ and notice that by Assumption 4*

$$\delta < \kappa/m < \frac{m-4}{2m} = 1/2 - 2/m.$$

This choice of ν thus guarantees that the former term dominates. Then for every $\kappa/m >$

$\delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t \mathbf{u}'_t \right\|_F = \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{V}_t \right\|_2 \leq C \sqrt{\frac{Kp \log(T)}{T}}.$$

□

Proposition A.13. *Suppose Assumptions 1*, 2, 3 and 4* are satisfied. Then for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{P}_t (Y_t - \mathbf{P}'_t \boldsymbol{\vartheta}^*) \right\|_2 \leq C \sqrt{\frac{K \log(T)}{T}},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. As in the proof of Proposition A.9, define $\mathbf{W}_t = (Y_t - \mathbf{P}'_t \boldsymbol{\vartheta}^*) \mathbf{P}_t$. It is easy to verify that $\mathbf{W}_t$ satisfies the conditions of Lemma A.9, by Assumptions 1* and 2 along with part (v) of Lemma A.4. Taking $\nu = 2/m + \delta$, the lemma implies that for every $\kappa/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C_1 > 0$ such that, for all T sufficiently large,

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \right\|_2 \leq C_1 \left(\sqrt{\frac{K \log(T)}{T}} + \frac{\sqrt{K} \log(T)^3}{T^{1-2/m-\delta}} \right),$$

with probability at least $1 - T^{-\eta}$. The former term dominates because $\delta < 1/2 - 2/m$ by Assumption 4*, which guarantees $1/2 < 1 - 2/m - \delta$. It thus follows that there exists a constant $C > 0$ such that

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \right\|_2 \leq C \sqrt{\frac{K \log T}{T}},$$

with probability at least $1 - T^{-\eta}$. □

Proposition A.14. *Suppose Assumptions 1* and 2 are satisfied. Then for every $1 - 2/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exists a constant $C > 0$ such that, for all T sufficiently large*

$$\|\hat{\boldsymbol{\vartheta}}\|_2^2 \leq \frac{1}{T} \|\mathbf{Y}\|_2^2 \leq C,$$

holds with probability at least $1 - T^{-\eta}$.

Proof. As in Proposition A.7, we have

$$\|\hat{\boldsymbol{\vartheta}}\|_2^2 \leq \frac{1}{T} \|\mathbf{Y}\|_2^2 = \mathbb{E}[Y_t^2] + \frac{1}{T} \sum_{t=1}^T (Y_t^2 - \mathbb{E}[Y_t^2]).$$

We thus wish to apply Lemma A.9 to $U_t = Y_t^2 - \mathbb{E}[Y_t^2]$. By the elementary inequality $|a - b|^q \leq 2^{q-1}(|a|^q + |b|^q)$ for two real numbers a, b and $q \geq 1$, Assumption 1* and Jensen's inequality imply that

$$\mathbb{E}[|U_t|^{m/2}] \leq 2^{m/2-1}(\mathbb{E}[|Y_t|^m] + |\mathbb{E}[Y_t^2]|^{m/2}) \leq 2^{m/2-1}(\mathbb{E}[|Y_t|^m] + \mathbb{E}[|Y_t^m|]) \leq C_1.$$

for some constant $C_1 > 0$, where we took $q = m/2 > 2$. It is easy to verify that $U_t = Y_t^2 - \mathbb{E}[Y_t^2]$ satisfies the remaining conditions of Lemma A.9. The lemma then implies that for every $1 - 2/m > \delta > 0$ and every $m\delta/2 > \eta > 0$ there exist constants $C_2, C_3 > 0$ such that, for all T sufficiently large,

$$\left| \frac{1}{T} \sum_{t=1}^T (Y_t^2 - \mathbb{E}[Y_t^2]) \right| \leq C_2 \left(\sqrt{\frac{\log T}{T}} + \frac{\log(T)^3}{T^{1-2/m-\delta}} \right) \leq C_3,$$

with probability at least $1 - T^{-\eta}$, where we took $\nu = 2/m + \delta$. The last inequality follows for sufficiently large T because $1 - 2/m - \delta > 0$. As $\mathbb{E}[Y_t^2] \leq (\mathbb{E}[|Y_t|^m])^{2/m} \leq C_Z^{2/m}$ by Lyapunov's inequality, it follows that for some constant $C > 0$ and all T sufficiently large,

$$\frac{1}{T} \|\mathbf{Y}\|_2^2 = \frac{1}{T} \sum_{t=1}^T Y_t^2 \leq C,$$

with probability at least $1 - T^{-\eta}$. □

A.4 Auxiliary Results

A.4.1 Thin-Tailed

Lemma A.1. *Suppose Assumptions 1 and 3 are satisfied.*

Then we have that (i) there exists a constant $C_1 > 0$ such that $\mathbf{P}_t$ is sub-Gaussian with

parameter C_1 ; Define $\mathbf{W}_t = \mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{V}_R' \mathbf{u}_t$. Then (ii) if $c_{K+1} > 0$, there exists a constant $C_2 > 0$ such that $\mathbf{W}_t$ is sub-Gaussian with parameter C_2 , otherwise $\mathbf{W}_t$ is degenerate at zero; (iii) if $c_{K+1} > 0$ then there exists a constant $C_3 > 0$ such that $\mathbf{u}_t$ is sub-Gaussian with parameter C_3 , otherwise $\mathbf{u}_t$ is degenerate at zero; (iv) there exists a constant $C_4 > 0$ such that $Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*$ is sub-Gaussian with parameter C_4 .

Proof. (i) Despite the fact that $\boldsymbol{\Sigma}$ may be singular, it still holds that $\mathbf{X}_t = \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{-1/2} \mathbf{X}_t = \boldsymbol{\Sigma}^{1/2} \mathbf{Z}_t$ almost surely since $\mathbf{X}_t \in \text{Range}(\boldsymbol{\Sigma})$ almost surely. Since $\mathbf{P}_t = \mathbf{\Lambda}_K^{-1/2} \mathbf{V}_K' \mathbf{X}_t = \mathbf{V}_K' \mathbf{Z}_t$ almost surely, it follows that for any $\varepsilon > 0$

$$\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{P}_t| > \varepsilon) \leq \sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{Z}_t| > \varepsilon) \leq C_0 \exp(-C_m \varepsilon^2),$$

so $\mathbf{P}_t$ is sub-Gaussian with parameter $C_1 = C_m$.

(ii) As $\mathbf{W}_t = \mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{V}_R' \mathbf{u}_t = \mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{\Lambda}_R^{1/2} \mathbf{V}_R' \mathbf{Z}_t$ almost surely, we have that for any $\varepsilon > 0$

$$\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{V}_R' \mathbf{u}_t| > \varepsilon) \leq \sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{Z}_t| > \varepsilon) \leq C_0 \exp(-C_m \varepsilon^2),$$

because $\mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{\Lambda}_R^{1/2} \mathbf{V}_R'$ is an orthogonal projector. $\mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{V}_R' \mathbf{u}_t$ is thus sub-Gaussian with parameter $C_2 = C_m$.

(iii) The conclusion is obvious if $c_{K+1} = 0$, so assume that $c_{K+1} > 0$. Since $\mathbf{u}_t = \mathbf{V}_R \mathbf{V}_R' \mathbf{X}_t = \mathbf{V}_R \mathbf{\Lambda}_R^{1/2} \mathbf{V}_R' \mathbf{Z}_t$ almost surely, we have that for any $\varepsilon > 0$

$$\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \mathbf{u}_t| > \varepsilon) \leq \sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(c_{K+1}^{1/2} |\mathbf{v}' \mathbf{Z}_t| > \varepsilon) \leq C_0 \exp\left(-\frac{C_m}{c_{K+1}} \varepsilon^2\right),$$

where we have used the fact that $\|\mathbf{V}_R \mathbf{\Lambda}_R^{1/2} \mathbf{V}_R'\|_2 \leq \|\mathbf{V}_R\|_2 \|\mathbf{\Lambda}_R^{1/2}\|_2 \|\mathbf{V}_R'\|_2 = c_{K+1}^{1/2}$. As a consequence, $\mathbf{u}_t$ is sub-Gaussian with parameter $C_3 = C_m / c_{K+1}$.

(iv) We first show that $\mathbf{P}_t' \boldsymbol{\vartheta}^*$ is sub-Gaussian. If $\boldsymbol{\vartheta}^* = \mathbf{0}$, the conclusion trivially follows, so we assume otherwise. Define $\mathbf{a} = \boldsymbol{\vartheta}^* / \|\boldsymbol{\vartheta}^*\|_2$ so that $\mathbf{P}_t' \boldsymbol{\vartheta}^* = \|\boldsymbol{\vartheta}^*\|_2 \mathbf{a}' \mathbf{P}_t$. As $\mathbf{P}_t$ is

sub-Gaussian, and because Assumption 1 implies there exists a constant $C' > 0$ such that

$$\begin{aligned}\mathbb{E}[Y_t^2] &= \int_0^\infty \mathbb{P}(Y_t^2 > s) ds = \int_0^\infty \mathbb{P}(|Y_t| > \sqrt{s}) ds \\ &= 2 \int_0^\infty \varepsilon \mathbb{P}(|Y_t| > \varepsilon) d\varepsilon \leq 2C_0 \int_0^\infty \varepsilon \exp(-C_m \varepsilon^2) d\varepsilon \leq (C')^2,\end{aligned}$$

so that $\|\boldsymbol{\vartheta}^*\|_2 \leq C'$ by Lemma 1, we have

$$\mathbb{P}(|\mathbf{P}_t' \boldsymbol{\vartheta}^*| > \varepsilon) = \mathbb{P}\left(|\mathbf{a}' \mathbf{P}_t| > \frac{\varepsilon}{\|\boldsymbol{\vartheta}^*\|_2}\right) \leq C_0 \exp(-C_m \varepsilon^2 / (C')^2),$$

so $\mathbf{P}_t' \boldsymbol{\vartheta}^*$ is sub-Gaussian with parameter $C'' = C_m / (C')^2$. Finally, we have

$$\begin{aligned}\mathbb{P}(|Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*| > \varepsilon) &\leq \mathbb{P}(|Y_t| > \varepsilon/2) + \mathbb{P}(|\mathbf{P}_t' \boldsymbol{\vartheta}^*| > \varepsilon/2) \\ &\leq C_0 \exp(-C_m \varepsilon^2 / 4) + C_0 \exp(-C_m \varepsilon^2 / (4(C')^2)) \\ &\leq 2C_0 \exp(-C_4 \varepsilon^2),\end{aligned}$$

for some constant $C_4 > 0$, which establishes the final claim. $\square$

Lemma A.2. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional random vectors and $\boldsymbol{\Sigma}_z = \mathbb{E}[\mathbf{Z}_t \mathbf{Z}_t']$. Suppose (i) for any $\varepsilon > 0$, $\sup_{\mathbf{v}: \|\mathbf{v}\|_2=1} \mathbb{P}(|\mathbf{v}' \boldsymbol{\Sigma}_z^{-1/2} \mathbf{Z}_t| > \varepsilon) \leq 2 \exp(-C_m \varepsilon^2)$ for some $C_m > 0$; (ii) the α -mixing coefficients of the sequence satisfy $\alpha(l) < \exp(-C_\alpha l^{r_\alpha})$ for some $C_\alpha > 0$ and $r_\alpha > 0$; and (iii) $d = \lfloor C_d T^{r_d} \rfloor$ for some $C_d > 0$ and $r_d \in [0, r_\alpha]$.*

Then for every $\eta > 0$ there exists a positive constant $C_\eta > 0$ such that, for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \mathbf{Z}_t' - \boldsymbol{\Sigma}_z \right\|_2 \leq C_\eta \|\boldsymbol{\Sigma}_z\|_2 \left(\frac{(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}}{T} + \sqrt{\frac{(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}}{T}} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. If $\boldsymbol{\Sigma}_z = \mathbf{0}$, $\mathbf{Z}_t = \mathbf{0}$ almost surely and the result follows trivially, so we assume otherwise. Let $\boldsymbol{\Sigma}_z = \mathbf{V}_z \boldsymbol{\Lambda}_z \mathbf{V}_z'$ be the eigendecomposition of $\boldsymbol{\Sigma}_z$. As $\boldsymbol{\Sigma}_z$ may be singular, we have the orthogonal projector $\boldsymbol{\Pi}_z \equiv \boldsymbol{\Sigma}_z^{-1/2} \boldsymbol{\Sigma}_z \boldsymbol{\Sigma}_z^{-1/2} = \mathbf{V}_z \boldsymbol{\Lambda}_z^{-1/2} \boldsymbol{\Lambda}_z \boldsymbol{\Lambda}_z^{-1/2} \mathbf{V}_z'$, with the

property that $\mathbf{\Pi}_z \Sigma_z^{1/2} = \Sigma_z^{1/2}$. Note also that since $\mathbf{Z}_t \in \text{Range}(\Sigma_z)$ almost surely, we have $\mathbf{Z}_t = \Sigma_z^{1/2} \Sigma_z^{-1/2} \mathbf{Z}_t$ almost surely. Then almost surely

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \mathbf{Z}_t' - \Sigma_z \right\|_2 &= \left\| \Sigma_z^{1/2} \left(\frac{1}{T} \sum_{t=1}^T \Sigma_z^{-1/2} \mathbf{Z}_t \mathbf{Z}_t' \Sigma_z^{-1/2} - \mathbf{\Pi}_z \right) \Sigma_z^{1/2} \right\|_2 \\ &\leq \|\Sigma_z\|_2 \left\| \frac{1}{T} \sum_{t=1}^T \Sigma_z^{-1/2} \mathbf{Z}_t \mathbf{Z}_t' \Sigma_z^{-1/2} - \mathbf{\Pi}_z \right\|_2. \end{aligned} \quad (\text{A.24})$$

It is therefore sufficient for us to bound the right hand side. Lemma 5.4 of Vershynin (2012) implies that if $\mathcal{N}$ is a $\frac{1}{4}$ -net of $\mathcal{S}^{d-1} \cap \text{Range}(\Sigma_z)$ then

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \Sigma_z^{-1/2} \mathbf{Z}_t \mathbf{Z}_t' \Sigma_z^{-1/2} - \mathbf{\Pi}_z \right\|_2 &= \max_{\mathbf{x} \in \mathcal{S}^{d-1} \cap \text{Range}(\Sigma_z)} \left| \frac{1}{T} \sum_{t=1}^T (\mathbf{Z}_t' \Sigma_z^{-1/2} \mathbf{x})^2 - 1 \right| \\ &\leq 2 \max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{T} \sum_{t=1}^T (\mathbf{Z}_t' \Sigma_z^{-1/2} \mathbf{x})^2 - 1 \right| \\ &\equiv 2 \max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{T} \sum_{t=1}^T W_{\mathbf{x},t} \right|, \end{aligned} \quad (\text{A.25})$$

where we defined $W_{\mathbf{x},t} = (\mathbf{Z}_t' \Sigma_z^{-1/2} \mathbf{x})^2 - 1$. To establish the claim we first show that for all T sufficiently large (A.25) can be bounded with high probability.

We begin by establishing a bound for a fixed vector $\mathbf{x} \in \mathcal{N}$. Note the following three properties of the sequence $\{W_{\mathbf{x},t}\}$. First, $\mathbb{E}(W_{\mathbf{x},t}) = 0$ since

$$\mathbb{E}[(\mathbf{Z}_t' \Sigma_z^{-1/2} \mathbf{x})^2] = \mathbf{x}' \mathbf{\Pi}_z \mathbf{x} = \|\mathbf{x}\|_2^2 = 1.$$

Second, $W_{\mathbf{x},t}$ is the de-meaned square of a sub-Gaussian random variable with parameter C_m (which is independent of $\mathbf{x}$) by (i). Third, the sequence $\{W_{\mathbf{x},t}\}_{t=1}^T$ inherits the mixing properties of $\{\mathbf{Z}_t\}_{t=1}^T$ from (ii). We may then apply Theorem 1.4 of Bosq (1998).⁴ Define

$$\varepsilon_T = \frac{C_1^*(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}}{T} + \left[\frac{C_1^*(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}}{T} \right]^{1/2}$$

⁴The C_r inequality and Boucheron, Lugosi, and Massart (2013, Theorem 2.1) imply that the condition in equation (1.33) of Bosq (1998) is satisfied.

and

$$q_T = \left\lceil \frac{T}{C_2^*(d + \log(T))^{\frac{1}{r_\alpha}}} \right\rceil - 1,$$

for positive constants C_1^* and C_2^* to be chosen below. Provided that $C_2^* \geq 2$ and T sufficiently large, we have $q_T \in [1, T/2]$ by (iii), as required by the theorem. Theorem 1.4 of Bosq (1998) then implies that for any $r \geq 3$, there exists a positive constant C_1 that depends on C_m and r such that, for all T sufficiently large

$$\mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T W_{\mathbf{x},t} \right| > \varepsilon_T \right) \leq a_{1T} \exp \left(-\frac{q_T \varepsilon_T^2}{C_1 + C_1 \varepsilon_T} \right) + a_{2T} \alpha \left(\left\lfloor \frac{T}{q_T + 1} \right\rfloor \right)^{\frac{2r}{2r+1}}, \quad (\text{A.26})$$

where $a_{1T} = 2T/q_T + 2[1 + \varepsilon_T^2/(C_1 + C_1 \varepsilon_T)]$ and $a_{2T} = 11T(1 + C_1/\varepsilon_T)$. We proceed by bounding the right hand side of (A.26).

For the first term, we have for all T sufficiently large

$$\begin{aligned} a_{1T} \exp \left(-\frac{q_T \varepsilon_T^2}{C_1 + C_1 \varepsilon_T} \right) &\leq \left(2T + 2 + 2\frac{\varepsilon_T}{C_1} \right) \exp \left(-\frac{\min(\varepsilon_T, \varepsilon_T^2)}{2C_1} q_T \right) \\ &\leq \exp \left(\log \left(3T + 2\frac{\varepsilon_T}{C_1} \right) - \frac{C_1^*(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}}{2TC_1} q_T \right) \\ &\leq \exp \left(\log \left(3T + 2\frac{\varepsilon_T}{C_1} \right) - \frac{C_1^*}{4C_1 C_2^*} (d + \log(T)) \right) \\ &\leq \exp \left(-\left(\frac{C_1^*}{4C_1 C_2^*} - 1 \right) (d + \log(T)) \right), \end{aligned} \quad (\text{A.27})$$

where the first inequality uses the fact that $x^2/(1+x) \geq \min(x, x^2)/2$ for $x > 0$, the second uses $\min(\varepsilon_T, \varepsilon_T^2) \geq C_1^*(d + \log(T))^{\frac{r_\alpha+1}{r_\alpha}}/T$ and the third uses the fact that $q_T \geq T/(2C_2^*(d + \log(T)))^{1/r_\alpha}$ by (iii) for sufficiently large T . For the second term in (A.26), we have for all T sufficiently large

$$\begin{aligned} a_{2T} \alpha \left(\left\lfloor \frac{T}{q_T + 1} \right\rfloor \right)^{\frac{2r}{2r+1}} &\leq \exp \left(2\log(T) - C_\alpha \frac{2r}{2r+1} \left(\frac{T}{q_T + 1} - 1 \right)^{r_\alpha} \right) \\ &\leq \exp \left(2\log(T) - C_\alpha (C_2^*/4)^{r_\alpha} \frac{2r}{2r+1} (d + \log(T)) \right) \\ &\leq \exp \left(-\left(C_\alpha (C_2^*/4)^{r_\alpha} \frac{2r}{2r+1} - 2 \right) (d + \log(T)) \right), \end{aligned} \quad (\text{A.28})$$

where the first inequality follows from (ii) and the fact that $\varepsilon_T \geq 1/T$ for sufficiently large

T and the second follows as $(T/(q_T + 1) - 1)^{r_\alpha} \geq (C_2^*/4)^{r_\alpha}(d + \log(T))$ for all sufficiently large T .

Combining (A.27) and (A.28) into (A.26) then yields for a given $\mathbf{x}$, for all T sufficiently large

$$\begin{aligned} \mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T W_{\mathbf{x},t} \right| > \varepsilon_T \right) &\leq \exp \left(- \left(\frac{C_1^*}{4C_1C_2^*} - 1 \right) (d + \log T) \right) \\ &\quad + \exp \left(- \left(C_\alpha (C_2^*/4)^{r_\alpha} \frac{2r}{2r+1} - 2 \right) (d + \log(T)) \right). \end{aligned} \quad (\text{A.29})$$

Finally, we take the union bound over all $\mathbf{x} \in \mathcal{N}$. The cardinality of a $\frac{1}{4}$ -net $\mathcal{N}$ of the unit sphere $\mathcal{S}^{d-1}$ is bounded by 9^d by Lemma 5.2 of Vershynin (2012). Setting

$$C_2^* = 4 \left(\frac{(\eta' + \log(9) + 3)(2r + 1)}{2rC_\alpha} \right)^{\frac{1}{r_\alpha}},$$

where $\eta' = \eta + \delta$ with $\delta > 0$ and $C_1^* = 4C_1C_2^*(\eta' + \log(9) + 2)$, in (A.29) we then obtain that, for all T sufficiently large

$$\begin{aligned} \mathbb{P} \left(\max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{T} \sum_{t=1}^T W_{\mathbf{x},t} \right| > \varepsilon_T \right) &\leq 9^d \max_{\mathbf{x} \in \mathcal{N}} \mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T W_{\mathbf{x},t} \right| > \varepsilon_T \right) \\ &\leq 2 \exp(\log(9)d - (\eta' + \log(9))(d + \log(T))) \\ &\leq 2 \exp(-\eta' \log(T)) \leq T^{-\eta}. \end{aligned}$$

Combining this with (A.25) and (A.24) then yields the result. $\square$

Lemma A.3. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional zero-mean random vectors. Suppose (i) for any $\varepsilon > 0$ and some $C_0, C_m > 0$, $\sup_{1 \leq i \leq d} \mathbb{P}(|Z_{it}| > \varepsilon) \leq C_0 \exp(-C_m \varepsilon)$; (ii) the α -mixing coefficients of the sequence satisfy $\alpha(l) < \exp(-C_\alpha l^{r_\alpha})$ for some $C_\alpha > 0$ and $r_\alpha > 0$; and (iii) $d = \lfloor C_d T^{r_d} \rfloor$ for some $C_d > 0$ and $r_d \in [0, 1]$.*

Then for every $\eta > 0$ there exists a positive constant C_η such that, for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \right\|_2 \leq C_\eta \sqrt{\frac{d \log(T)}{T}},$$

holds with probability at least $1 - T^{-\eta}$.

Proof. Fix $\eta' = \eta + \delta$ with $\delta > 0$. Let C_* denote a positive constant to be chosen below.

By the union bound

$$\mathbb{P} \left(\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \right\|_2 \geq C^* \sqrt{\frac{d \log(T)}{T}} \right) \leq d \max_{1 \leq i \leq d} \mathbb{P} \left(\left| \sum_{t=1}^T Z_{it} \right| \geq C^* \sqrt{T \log(T)} \right).$$

Define the truncation threshold

$$b_T = 2 \left(r_d + \frac{1}{2} + \eta' \right) \frac{1}{C_m} \log(T),$$

and decompose $\sum_{t=1}^T Z_{it} = \sum_{t=1}^T Z'_{it} + \sum_{t=1}^T Z''_{it}$ where

$$Z'_{it} = Z_{it} \mathbb{1}_{\{|Z_{it}| \leq b_T\}} - \mathbb{E}(Z_{it} \mathbb{1}_{\{|Z_{it}| \leq b_T\}}),$$

$$Z''_{it} = Z_{it} \mathbb{1}_{\{|Z_{it}| > b_T\}} - \mathbb{E}(Z_{it} \mathbb{1}_{\{|Z_{it}| > b_T\}}).$$

If we define $\varepsilon_T = (C_*/2) \sqrt{T \log(T)}$, we have

$$\mathbb{P} \left(\left| \sum_{t=1}^T Z_{it} \right| > C_* \sqrt{T \log(T)} \right) \leq \mathbb{P} \left(\left| \sum_{t=1}^T Z'_{it} \right| > \varepsilon_T \right) + \mathbb{P} \left(\left| \sum_{t=1}^T Z''_{it} \right| > \varepsilon_T \right).$$

First consider the bounded part. The sequence $\{Z'_{it}\}_{t=1}^T$ has the same mixing properties as $\{\mathbf{Z}_t\}_{t=1}^T$ and $\sup_{1 \leq i \leq d} \|Z'_{it}\|_\infty < 2b_T$. Define $M_T = \lfloor b_T^{-1} \sqrt{T / \log(T)} \rfloor$. Then, for all T sufficiently large we have $M_T \in [1, T]$ and

$$4(2b_T)M_T = 8 \sqrt{\frac{T}{\log(T)}} < \varepsilon_T,$$

so the conditions of Theorem 2.1 of Liebscher (1996) are satisfied. The theorem gives us

$$\mathbb{P} \left(\left| \sum_{t=1}^T Z'_{it} \right| > \varepsilon_T \right) < 4 \exp \left(- \frac{\varepsilon_T^2}{64(T/M_T)D_T + (16/3)b_T M_T \varepsilon_T} \right) + 4 \frac{T}{M_T} \exp(-C_\alpha M_T^{\alpha}), \quad (\text{A.30})$$

with $D_T = \sup_{1 \leq i \leq d} \mathbb{E}[(\sum_{t=1}^{M_T} Z'_{it})^2]$. To bound D_T , define $\gamma(l) = \sup_{1 \leq i \leq d} |\text{Cov}(Z'_{it}, Z'_{it+l})|$

for $l = 0, 1, \dots$ and note that $D_T \leq M_T \sum_{l=-M_T+1}^{M_T-1} \gamma(l)$ by stationarity. For $l = 1, 2, \dots$ we obtain

$$\gamma(l) \leq 12\alpha(l)^{\frac{1}{2}} \sup_{1 \leq i \leq d} \|Z'_{it}\|_{L_4}^2 \leq C_1 \alpha(l)^{\frac{1}{2}} \sup_{1 \leq i \leq d} \|Z_{it}\|_{L_4}^2,$$

for some constant $C_1 > 0$, where the first inequality is Davydov's inequality (Bosq, 1998, Corollary 1.1) and the second follows from the fact that $\|Z'_{it}\|_{L_4} \leq 2\|Z_{it}\mathbb{1}_{\{|Z_{it}| \leq b_T\}}\|_{L_4} \leq 2\|Z_{it}\|_{L_4}$. As $\sup_{1 \leq i \leq d} \|Z_{it}\|_{L_4}$ is also bounded, Assumption (ii) then implies that $D < C_2 M_T$ where $C_2 > 0$ is a constant. The definition of ε_T and the fact that $b_T M_T \leq \sqrt{T/\log(T)}$ together imply $(16/3)b_T M_T \varepsilon_T \leq (8/3)C_* T$. As a consequence, (A.30) becomes

$$\mathbb{P}\left(\left|\sum_{t=1}^T Z'_{it}\right| > \varepsilon_T\right) \leq 4 \exp\left(-\frac{C_*^2 \log(T)}{256C_2 + (32/3)C_*}\right) + 4T \exp(-C_\alpha M_T^{r_\alpha}),$$

and using Assumption (iii), we finally obtain

$$\begin{aligned} d \max_{1 \leq i \leq d} \mathbb{P}\left(\left|\sum_{t=1}^T Z'_{it}\right| > \varepsilon_T\right) &\leq 4C_d \left(\exp\left(r_d \log(T) - \frac{C_*^2 \log(T)}{256C_2 + (32/3)C_*}\right) + T^{1+r_d} \exp(-C_\alpha M_T^{r_\alpha})\right) \\ &\leq 4C_d (\exp(-\eta' \log(T)) + \exp(2 \log(T) - C_\alpha M_T^{r_\alpha})) \\ &\leq 5C_d T^{-\eta'} \leq T^{-\eta}, \end{aligned} \tag{A.31}$$

where the second inequality follows from choosing C_* large enough to ensure $C_*^2/(256C_2 + (32/3)C_*) \geq r_d + \eta'$.

Next we bound the tail component sequence $\{Z''_{it}\}_{t=1}^T$. By Markov's inequality and the triangle inequality

$$S \equiv d \max_{1 \leq i \leq d} \mathbb{P}\left(\left|\sum_{t=1}^T Z''_{it}\right| > \varepsilon_T\right) \leq \frac{d}{\varepsilon_T} \max_{1 \leq i \leq d} \mathbb{E}\left|\sum_{t=1}^T Z''_{it}\right| \leq \frac{d}{\varepsilon_T} \sum_{t=1}^T \max_{1 \leq i \leq d} \mathbb{E}|Z''_{it}|.$$

Using the definition of Z''_{it} and Cauchy-Schwarz we continue with

$$S \leq 2 \frac{dT}{\varepsilon_T} \max_{1 \leq i \leq d} \mathbb{E}(|Z_{it}\mathbb{1}_{\{|Z_{it}| > b_T\}}|) \leq 2 \frac{dT}{\varepsilon_T} \max_{1 \leq i \leq d} \|Z_{it}\|_{L_2} \|\mathbb{1}_{\{|Z_{it}| > b_T\}}\|_{L_2}.$$

If we define $\sigma^2 = \sup_{1 \leq i \leq d} \|Z_{it}\|_{L_2}$, Assumption (i) implies

$$S \leq 2 \frac{dT}{\varepsilon_T} \max_{1 \leq i \leq d} \|Z_{it}\|_{L_2} \mathbb{P}(|Z_{it}| > b_T)^{1/2} < 2 \frac{dT}{\varepsilon_T} \sigma^2 \exp\left(-\frac{C_m}{2} b_T\right).$$

Finally, using Assumption (iii) and the definitions of ε_T and b_T

$$S \leq \frac{4C_d \sigma^2}{C_* \sqrt{\log(T)}} \exp((r_d + (1/2)) \log(T) - (r_d + (1/2) + \eta') \log(T)) \leq T^{-\eta}. \quad (\text{A.32})$$

Equations (A.31) and (A.32) together then imply the claim. $\square$

A.4.2 Heavy-tailed

Lemma A.4. *Suppose Assumptions 1* and 3 are satisfied. Then we have that (i) $\mathbb{E}(\|\mathbf{Z}_t\|_2^m) \leq C_Z p^{m/2}$; (ii) $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[\|\mathbf{v}' \mathbf{P}_t\|^m] \leq C_Z$ and $\mathbb{E}[\|\mathbf{P}_t\|_2^m] \leq C_Z K^{m/2}$; Define $\mathbf{W}_t = \mathbf{V}_R \mathbf{\Lambda}_R^{-1/2} \mathbf{V}_R \mathbf{u}_t$. Then (iii) if $c_{K+1} > 0$, we have $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[\|\mathbf{v}' \mathbf{W}_t\|^m] \leq C_Z$ and $\mathbb{E}[\|\mathbf{W}_t\|_2^m] \leq C_Z p^{m/2}$, otherwise $\mathbf{W}_t$ is degenerate at zero; (iv) if $c_{K+1} > 0$ there exists a constant $C'_Z > 0$ such that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[\|\mathbf{v}' \mathbf{u}_t\|^m] \leq C'_Z$ and $\mathbb{E}[\|\mathbf{u}_t\|_2^m] \leq C'_Z p^{m/2}$, otherwise $\mathbf{u}_t$ is degenerate at zero; (v) there exists a constant $C''_Z > 0$ such that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[\|\mathbf{v}'(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t\|^{m/2}] \leq C''_Z$ and $\mathbb{E}[\|(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t\|_2^{m/2}] \leq C''_Z K^{m/4}$.*

Proof. (i) By the standard inequality $\|\mathbf{Z}_t\|_2 \leq p^{1/2-1/m} \|\mathbf{Z}_t\|_m$ we obtain

$$\|\mathbf{Z}_t\|_2^m \leq p^{m/2-1} \|\mathbf{Z}_t\|_m^m = p^{m/2-1} \sum_{i=1}^p |Z_{it}|^m.$$

Taking expectations and using $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[\|\mathbf{v}' \mathbf{Z}_t\|^m] \leq C_Z$ from Assumption 1* then implies

$$\mathbb{E}[\|\mathbf{Z}_t\|_2^m] \leq p^{m/2-1} \sum_{i=1}^p \mathbb{E}[|Z_{it}|^m] \leq C_Z p^{m/2}.$$

(ii) As $\mathbf{X}_t \in \text{Range}(\boldsymbol{\Sigma})$ almost surely, we have $\mathbf{X}_t = \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{-1/2} \mathbf{X}_t = \boldsymbol{\Sigma}^{1/2} \mathbf{Z}_t$ almost surely. We therefore have $\mathbf{P}_t = \boldsymbol{\Lambda}_K^{-1/2} \mathbf{V}_K' \boldsymbol{\Sigma}^{1/2} \mathbf{Z}_t \equiv \mathbf{Q}_P \mathbf{Z}_t$ almost surely, where

$$\mathbf{Q}_P \mathbf{Q}_P' = \boldsymbol{\Lambda}_K^{-1/2} \mathbf{V}_K' \boldsymbol{\Sigma} \mathbf{V}_K \boldsymbol{\Lambda}_K^{-1/2} = \mathbf{I}_K,$$

and hence $\|\mathbf{Q}_P\|_2 = 1$. For any unit vector $\mathbf{x}$ we have $|\mathbf{x}'\mathbf{P}_t| = |(\mathbf{Q}'_P\mathbf{x})'\mathbf{Z}_t|$, so Assumption 1* implies $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{v}'\mathbf{P}_t|^m] \leq C_Z$. By the same arguments as in (i), we also immediately obtain $\mathbb{E}[\|\mathbf{P}_t\|_2^m] \leq C_Z K^{m/2}$.

(iii) Let $\Sigma_u = \mathbb{E}[\mathbf{u}_t\mathbf{u}_t']$. Similar to before, we have $\mathbf{W}_t = \Sigma_u^{-1/2}\mathbf{u}_t = \Sigma_u^{-1/2}\mathbf{V}_R\mathbf{V}_R'\Sigma_u^{1/2}\mathbf{Z}_t \equiv \mathbf{Q}_W\mathbf{Z}_t$ almost surely, where

$$\mathbf{Q}_W\mathbf{Q}_W' = \Sigma_u^{-1/2}\mathbf{V}_R\mathbf{V}_R'\Sigma\mathbf{V}_R\mathbf{V}_R'\Sigma_u^{-1/2} = \Pi_u,$$

where Π_u is the orthogonal projector onto the range of Σ_u , and hence $\|\mathbf{Q}_W\|_2 \leq 1$. For any unit vector $\mathbf{x}$ we have $|\mathbf{x}'\mathbf{W}_t| = |(\mathbf{Q}'_W\mathbf{x})'\mathbf{Z}_t|$, where $\|\mathbf{Q}'_W\mathbf{x}\|_2 \leq 1$. We may take $\mathbf{Q}'_W\mathbf{x} = \|\mathbf{Q}'_W\mathbf{x}\|_2\mathbf{v}$ with $\|\mathbf{v}\|_2 = 1$, obtaining $\mathbb{E}[|\mathbf{x}'\mathbf{W}_t|^m] = \|\mathbf{Q}'_W\mathbf{x}\|_2^m \mathbb{E}[|\mathbf{v}'\mathbf{Z}_t|^m]$. As a consequence, Assumption 1* implies $\sup_{\|\mathbf{x}\|_2=1} \mathbb{E}[|\mathbf{x}'\mathbf{W}_t|^m] \leq C_Z$ and $\mathbb{E}[\|\mathbf{W}_t\|_2^m] \leq C_Z p^{m/2}$.

(iv) The conclusion is obvious if $c_{K+1} = 0$, so assume that $c_{K+1} > 0$. As $\mathbf{u}_t = \Sigma_u^{1/2}\mathbf{W}_t$ almost surely, we have for any unit vector $\mathbf{x}$ that $|\mathbf{x}'\mathbf{u}_t| = |(\Sigma_u^{1/2}\mathbf{x})'\mathbf{W}_t|$. Since Assumption 3 implies $\|\Sigma_u^{1/2}\mathbf{x}\|_2 \leq \|\Sigma_u\|_2^{1/2} \leq \sqrt{c_{K+1}}$, we have by Assumption 1* that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{v}'\mathbf{u}_t|^m] \leq C_Z c_{K+1}^{m/2}$ and $\mathbb{E}[\|\mathbf{u}_t\|_2^m] \leq C_Z c_{K+1}^{m/2} p^{m/2}$.

(v) We first consider $\mathbf{P}_t'\boldsymbol{\vartheta}^*$. If $\boldsymbol{\vartheta}^* = \mathbf{0}$, the conclusion trivially follows, so we assume otherwise. We can take $\mathbf{v} = \boldsymbol{\vartheta}^*/\|\boldsymbol{\vartheta}^*\|_2$ and write $\mathbb{E}[|\mathbf{P}_t'\boldsymbol{\vartheta}^*|^m] = \|\boldsymbol{\vartheta}^*\|_2^m \mathbb{E}[|\mathbf{v}'\mathbf{P}_t|^m]$. As Lemma 1 and Assumption 1* imply that $\|\boldsymbol{\vartheta}^*\|_2 \leq \sqrt{C_Z}$, we obtain from part (i) that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{P}_t'\boldsymbol{\vartheta}^*|^m] \leq C_Z^{1+m/2}$. Next use the elementary convexity inequality $|a - b|^q \leq 2^{q-1}(|a|^q + |b|^q)$ for $q \geq 1$ to obtain

$$|Y_t - \mathbf{P}_t'\boldsymbol{\vartheta}^*|^m \leq 2^{m-1}(|Y_t|^m + |\mathbf{P}_t'\boldsymbol{\vartheta}^*|^m).$$

As a consequence

$$\mathbb{E}[|Y_t - \mathbf{P}_t'\boldsymbol{\vartheta}^*|^m] \leq 2^{m-1}(\mathbb{E}[|Y_t|^m] + \mathbb{E}[|\mathbf{P}_t'\boldsymbol{\vartheta}^*|^m]) \leq 2^{m-1}C_Z(1 + C_Z^{m/2}). \quad (\text{A.33})$$

Finally, we see that by Cauchy-Schwarz

$$\mathbb{E}[|\mathbf{v}'(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t|^{m/2}] = \mathbb{E}[|Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*|^{m/2} |\mathbf{v}' \mathbf{P}_t|^{m/2}] \leq \mathbb{E}[|Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*|^m]^{1/2} \mathbb{E}[|\mathbf{v}' \mathbf{P}_t|^m]^{1/2}.$$

As a consequence, we obtain from (i) and (A.33)

$$\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}[|\mathbf{v}'(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t|^{m/2}] \leq 2^{(m-1)/2} C_Z \sqrt{1 + C_Z^{m/2}},$$

and then also immediately $\mathbb{E}[\|(Y_t - \mathbf{P}_t' \boldsymbol{\vartheta}^*) \mathbf{P}_t\|_2^{m/2}] \leq 2^{(m-1)/2} C_Z \sqrt{1 + C_Z^{m/2}} K^{m/4}$. $\square$

Lemma A.5. *Let $\{\mathbf{W}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional zero-mean random vectors with beta-mixing coefficients $\beta(l)$. Fix $1 \leq t < s \leq T$ such that $h = s - t$. Then for every $r > 2$ such that $\|\mathbf{W}_t\|_{L_r} < \infty$, we have*

$$|\mathbb{E}[\mathbf{W}_t' \mathbf{W}_s]| \leq 2 \|\mathbf{W}_t\|_{L_r}^2 \beta(h)^{1-2/r}.$$

Proof. By Berbee's Lemma (Bosq, 1998, Lemma 1.1), there exists a copy $\mathbf{W}_s^*$ that satisfies (i) it has the same marginal distribution as $\mathbf{W}_s$, (ii) it is independent of $\mathbf{W}_t$ and (iii) $\mathbb{P}(\mathbf{W}_s^* \neq \mathbf{W}_s) \leq \beta(h)$. As $\mathbb{E}[\mathbf{W}_s^*] = \mathbf{0}$ and $\mathbf{W}_s^*$ is independent of $\mathbf{W}_t$, we have $\mathbb{E}[\mathbf{W}_t' \mathbf{W}_s^*] = 0$, implying $\mathbb{E}[\mathbf{W}_t' \mathbf{W}_s] = \mathbb{E}[\mathbf{W}_t'(\mathbf{W}_s - \mathbf{W}_s^*)]$. Then note that by Cauchy-Schwarz and the triangle inequality

$$\begin{aligned} |\mathbb{E}[\mathbf{W}_t'(\mathbf{W}_s - \mathbf{W}_s^*)]| &= |\mathbb{E}[\mathbf{W}_t'(\mathbf{W}_s - \mathbf{W}_s^*) \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}]| \\ &\leq \mathbb{E}[\|\mathbf{W}_t\|_2 (\|\mathbf{W}_s\|_2 + \|\mathbf{W}_s^*\|_2) \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}] \\ &= \mathbb{E}[\|\mathbf{W}_t\|_2 \|\mathbf{W}_s\|_2 \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}] + \mathbb{E}[\|\mathbf{W}_t\|_2 \|\mathbf{W}_s^*\|_2 \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}]. \end{aligned} \quad (\text{A.34})$$

Fix $r > 2$, let $q = r/(r-2)$ such that $1/r + 1/r + 1/q = 1$, and apply Hölder's inequality

to the first term

$$\begin{aligned}
\mathbb{E}[\|\mathbf{W}_t\|_2 \|\mathbf{W}_s\|_2 \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}] &\leq \|\mathbf{W}_t\|_2 \| \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}} \|_{L_q} \\
&= \|\mathbf{W}_t\|_{L_r} \|\mathbf{W}_s\|_{L_r} \|\mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}\|_{L_q} \\
&\leq \|\mathbf{W}_t\|_{L_r} \|\mathbf{W}_s\|_{L_r} \beta(h)^{1-2/r},
\end{aligned}$$

where the last inequality follows as

$$\|\mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}\|_{L_q} = \mathbb{E}[\mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}^q]^{1/q} = \mathbb{P}[\{\mathbf{W}_s \neq \mathbf{W}_s^*\}]^{1-2/r} \leq \beta(h)^{1-2/r}.$$

Analogous arguments apply to the second term in (A.34) as $\|\mathbf{W}_s\|_{L_r} = \|\mathbf{W}_s^*\|_{L_r}$, yielding

$$\begin{aligned}
|\mathbb{E}[\mathbf{W}_t' \mathbf{W}_s]| &\leq \mathbb{E}[\|\mathbf{W}_t\|_2 \|\mathbf{W}_s\|_2 \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}] + \mathbb{E}[\|\mathbf{W}_t\|_2 \|\mathbf{W}_s^*\|_2 \mathbb{1}_{\{\mathbf{W}_s \neq \mathbf{W}_s^*\}}] \\
&\leq 2 \|\mathbf{W}_t\|_{L_r} \|\mathbf{W}_s\|_{L_r} \beta(h)^{1-2/r}.
\end{aligned}$$

The result then follows from stationarity. $\square$

Lemma A.6. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional random vectors and define $\mathbb{E}(\mathbf{Z}_t \mathbf{Z}_t') = \Sigma_z$. Suppose that (i) there exists a constant $C_0 > 0$ and a deterministic quantity $B \geq 1$ that may depend on T , such that $\|\Sigma_z\|_2 \leq C_0$ and $\|\mathbf{Z}_t\|_2 \leq B\sqrt{d}$ almost surely; (ii) there exists a constant $m > 4$ and a positive constant C_Z such that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}(|\mathbf{v}' \mathbf{Z}_t|^m) \leq C_Z$; (iii) the beta-mixing coefficients of the sequence satisfy $\beta(l) \leq \exp(-C_\beta l^{r_\beta})$ for some $C_\beta > 0$ and $r_\beta \geq 1$.*

Then for every $\eta > 0$ there exists a positive constant $C_\eta > 0$ such that for any $T \geq 2$

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \mathbf{Z}_t' - \Sigma_z \right\|_2 \leq C_\eta \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + B^2 \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. Define $\mathbf{M}_t = \mathbf{Z}_t \mathbf{Z}_t' - \Sigma_z$. We wish to apply Theorem 1 of Banna *et al.* (2016) to this sequence of matrices. We begin by verifying the conditions of the theorem. First, $\mathbf{M}_t$

is symmetric and mean zero. Second, we have that

$$\|\mathbf{M}_t\|_2 \leq \|\mathbf{Z}_t \mathbf{Z}_t'\|_2 + \|\boldsymbol{\Sigma}_z\|_2 \leq B^2 d + C_0 \leq C_1 B^2 d \equiv R, \quad (\text{A.35})$$

almost surely by (i), as $\|\mathbf{Z}_t \mathbf{Z}_t'\|_2 = \|\mathbf{Z}_t\|_2^2$, and the last step follows for a suitable constant $C_1 > 0$ as $B, d \geq 1$. Third, (iii) guarantees that the β -mixing conditions for the theorem are satisfied, since $\beta(l) \leq \exp(-C_\beta l^{r_\beta}) \leq \exp(-C_\beta(l-1))$ for $l \geq 1$. Theorem 1 of Banna *et al.* (2016) and the union bound then imply that there exists a constant $C_2 > 0$ such that for any $x > 0$ we have

$$\mathbb{P}\left(\left\|\sum_{t=1}^T \mathbf{M}_t\right\|_2 \leq x\right) \geq 1 - 2d \exp\left(-\frac{C_2 x^2}{v^2 T + C_\beta^{-1} R^2 + x R \gamma(C_\beta, T)}\right),$$

where

$$v^2 = \sup_{S \subset \{1, \dots, T\}} \frac{1}{|S|} \lambda_{\max}\left(\mathbb{E}\left[\left(\sum_{t \in S} \mathbf{M}_t\right)' \left(\sum_{t \in S} \mathbf{M}_t\right)\right]\right), \quad (\text{A.36})$$

and

$$\gamma(C_\beta, T) = \frac{\log(T)}{\log(2)} \max\left(2, \frac{32 \log(T)}{C_\beta \log(2)}\right). \quad (\text{A.37})$$

We wish to obtain an upper bound that holds with probability at least $1 - T^{-\eta}$. To this end, we will choose x large enough to ensure

$$\frac{C_2 x^2}{v^2 T + C_\beta^{-1} R^2 + x R \gamma(C_\beta, T)} \geq (1 + \eta) \log(T) + \log(2d) \equiv \eta_T. \quad (\text{A.38})$$

It is sufficient to choose

$$x = A \left(\sqrt{v^2 T \eta_T} + R \sqrt{\eta_T} + R \gamma(C_\beta, T) \eta_T \right),$$

where $A > 0$, which guarantees that $v^2 T \leq C_2 x^2 / (3\eta_T)$, $C_\beta^{-1} R^2 \leq C_2 x^2 / (3\eta_T)$ and $x R \gamma(C_\beta, T) \leq C_2 x^2 / (3\eta_T)$ if we choose A sufficiently large, implying that

$$v^2 T + C_\beta^{-1} R^2 + x R \gamma(C_\beta, T) \leq C_2 \frac{x^2}{\eta_T}.$$

With this choice, the theorem implies, after dividing by T , that there exists a constant C_3 such that

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{M}_t \right\|_2 &\leq \sqrt{\frac{3}{C_2}} \left(\sqrt{\frac{v^2 \eta_T}{T}} + R \frac{\sqrt{\eta_T}}{T} + R \frac{\gamma(C_\beta, T) \eta_T}{T} \right) \\ &\leq C_3 \left(\sqrt{\frac{v^2 \eta_T}{T}} + R \frac{\gamma(C_\beta, T) \eta_T}{T} \right), \end{aligned} \quad (\text{A.39})$$

with probability at least $1 - 2de^{-\eta_T} = 1 - T^{-1-\eta} \geq 1 - T^{-\eta}$, because $\gamma(C_\beta, T) \geq 1$ and $\eta_T \geq 1$.

It remains to bound v^2 . Consider (A.36), fix a non-empty set $S \subset \{1, \dots, T\}$ and fix a unit vector $\mathbf{x}$. Then

$$\begin{aligned} \mathbf{x}' \mathbb{E} \left[\left(\sum_{t \in S} \mathbf{M}_t \right)' \left(\sum_{t \in S} \mathbf{M}_t \right) \right] \mathbf{x} &= \mathbb{E} \left(\left\| \sum_{t \in S} \mathbf{M}_t \mathbf{x} \right\|_2^2 \right) \\ &= \sum_{t \in S} \mathbb{E}[\|\mathbf{M}_t \mathbf{x}\|_2^2] + 2 \sum_{t < s \in S} \mathbb{E}[\mathbf{x}' \mathbf{M}_t' \mathbf{M}_s \mathbf{x}]. \end{aligned} \quad (\text{A.40})$$

To bound the diagonal part of (A.40), first note that

$$\mathbf{M}_t' \mathbf{M}_t = \mathbf{Z}_t \mathbf{Z}_t' \mathbf{Z}_t \mathbf{Z}_t' - \mathbf{Z}_t \mathbf{Z}_t' \Sigma_z - \Sigma_z \mathbf{Z}_t \mathbf{Z}_t' + \Sigma_z' \Sigma_z,$$

which implies

$$\mathbb{E}(\mathbf{M}_t' \mathbf{M}_t) = \mathbb{E}(\mathbf{Z}_t \mathbf{Z}_t' \mathbf{Z}_t \mathbf{Z}_t') - \Sigma_z' \Sigma_z = \mathbb{E}(\|\mathbf{Z}_t\|_2^2 \mathbf{Z}_t \mathbf{Z}_t') - \Sigma_z' \Sigma_z.$$

We then have

$$\mathbb{E}[\|\mathbf{M}_t \mathbf{x}\|_2^2] = \mathbf{x}' \mathbb{E}[\mathbf{M}_t' \mathbf{M}_t] \mathbf{x} = \mathbf{x}' \mathbb{E}(\|\mathbf{Z}_t\|_2^2 \mathbf{Z}_t \mathbf{Z}_t') \mathbf{x} - \mathbf{x}' \Sigma_z' \Sigma_z \mathbf{x}.$$

By Cauchy-Schwarz

$$\mathbf{x}' \mathbb{E}(\|\mathbf{Z}_t\|_2^2 \mathbf{Z}_t \mathbf{Z}_t') \mathbf{x} = \mathbb{E}(\|\mathbf{Z}_t\|_2^2 (\mathbf{x}' \mathbf{Z}_t)^2) \leq \mathbb{E}(\|\mathbf{Z}_t\|_2^4)^{1/2} \mathbb{E}((\mathbf{x}' \mathbf{Z}_t)^4)^{1/2} \leq C_Z d,$$

because (ii) implies $\mathbb{E}(\|\mathbf{Z}_t\|_2^m) \leq C_Z d^{m/2}$ by the same arguments as in part (i) of Lemma A.4. As a consequence we obtain

$$\mathbb{E}[\|\mathbf{M}_t \mathbf{x}\|_2^2] \leq C_Z d + \|\boldsymbol{\Sigma}_z \mathbf{x}\|^2 \leq C_4 d, \quad (\text{A.41})$$

for a suitable constant $C_4 > 0$, by (i). As a consequence, we see that the diagonal part is bounded as

$$\sum_{t \in S} \mathbb{E}[\|\mathbf{M}_t \mathbf{x}\|_2^2] \leq C_4 |S| d. \quad (\text{A.42})$$

Now we turn to the off-diagonal part of (A.40). Define $\mathbf{W}_r = \mathbf{M}_r \mathbf{x}$ for all $r \in S$ and consider two values $\mathbf{W}_s$ and $\mathbf{W}_t$, spaced $h = s - t$ time periods apart. Note that $\mathbb{E}[\mathbf{W}_r] = \mathbb{E}[\mathbf{M}_r \mathbf{x}] = 0$. By Lemma A.5 with $r = m/2$, we obtain

$$|\mathbb{E}[\mathbf{W}_t' \mathbf{W}_s]| \leq 2 \|\mathbf{W}_t\|_{L_{m/2}}^2 \beta(h)^{1-4/m} \leq C_5 d \beta(h)^{1-4/m},$$

for some constant $C_5 > 0$, where the latter inequality follows as

$$\begin{aligned} \|\mathbf{W}_t\|_{L_{m/2}} &= \|\mathbf{Z}_t \mathbf{Z}_t' \mathbf{x} - \boldsymbol{\Sigma}_z \mathbf{x}\|_{L_{m/2}} \leq \|\|\mathbf{Z}_t\|_2 (\mathbf{Z}_t' \mathbf{x})\|_{L_{m/2}} + \|\boldsymbol{\Sigma}_z \mathbf{x}\|_2 \\ &\leq \|\|\mathbf{Z}_t\|_2\|_{L_m} \|\mathbf{Z}_t' \mathbf{x}\|_{L_m} + C_0, \end{aligned}$$

by Hölder's inequality and (i), and then applying (ii) to the first term. We can then obtain a bound on the off-diagonal part of (A.40) by noting that there are at most $|S|$ pairs with $h = s - t$

$$\begin{aligned} \sum_{t < s \in S} \mathbb{E}[\mathbf{x}' \mathbf{M}_t' \mathbf{M}_s \mathbf{x}] &\leq C_5 |S| d \sum_{h=1}^{\infty} \beta(h)^{1-4/m} \\ &\leq C_5 |S| d \sum_{h=1}^{\infty} \exp(-(1 - 4/m) C_\beta h^{r_\beta}) \\ &\leq C_6 |S| d, \end{aligned}$$

where we used (iii). Combining this with the diagonal bound from (A.42) and putting

into (A.40)

$$\begin{aligned}
\mathbf{x}' \mathbb{E} \left[\left(\sum_{t \in S} \mathbf{M}_t \right)' \left(\sum_{t \in S} \mathbf{M}_t \right) \right] \mathbf{x} &= \sum_{t \in S} \mathbb{E}[\|\mathbf{M}_t \mathbf{x}\|_2^2] + 2 \sum_{t < s \in S} \mathbb{E}[\mathbf{x}' \mathbf{M}_t' \mathbf{M}_s \mathbf{x}] \\
&\leq C_4 |S| d + 2C_6 |S| d \\
&\leq C_v |S| d,
\end{aligned}$$

with $C_v = C_4 + 2C_6$. Finally, dividing both sides by $|S|$, and taking the supremum over all unit vectors $\mathbf{x}$ and all non-empty subsets $S \subset \{1, \dots, T\}$, we obtain

$$v^2 \leq C_v d. \quad (\text{A.43})$$

Finally, going back to (A.39) and inserting (A.35) along with (A.43) implies that there exists a constant $C > 0$ such that

$$\begin{aligned}
\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{M}_t \right\|_2 &\leq C_3 \left(\sqrt{\frac{v^2 \eta_T}{T}} + R \frac{\gamma(C_\beta, T) \eta_T}{T} \right) \\
&\leq C_3 \left(\sqrt{C_v} \sqrt{\frac{d \eta_T}{T}} + C_1 B^2 \frac{d \gamma(C_\beta, T) \eta_T}{T} \right) \\
&\leq C \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + B^2 \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \right),
\end{aligned}$$

with probability at least $1 - T^{-\eta}$, where we used (A.38) and (A.37) in the final step. $\square$

Lemma A.7. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional random vectors and define $\Sigma_z = \mathbb{E}[\mathbf{Z}_t \mathbf{Z}_t']$. Suppose (i) there exists a constant $m > 4$ and a positive constant C_Z such that $\sup_{\|\mathbf{v}\|_2=1} \mathbb{E}(|\mathbf{v}' \Sigma_z^{-1/2} \mathbf{Z}_t|^m) \leq C_Z$; (ii) the beta-mixing coefficients of the sequence satisfy $\beta(l) \leq \exp(-C_\beta l^{r_\beta})$ for some $C_\beta > 0$ and $r_\beta \geq 1$.*

Then for every $\eta > 0$ and every $\nu > (1 + \eta)/m$ there exists a constant $C_\eta > 0$ such that, for all sufficiently large T

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \mathbf{Z}_t' - \Sigma_z \right\|_2 \leq C_\eta \|\Sigma_z\|_2 \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^{2\nu} \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. If $\Sigma_z = \mathbf{0}$, $\mathbf{Z}_t = 0$ almost surely and the result follows trivially, so we assume otherwise. Define $\mathbf{W}_t = \Sigma_z^{-1/2} \mathbf{Z}_t$ and let Π_z be the orthogonal projector onto the range of Σ_z as in the proof of Lemma A.2. As $\mathbf{Z}_t = \Sigma_z^{1/2} \Sigma_z^{-1/2} \mathbf{Z}_t$ almost surely, we have almost surely

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \mathbf{Z}_t' - \Sigma_z \right\|_2 &= \left\| \Sigma_z^{1/2} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \Pi_z \right) \Sigma_z^{1/2} \right\|_2 \\ &\leq \|\Sigma_z\|_2 \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \Pi_z \right\|_2. \end{aligned} \quad (\text{A.44})$$

It is therefore sufficient for us to bound the right hand side.

Define

$$\overline{\mathbf{W}}_t = \mathbf{W}_t \mathbb{1}_{\{\|\mathbf{W}_t\|_2 \leq \sqrt{dT^\nu}\}}, \quad (\text{A.45})$$

so that $\|\overline{\mathbf{W}}_t\|_2 \leq \sqrt{dT^\nu}$. Then define the event $\mathcal{E}_T = \{\max_{1 \leq t \leq T} \|\mathbf{W}_t\|_2 \leq \sqrt{dT^\nu}\}$ and note that the union bound followed by Markov's inequality implies

$$\mathbb{P}(\mathcal{E}_T^c) \leq \sum_{t=1}^T \mathbb{P}(\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}) \leq \sum_{t=1}^T \frac{\mathbb{E}[\|\mathbf{W}_t\|_2^m]}{d^{m/2} T^{\nu m}}.$$

Since $\nu m - 1 > \eta$, define $\eta' = (\eta + \nu m - 1)/2$ so that $\eta < \eta' < \nu m - 1$. Then, as (i) implies $\mathbb{E}(\|\Sigma_z^{-1/2} \mathbf{Z}_t\|_2^m) \leq C_Z d^{m/2}$ by the same argument as in part (i) of Lemma A.4, we have

$$\mathbb{P}(\mathcal{E}_T^c) \leq \sum_{t=1}^T C_Z T^{-\nu m} = C_Z T^{1-\nu m} \leq T^{-\eta'}. \quad (\text{A.46})$$

Next add and subtract to decompose the object of interest as

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \Pi_z &= \frac{1}{T} \sum_{t=1}^T \overline{\mathbf{W}}_t \overline{\mathbf{W}}_t' - \mathbb{E}[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t'] \\ &\quad + \mathbb{E}[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t'] - \mathbb{E}[\mathbf{W}_t \mathbf{W}_t'] \\ &\quad + \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \frac{1}{T} \sum_{t=1}^T \overline{\mathbf{W}}_t \overline{\mathbf{W}}_t', \end{aligned}$$

where $\mathbb{E}[\mathbf{W}_t \mathbf{W}_t'] = \mathbf{\Pi}_z$. Note that on the event $\mathcal{E}_T$, $\mathbf{W}_t = \overline{\mathbf{W}}_t$ for all t so the last difference is zero. Hence, on $\mathcal{E}_T$, we have

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \mathbf{\Pi}_z \right\|_2 &\leq \left\| \frac{1}{T} \sum_{t=1}^T \overline{\mathbf{W}}_t \overline{\mathbf{W}}_t' - \mathbb{E}[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t'] \right\|_2 \\ &\quad + \left\| \mathbb{E}[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t'] - \mathbb{E}(\mathbf{W}_t \mathbf{W}_t') \right\|_2 \\ &\equiv \|\mathbf{A}_T\|_2 + \|\mathbf{B}_T\|_2. \end{aligned} \tag{A.47}$$

We first bound $\|\mathbf{B}_T\|_2$. By (A.45)

$$\mathbf{W}_t \mathbf{W}_t' - \overline{\mathbf{W}}_t \overline{\mathbf{W}}_t' = \mathbf{W}_t \mathbf{W}_t' \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}},$$

so that

$$\mathbf{B}_T = \mathbb{E}[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t' - \mathbf{W}_t \mathbf{W}_t'] = -\mathbb{E}[\mathbf{W}_t \mathbf{W}_t' \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}]$$

and thus

$$\|\mathbf{B}_T\|_2 = \left\| \mathbb{E}[\mathbf{W}_t \mathbf{W}_t' \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}] \right\|_2 = \sup_{\|\mathbf{x}\|_2=1} \mathbb{E}[(\mathbf{x}' \mathbf{W}_t)^2 \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}].$$

For a given $\mathbf{x}$ such that $\|\mathbf{x}\|_2 = 1$, Hölder's inequality gives

$$\begin{aligned} \mathbb{E}[(\mathbf{x}' \mathbf{W}_t)^2 \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}] &\leq \|(\mathbf{x}' \mathbf{W}_t)^2\|_{L_r} \|\mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}\|_{L_s} \\ &= (\mathbb{E}[|\mathbf{x}' \mathbf{W}_t|^m])^{2/m} (\mathbb{P}(\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}))^{1-2/m}, \end{aligned}$$

with $r = m/2$ and $s = m/(m-2)$. Using Markov's inequality as before followed by (i), we obtain

$$\begin{aligned} \mathbb{E}[(\mathbf{x}' \mathbf{W}_t)^2 \mathbb{1}_{\{\|\mathbf{W}_t\|_2 > \sqrt{dT^\nu}\}}] &\leq (\mathbb{E}[|\mathbf{x}' \mathbf{W}_t|^m])^{2/m} \left(\frac{\mathbb{E}[\|\mathbf{W}_t\|_2^m]}{d^{m/2} T^{\nu m}} \right)^{1-2/m} \\ &\leq C_Z^{2/m} \left(\frac{C_Z}{T^{\nu m}} \right)^{1-2/m}. \end{aligned}$$

As a consequence, we obtain

$$\|\mathbf{B}_T\|_2 \leq C_Z^{2/m} \left(\frac{C_Z}{T^{\nu m}} \right)^{1-2/m} = C_Z T^{-\nu(m-2)}. \quad (\text{A.48})$$

Next we bound $\|\mathbf{A}_T\|_2$. Before applying Lemma A.6 to $\overline{\mathbf{W}}_t$, we verify that its conditions are satisfied. First, we may note that

$$\left\| \mathbb{E} \left[\overline{\mathbf{W}}_t \overline{\mathbf{W}}_t' \right] \right\|_2 \leq \left\| \mathbb{E} [\mathbf{W}_t \mathbf{W}_t'] \right\|_2 = \|\boldsymbol{\Pi}_z\|_2 \leq 1.$$

Second, we saw before that $\|\overline{\mathbf{W}}_t\|_2 \leq \sqrt{d}T^\nu$ so that we may take $B = T^\nu$ in the lemma. Third, (i) implies the necessary moment conditions for $\overline{\mathbf{W}}_t$. Finally, the beta-mixing coefficients of $\overline{\mathbf{W}}_t$ are bounded by those of $\mathbf{W}_t$. Recall that $\eta' = (\eta + \nu m - 1)/2$. Lemma A.6 then implies that there exists a positive constant $C > 0$ such that

$$\|\mathbf{A}_T\|_2 \leq C \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^{2\nu} \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \right), \quad (\text{A.49})$$

holds with probability at least $1 - T^{-\eta'}$.

Hence, on the intersection of $\mathcal{E}_T$ and the event of Lemma A.6, we have by combining (A.49) and (A.48) with (A.47)

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \boldsymbol{\Pi}_z \right\|_2 &\leq \|\mathbf{A}_T\|_2 + \|\mathbf{B}_T\|_2 \\ &\leq C \sqrt{\frac{d(\log(2d) + \log(T))}{T}} \\ &\quad + C T^{2\nu} \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \\ &\quad + C_Z T^{-\nu(m-2)}. \end{aligned}$$

Notice that the last term is dominated by the second, since $d \log(T)^2 \geq 1$ and $\log(2d) + \log(T) \geq 1$ for all sufficiently large T and

$$T^{-\nu(m-2)} \leq T^{2\nu} T^{-1},$$

because $\nu > 1/m$ implies $\nu(m-2) + 2\nu = \nu m > 1$. We therefore have, on the same events, that there exists a constant $C_\eta > 0$ such that for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{W}_t \mathbf{W}_t' - \boldsymbol{\Pi}_z \right\|_2 \leq C_\eta \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^{2\nu} \frac{d \log(T)^2 (\log(2d) + \log(T))}{T} \right).$$

It only remains to work out the probability of the intersection of the two events. The truncation event fails with probability at most $T^{-\eta'}$ from (A.46), and the bound from (A.49) fails with probability at most $T^{-\eta'}$. Both events thus occur with probability at least $1 - 2T^{-\eta'} \geq 1 - T^{-\eta}$. Combining this result with (A.44) then proves the claim. $\square$

Lemma A.8. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional zero-mean random vectors. Suppose that (i) there exists a deterministic quantity $B \geq 1$ that may depend on T such that $\|\mathbf{Z}_t\|_2 \leq B\sqrt{d}$ almost surely; (ii) there exists a constant $m > 2$ and a positive constant C_Z such that $\mathbb{E}(\|\mathbf{Z}_t\|_2^m) \leq C_Z d^{m/2}$; (iii) the beta-mixing coefficients of the sequence satisfy $\beta(l) \leq \exp(-C_\beta l^{r_\beta})$ for some $C_\beta > 0$ and $r_\beta \geq 1$.*

Then for every $\eta > 0$ there exists a positive constant $C_\eta > 0$ such that for any $T \geq 2$

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \right\|_2 \leq C \left(\sqrt{d \frac{(\log(2d) + \log(T))}{T}} + B\sqrt{d} \frac{\log(T)^2 (\log(2d) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. We follow a similar strategy as in the proof of Lemma A.6. Define the $(d+1) \times (d+1)$ matrix

$$\mathbf{M}_t = \begin{pmatrix} \mathbf{0}_{d \times d} & \mathbf{Z}_t \\ \mathbf{Z}_t' & 0 \end{pmatrix}.$$

We wish to apply Theorem 1 of Banna *et al.* (2016) to this sequence of matrices. We begin by verifying the conditions of the theorem. First, $\mathbf{M}_t$ is symmetric and mean zero. Second, we have that

$$\|\mathbf{M}_t\|_2 = \|\mathbf{Z}_t\|_2 \leq B\sqrt{d} \equiv R, \tag{A.50}$$

almost surely by (i). Third, (iii) guarantees that the β -mixing conditions for the theorem are satisfied, since $\beta(l) \leq \exp(-C_\beta l^{r_\beta}) \leq \exp(-C_\beta(l-1))$ for $l \geq 1$. Following the proof

of Lemma A.6 and applying Theorem 1 of Banna *et al.* (2016) then implies that there exists a constant $C_1 > 0$ such that

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{M}_t \right\|_2 \leq C_1 \left(\sqrt{\frac{v^2(\log(2d) + \log(T))}{T}} + \frac{R \log(T)^2(\log(2d) + \log(T))}{T} \right), \quad (\text{A.51})$$

with probability at least $1 - T^{-\eta}$, where we used $d + 1 \leq 2d$ and v^2 is as (A.36).

We then require a bound on v^2 . Let $\mathbf{Q} = \sum_{t \in S} \mathbf{Z}_t$ and notice that

$$\left(\sum_{t \in S} \mathbf{M}_t \right)' \left(\sum_{t \in S} \mathbf{M}_t \right) = \begin{pmatrix} \mathbf{Q}\mathbf{Q}' & \mathbf{0} \\ \mathbf{0} & \|\mathbf{Q}\|_2^2 \end{pmatrix}.$$

As a consequence,

$$\lambda_{\max} \left(\mathbb{E} \left[\left(\sum_{t \in S} \mathbf{M}_t \right)' \left(\sum_{t \in S} \mathbf{M}_t \right) \right] \right) = \max\{\lambda_{\max}(\mathbb{E}[\mathbf{Q}\mathbf{Q}']), \mathbb{E}[\|\mathbf{Q}\|_2^2]\} = \mathbb{E}[\|\mathbf{Q}\|_2^2],$$

as $\lambda_{\max}(\mathbf{Q}\mathbf{Q}') \leq \text{tr}(\mathbf{Q}\mathbf{Q}') = \|\mathbf{Q}\|_2^2$. It then follows from (A.36) that

$$v^2 \leq \sup_{S \subset \{1, \dots, T\}} \frac{1}{|S|} \mathbb{E} \left[\left\| \sum_{t \in S} \mathbf{Z}_t \right\|_2^2 \right]. \quad (\text{A.52})$$

We thus consider

$$\mathbb{E} \left[\left\| \sum_{t \in S} \mathbf{Z}_t \right\|_2^2 \right] = \sum_{t \in S} \mathbb{E}[\|\mathbf{Z}_t\|_2^2] + 2 \sum_{t < s \in S} \mathbb{E}[\mathbf{Z}_t' \mathbf{Z}_s] \leq |S| C_Z d + 2 \sum_{t < s \in S} |\mathbb{E}[\mathbf{Z}_t' \mathbf{Z}_s]|, \quad (\text{A.53})$$

by (ii). We now turn to the off-diagonal part of (A.53). Fix $h = s - t$. By Lemma A.5 with $r = m$, we obtain

$$|\mathbb{E}[\mathbf{Z}_t' \mathbf{Z}_s]| \leq 2 \|\mathbf{Z}_t\|_{L_m}^2 \beta(h)^{1-2/m} \leq 2 C_Z^{2/m} d \beta(h)^{1-2/m},$$

as $\|\mathbf{Z}_t\|_{L_m} \leq C_Z^{1/m} d^{1/2}$ by (ii). As there are at most $|S|$ pairs with $h = s - t$, we thus obtain by (iii)

$$\sum_{t < s \in S} \mathbb{E}[\mathbf{Z}_t' \mathbf{Z}_s] \leq 2 C_Z^{2/m} |S| d \sum_{h=1}^{\infty} \beta(h)^{1-2/m} \leq C_2 |S| d,$$

for some constant $C_2 > 0$. We then have by (A.53)

$$\mathbb{E} \left[\left\| \sum_{t \in S} \mathbf{Z}_t \right\|_2^2 \right] \leq |S|C_Z d + 2C_2 |S|d \leq C_v |S|d,$$

with $C_v = C_Z + 2C_2$. Finally, dividing both sides by $|S|$, and taking the supremum over all non-empty subsets $S \subset \{1, \dots, T\}$, we obtain from (A.52)

$$v^2 \leq C_v d. \quad (\text{A.54})$$

Finally, going back to (A.51) and inserting (A.50) along with (A.54) implies that there exists a constant $C > 0$ such that

$$\begin{aligned} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{M}_t \right\|_2 &\leq C_1 \left(\sqrt{\frac{v^2(\log(2d) + \log(T))}{T}} + \frac{R \log(T)^2(\log(2d) + \log(T))}{T} \right) \\ &\leq C \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + B \frac{\sqrt{d} \log(T)^2(\log(2d) + \log(T))}{T} \right), \end{aligned}$$

with probability at least $1 - T^{-\eta}$, which implies the claim. $\square$

Lemma A.9. *Let $\{\mathbf{Z}_t\}_{t=1}^T$ be a stationary sequence of d -dimensional zero-mean random vectors. Suppose (i) there exists a constant $m > 2$ and a positive constant C_Z such that $\mathbb{E}(\|\mathbf{Z}_t\|_2^m) \leq C_Z d^{m/2}$; (ii) the beta-mixing coefficients of the sequence satisfy $\beta(l) \leq \exp(-C_\beta l^{r_\beta})$ for some $C_\beta > 0$ and $r_\beta \geq 1$.*

Then for every $\eta > 0$ and every $\nu > (1 + \eta)/m$ there exists a constant $C_\eta > 0$ such that, for all sufficiently large T

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \right\|_2 \leq C_\eta \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^\nu \frac{\sqrt{d} \log(T)^2(\log(2d) + \log(T))}{T} \right),$$

holds with probability at least $1 - T^{-\eta}$.

Proof. We follow the proof of Lemma A.7 closely. Define

$$\overline{\mathbf{Z}}_t = \mathbf{Z}_t \mathbb{1}_{\{\|\mathbf{Z}_t\|_2 \leq \sqrt{dT}^\nu\}}, \quad (\text{A.55})$$

which implies $\|\bar{\mathbf{Z}}_t\|_2 \leq \sqrt{dT^\nu}$. Then define the event $\mathcal{E}_T = \{\max_{1 \leq t \leq T} \|\mathbf{Z}_t\|_2 \leq \sqrt{dT^\nu}\}$ and note that

$$\mathbb{P}(\mathcal{E}_T^c) \leq \sum_{t=1}^T \mathbb{P}(\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu}) \leq \sum_{t=1}^T \frac{\mathbb{E}[\|\mathbf{Z}_t\|_2^m]}{d^{m/2}T^{\nu m}}.$$

Since $\nu m - 1 > \eta$, define $\eta' = (\eta + \nu m - 1)/2$ so that $\eta < \eta' < \nu m - 1$. Then using (i), we obtain

$$\mathbb{P}(\mathcal{E}_T^c) \leq \sum_{t=1}^T C_Z T^{-\nu m} = C_Z T^{1-\nu m} \leq T^{-\eta'}, \quad (\text{A.56})$$

for all sufficiently large T .

Next add and subtract to decompose the object of interest as

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t &= \frac{1}{T} \sum_{t=1}^T \bar{\mathbf{Z}}_t - \mathbb{E}[\bar{\mathbf{Z}}_t] \\ &\quad + \mathbb{E}[\bar{\mathbf{Z}}_t] \\ &\quad + \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t - \frac{1}{T} \sum_{t=1}^T \bar{\mathbf{Z}}_t. \end{aligned}$$

Note that on the event $\mathcal{E}_T$, $\mathbf{Z}_t = \bar{\mathbf{Z}}_t$ for all t so the last difference is zero. Hence, on $\mathcal{E}_T$, we have

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t \right\|_2 \leq \left\| \frac{1}{T} \sum_{t=1}^T \bar{\mathbf{Z}}_t - \mathbb{E}[\bar{\mathbf{Z}}_t] \right\|_2 + \|\mathbb{E}[\bar{\mathbf{Z}}_t]\|_2 \equiv \|\mathbf{A}_T\|_2 + \|\mathbf{B}_T\|_2. \quad (\text{A.57})$$

We first bound $\|\mathbf{B}_T\|_2$. By (A.55), $\mathbf{Z}_t - \bar{\mathbf{Z}}_t = \mathbf{Z}_t \mathbb{1}_{\{\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu}\}}$. As $\mathbb{E}[\mathbf{Z}_t] = 0$, we thus obtain

$$\mathbf{B}_T = \mathbb{E}[\bar{\mathbf{Z}}_t] = \mathbb{E}[\bar{\mathbf{Z}}_t - \mathbf{Z}_t] = -\mathbb{E}[\mathbf{Z}_t \mathbb{1}_{\{\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu}\}}]$$

and Hölder's inequality therefore gives us

$$\begin{aligned} \|\mathbf{B}_T\|_2 &= \left\| \mathbb{E}[\mathbf{Z}_t \mathbb{1}_{\{\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu}\}}] \right\|_2 \leq \left\| \|\mathbf{Z}_t\|_{L_r} \|\mathbb{1}_{\{\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu}\}}\|_{L_s} \right\|_2 \\ &= \mathbb{E}[\|\mathbf{Z}_t\|_2^m]^{1/m} \mathbb{P}(\|\mathbf{Z}_t\|_2 > \sqrt{dT^\nu})^{1-1/m}, \end{aligned}$$

with $r = m$ and $s = m/(m-1)$. Using Markov's inequality as before followed by (i), we

obtain

$$\begin{aligned} \left\| \mathbb{E} \left[\mathbf{Z}_t \mathbb{1}_{\{\|\mathbf{Z}_t\|_2 > \sqrt{d}T^\nu\}} \right] \right\|_2 &\leq \mathbb{E} [\|\mathbf{Z}_t\|_2^m]^{1/m} \left(\frac{\mathbb{E} [\|\mathbf{Z}_t\|_2^m]}{d^{m/2} T^{\nu m}} \right)^{1-1/m} \\ &\leq C_Z^{1/m} \sqrt{d} \left(\frac{C_Z}{T^{\nu m}} \right)^{1-1/m}. \end{aligned}$$

As a consequence, we obtain

$$\|\mathbf{B}_T\|_2 \leq C_Z^{1/m} \sqrt{d} \left(\frac{C_Z}{T^{\nu m}} \right)^{1-1/m} = C_Z \sqrt{d} T^{-\nu(m-1)}. \quad (\text{A.58})$$

Next we bound $\|\mathbf{A}_T\|_2$. Let $\mathbf{W}_t \equiv \bar{\mathbf{Z}}_t - \mathbb{E}[\bar{\mathbf{Z}}_t]$. Before applying Lemma A.8 to $\mathbf{W}_t$, we verify that its conditions are satisfied. First, $\mathbb{E}[\mathbf{W}_t] = \mathbf{0}$. Second, we may note that

$$\|\mathbb{E}[\bar{\mathbf{Z}}_t]\|_2 \leq \mathbb{E} [\|\bar{\mathbf{Z}}_t\|_2] \leq \mathbb{E} [\|\bar{\mathbf{Z}}_t\|_2^2]^{1/2} \leq \mathbb{E} [\|\mathbf{Z}_t\|_2^2]^{1/2}$$

by Jensen's inequality and then Cauchy-Schwarz. It then follows by (i) that $\mathbb{E} [\|\mathbf{Z}_t\|_2^2]^{1/2} \leq \mathbb{E} [\|\mathbf{Z}_t\|_2^m]^{1/m} \leq C_Z^{1/m} \sqrt{d}$. We thus obtain

$$\|\mathbf{W}_t\|_2 \leq \|\bar{\mathbf{Z}}_t\|_2 + \|\mathbb{E}[\bar{\mathbf{Z}}_t]\|_2 = \sqrt{d}T^\nu + C_Z^{1/m} \sqrt{d} \leq C_1 \sqrt{d}T^\nu,$$

for some constant $C_1 > 0$, because $\|\bar{\mathbf{Z}}_t\|_2 \leq \sqrt{d}T^\nu$ as we saw before. Third, (i) implies the necessary moment conditions on $\mathbf{W}_t$. Finally, the beta-mixing coefficients of $\mathbf{W}_t$ follow from those of $\mathbf{Z}_t$. Recalling our choice of η' above and taking $B = C_1 T^\nu$, Lemma A.8 then implies that there exists a positive constant $C > 0$ such that

$$\|\mathbf{A}_T\|_2 \leq C \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^\nu \frac{\sqrt{d} \log(T)^2 (\log(2d) + \log(T))}{T} \right), \quad (\text{A.59})$$

holds with probability at least $1 - T^{-\eta'}$.

Hence, on the intersection of $\mathcal{E}_T$ and the event of Lemma A.8, we have

$$\begin{aligned}
\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{z}_t \right\|_2 &\leq \|\mathbf{A}_T\|_2 + \|\mathbf{B}_T\|_2 \\
&\leq C\sqrt{d} \sqrt{\frac{(\log(2d) + \log(T))}{T}} \\
&\quad + CT^\nu \sqrt{d} \frac{\log(T)^2 (\log(2d) + \log(T))}{T} \\
&\quad + C_Z \sqrt{d} T^{-\nu(m-1)}.
\end{aligned}$$

Note that the last term is dominated by the second by analogous arguments as in the proof of Lemma A.7. We therefore have, on the same events, that there exists a constant $C_\eta > 0$ such that for all T sufficiently large

$$\left\| \frac{1}{T} \sum_{t=1}^T \mathbf{z}_t \right\|_2 \leq C_\eta \left(\sqrt{\frac{d(\log(2d) + \log(T))}{T}} + T^\nu \frac{\sqrt{d} \log(T)^2 (\log(2d) + \log(T))}{T} \right).$$

It only remains to work out the probability of the intersection of the two events. The truncation event fails with probability at most $T^{-\eta'}$ from (A.56), and the bound from (A.59) fails with probability at most $T^{-\eta'}$. Both events thus occur with probability at least $1 - 2T^{-\eta'} \geq 1 - T^{-\eta}$. $\square$

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